



SED

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CHANGE RECORD

Version	Date	Changed chapters	Remarks
1.0	2026-Jan-14	All added	PDR
2.0	2026-May-24	4.9.3 *Dictyostelium* Culture Conditions and Biological Constraints Table 4.11 : *Dictyostelium* Biological Constraints for Experiment Design (based on [13]) 4.9.5 Gravity-Dependent Cellular Mechanisms 4.9.7 Laboratory Facilities and Implementation Plan Verification Matrix Table 5.1 : Verification table Abbreviations and References Appendix/Annex 1.5.2 Team members 4.5 Electronics design 4.7 Power system 4.4.4 Microfludic assembly 4.8 software design 4.9 ground support equipment	CDR
3.0	2026-Aug-19	1.5.2 Outreach lead changed 4.4 Wire-ropes removed 4.4.2 Pressure vessel internal	IPR

		assembly details given 4.8 Imaging characteristics specified, data storage details	
3.1	2026-Sep-14	4.4 EMI box changed. Hard-end stops added for PV insertion. 4.8 ESP/Pi interface changed. 5.2 Specific biology tests added 6.3 Many operational details added for biology.	IPR follow-up

Keywords: REXUS, SED - Student Experiment Documentation, STAR-D, Biology during microgravity

TABLE OF CONTENTS

Contents

TEAM LIST 2
CHANGE RECORD 3
TABLE OF CONTENTS 5
LIST OF FIGURES 9
LIST OF TABLES 11
PREFACE 14
ABSTRACT 15
1 INTRODUCTION 16
1.1 Scientific/Technical Background 16
1.2 Mission Statement 19
1.3 Experiment Objectives 20
1.4 Experiment Concept 20
1.5 Team Details 22
1.5.1 Contact Point 22
1.5.2 Team Members 22
2 EXPERIMENT REQUIREMENTS AND CONSTRAINTS 24
2.1 Functional Requirements 24
2.2 Performance Requirements 25
2.3 Design Requirements 27
2.4 Operational Requirements 29
2.5 Constraints 30
3 PROJECT PLANNING 31
3.1 Work Breakdown Structure (WBS) 31
3.2 Schedule 32
3.2.1 Workforce 34
3.2.2 Budget 40
Funding: 42
3.2.3 External Support 42
3.3 Outreach Approach 43
3.4 Risk Register 43
4 EXPERIMENT DESCRIPTION 51
4.1 Experiment Setup 51

4.2 Experiment Interfaces 51
4.2.1 Mechanical 51
Cable routing and connector accessibility 51
4.2.2 Electrical 52
4.3 Experiment Components 55
4.4 Mechanical Design 56
4.4.1 Mechanical Structure Design Overall mechanical architecture 56
Early access hatch and integration concept 59
Bottom bulkhead arrangement 60
Rigid support structure 62
Load-carrying structure (Experiment Support Frame) 63
General Mechanical Assembly 65
4.4.2 Pressure Vessel 69
Pressure Vessel Design 69
Pressure Vessel guidance and locking 71
Load path definition 74
Pressure vessel guidance and retention 75
4.4.4 Microfluidic Assembly (MA) 76
Microscope design 77
Microfluidic System Design 81
Pressure vessel assembly Tolerances 83
Electronics integration 88
4.5 Electronics Design 88
4.6 Thermal Design 99
4.7 Power System 108
4.8 Software Design 112
4.8.1 – Mission overview relevant to software operations 112
4.8.2 – Software operational concept – State machine 114
4.8.3 – Data flow and management 118
Table 4.15 - Data storage formats and logging frequency 123
4.8.3.3 Data output and telemetry 124
4.8.4 – Software constraints and requirements overview 126
4.8.5 – Verification and validation overview 126
4.9 Ground Support Equipment 126
4.9.1 Purpose and scope 126
4.9.2 Uplink commands 127
4.9.3 Downlink handling 128
4.9.4 Interface, operator display 128

4.10 Biology 128
4.10.1 Overview of the Biological subsystem 128
4.10.2 Biological Model 129
4.10.4 State of the Art of Biological Experiments under Altered Gravity 134
4.10.5 Gravity-Dependent Cellular Mechanisms 135
4.10.6 Bio–Microfluidic Interface 137
4.10.7 Laboratory Facilities and Implementation Plan 139
5 EXPERIMENT VERIFICATION AND TESTING 142
5.1 Verification Matrix 142
5.2 Verification Plan 145
5.3 Verification Results 150
6 LAUNCH CAMPAIGN PREPARATION 154
6.1 Input for the Campaign / Flight Requirement Plans 154
6.1.1 Dimensions and Mass 154
6.1.2 Ground and Flight Safety Risks 154
6.1.3 Electrical Interfaces 155
6.1.4 Launch Site Requirements 156
6.1.5 Flight Requirements 157
6.1.6 Accommodation Requirements 157
6.2 Preparation and Test Activities at Esrange 159
6.3 Timeline for Countdown and Flight 165
Sample Preparation and Operational Timeline 165
Biological Validity Window 181
Biological Sample Preparation Time and Operational Lifetime 182
Biological Material Transport to Esrange 187
6.4 Post-Flight Activities 190
7 DATA ANALYSIS AND RESULTS 191
7.1 Data Analysis Plan 191
Nature of the Data 191
Definition of a Phagocytosis Event 191
Primary Analysis Metrics 191
Image Acquisition Parameters 193
Analysis Methodology 194
Ground Control Comparison 194
Data Quality Assessment 195
Expected Outcomes 195
Experiment Recovery 195

8 ABBREVIATIONS AND REFERENCES 197
8.1 Abbreviations 197
8.2 References 198
Appendix D – CHECKLISTS 204

Please find attached in appendix X the preliminary mechanical simulation conducted in the isolation frame.

LIST OF FIGURES

Figure 1: Schematic Representation of Phagosome Maturation in Dictyostelium discoideum

Figure 2: Breakdown Structure

Figure 3: WBS

Figure 4: Gantt Chart

Figure 5: Team Structure

Figure 6: Project Schedule and Task Overview

Figure 7: Simplified Internal Electronics Architecture

Figure 8: Module Perspective View

Figure 9: Module Perspective View without REXUS module

Figure 10: Experiment High Rear View

Figure 11: Experiment Assembly In Front of the Early Access Hatch

Figure 12: Bottom View of the BulkHead

Figure 13: Back View of the Experiment Assembly

Figure 14: Front View of the Experiment Assembly

Figure 15: Perspective View of the Experiment Support Frame

Figure 16: Perspective View of the Experiment Support Frame

Figure 17: Front View of the Experiment Assembly

Figure 17.1: Perspective vision

Figure 17.2: Locking mechanism component

Figure 17.3: Locking mechanism

Figure 17.4: PCB Box

Figure 18: Pressure vessel parts

Figure 19: Front Perspective View of the Pressure Vessel inside the Experiment Support Frame

Figure 20: Front Perspective View of the Experiment Assembly

Figure 21: Back View of the Pressure Vessel

Figure 22: Front View of the Experiment Assembly

Figure 23: Back View of the Experiment Assembly

Figure 24: Rails

Figures 25: Whole microfluidic assembly - two view angles

Figure 26: Diagram of the microscope setup (before camera modification)

Figure 27: Representation of the microscope system

Figure 28: Light path inside of the microscope focus

Figure 29: Camera mounting and ribbon securer

Figure 30: Microscope and microfluidic assembly

Figure 31: Microfluidic system diagram

Figure 32: Prototype of the microfluidic system

Figure 33: Microfluidic pump setup

Figure 34: Objective assembly

Figures 34.1: Insert dichroic mirror into the internal filter placement hole

Figures 34.2: Insert filter into the external placement hole

Figures 34.3: Add the filter cap

Figure 34.4: Using a M2x6 screw tighten filter cap

Figure 34.5: In the side of the assembly insert the mirror and align its cap

Figure 34.6: Using a M2x8 tighten the mirror cap

Figures 34.7, 34.8: Push the 12.5mm lens until its blocked

Figures 34.9, 34.10: Push the second 12.5mm lens (identical) until its blocked

Figures 34.11: Screw the 40x objective onto the thread

Figure 34.12: Align the LED PCB over the lens and Pi cam and prepare 4 M2x3 and 4 M2x6

Figures 34.13: Tighten the screws

Figure 35: Updated STAR-D Main Board layout integrating the TMC2209 stepper motor driver and locking connectors

Figure 36: Simplified Schematic of ESP32 PCB

Figure 37: Optocoupler interface between REXUS service signals and onboard control electronics

Figure 38: Potentiometer interface with analog input (ADC)

Figure 39: ESP32-S3 microcontroller integration and peripheral interfaces

Figure 40: RS-422 transceiver interface for experiment telemetry communication

Figure 41: LSM6DSM IMU integration and SPI interface

Figure 42: Sensor interface connectors for ABP pressure sensor (SPI) and SLF3S flow sensor (I2C)

Figure 43: Power PCB schematic

Figure 44: Power PCB Layout

Figure 45: Power PCB top view

Figure 46: D-Sub interfaces located on the Power Board

Figure 47: PCB LED Matrix

Figure 48: Top and Perspective View of the PCB

Figure 49: STAR-D 5 V conversion board providing the regulated 5 V rail to the experiment subsystems

Figure 50: 5 V Board PCB layout

Figure 51: Thermal dissipation on ground with no power

Figures 52: Thermal analysis in PV when on ground with no power

Figure 53: Thermal dissipation on ground with power

Figures 54: Thermal analysis on ground with power

Figure 55: In-flight thermal analysis

Figures 56: Thermal analysis in the EM during flight

Figure 57: Software architecture overview

Figure 58: State-transitions

Figure 59: System interfaces visual diagram

Figure 60: Data flow diagram

Figure 61: Camera health software verifications

Figure 62: Telemetry downlink protocol

Figure 63: Error flag representing an error indication based on bit status

Figure 64: Overview of Dictyostelium discoideum as a biomedical model, Reproduced from [12]

Figure 65: Schematic overview of our FITC-based fluorescence strategy

Figure 66: Pressure testing

Figures 67: Mechanical resistance analysis on various module components

Figure 68: Estimated Operational Timeline

LIST OF TABLES

Table 1.1: Experiment Objectives

Table 2.1: Functional Requirements

Table 2.2: Performance Requirements

Table 2.3: Design Requirements

Table 2.4: Operational Requirements

Table 2.5: Constraints
Table 3.1: Schedule/overview of work in progress until EAR
Table 3.2: Budget
Table 3.3: Funding Sources
Table 3.4: Team social media links
Table 3.5: Risk Register
Table 4.1: Part of the mechanical Interfaces
Table 4.2: REXUS connector pin out usage
Table 4.3: Experiment summary table
Table 4.4: Weight per Mechanical Part (module included)
Table 4.5: Electronic components description
Table 4.6: Power consumption of the experiment's electrical components
Table 4.7.1: Regulator power distribution for the average scenario
Table 4.7.2: Power budget for the average scenario
Table 4.8.1: Regulator power distribution for the maximum scenario
Table 4.8.2: Power budget for the maximum scenario
Table 4.9: Mission Phases Overview
Table 4.10: Software Success Criteria
Table 4.11: Definition of operational modes
Table 4.12: Activation criteria to transition into a given operational mode
Table 4.13: ESP-Pi SPI link parameters
Table 4.14: Data sources, type and purpose
Table 4.15: Data storage formats and logging frequency
Table 4.16: Uplink commands and scope
Table 4.17: Dictyostelium Biological Constraints for Experiment Design (based on [13])
Table 4.18: Biology Experiment Interfaces Overview
Table 5.1: Verification table
Table 5.2: Verification plan
Table 5.3: Example of a test result overview

Table 6.1: Experiment Mass and Volume
Table 6.2: Ground and flight safety risks
Table 6.3: Electrical interfaces applicable to REXUS
Table 6.4: Launch Site Requirements
Table 6.5: Flight Requirements applicable to REXUS
Table 6.6: Flight Requirements on timeline events applicable to REXUS
Table 6.7: List of planned activities
Table 6.8: Example of timeline of the experiment events
Table 6.9: Preliminary Biological Operational Timeline for the Launch Campaign
Table 6.10: Preliminary Biological Material and Consumables Required for the Launch Campaign
Table 6.11: Preliminary Biological Sample Preparation and Operational Time Estimates
Table 6.12: Preliminary Biological Material Transport Plan

PREFACE

The REXUS/BEXUS program is realized under a bilateral Agency Agreement between the German Aerospace Center (DLR) and the Swedish National Space Agency (SNSA). The Swedish share of the payload has been made available to students from other European countries through a collaboration with the European Space Agency (ESA). EuroLaunch, a cooperation between the Swedish Space Corporation (SSC) and the Mobile Rocket Base (MORABA) of DLR, is responsible for the campaign management and operations of the launch vehicles. Experts from DLR, SSC, ZARM and ESA provide technical support to the student teams throughout the project. REXUS and BEXUS are launched from SSC, Esrange Space Center in northern Sweden.

STAR-D (Space Test Amoeba Response - *Dictyostelium*) is an experiment developed by a team of students from the following universities: Ecole Supérieure d'Ingénieurs Léonard de Vinci, AgroParisTech, Université de Bordeaux, University of Newcastle, and Aalto University. Its aim is to test the impact of microgravity on the phagocytosis of *Dictyostelium*. STAR-D is scheduled to fly aboard REXUS 38 as part of cycle 17 of the Swedish - German REXUS/BEXUS program.

This document contains all relevant available information regarding STAR-D, including the experiment's objectives and requirements, a breakdown of the experimental module's design, design verification procedures, a timeline and plans for pre-launch campaign activities. Technical documents that pertain to the design are also included for reference, as appendix sections.

ABSTRACT

Microgravity alters fundamental physical constraints governing cellular transport, adhesion, and mechanotransduction, potentially impacting immune-related cellular functions. The STAR-D experiment investigates the influence of microgravity on phagocytosis using *Dictyostelium discoideum* (AX2), a well-established non-pathogenic model organism sharing key functional similarities with mammalian macrophages, including actin-driven motility, chemotaxis, and phagosome maturation.

Phagocytic activity is quantified using heat-inactivated *Escherichia coli* K12 cells labelled with Fluorescein isothiocyanate (FITC). FITC-labelled bacteria are fluorescent upon excitation and can therefore be continuously visualized throughout the experiment. Phagocytic events are identified by tracking the interaction, engulfment, and persistent intracellular localization of fluorescent bacterial targets within *Dictyostelium discoideum* cells. This approach enables real-time monitoring of phagocytic dynamics during the short microgravity phase without relying on phagosome acidification for fluorescence activation.

The biological samples are housed within an autonomous microfluidic lab-on-chip system integrating live-cell culture chambers, a controlled particle injection, and onboard optical microscopy. Upon entry into microgravity during a sounding rocket flight, fluorescent bacterial particles are injected into the observation chamber, where transport occurs solely via diffusion and cellular activity. Bright-field and fluorescence imaging are performed continuously throughout the microgravity window to capture phagocytosis dynamics in real time.

Post-flight image analysis will quantify phagocytosis rate, fluorescence intensity kinetics, and cell morphology, and results will be compared to identical ground controls. By isolating gravity-dependent effects on cell–particle interactions, STAR-D provides a robust and quantitative assessment of immune-like cellular behavior in microgravity. This experiment demonstrates a safe, automated, and scalable platform for studying cellular immunity in space and contributes to a broader understanding of how altered gravity affects fundamental biological processes relevant to human spaceflight.

1 INTRODUCTION

1.1 Scientific/Technical Background

Microgravity is known to affect a wide range of biological processes by altering the physical environment in which cells operate. Immune-related cellular functions, such as motility, adhesion, cytoskeletal organization, and phagocytosis, are sensitive to changes in gravity-driven forces. Previous spaceflight and parabolic flight experiments on mammalian immune cells have reported modified activation states, altered migration behavior, and in some cases reduced phagocytic efficiency. However, the underlying mechanisms remain only partially understood, due in part to experimental constraints, short microgravity durations, and the complexity of working with mammalian cells in autonomous flight hardware.

Despite these observations, a major gap remains in understanding how short-duration microgravity (on the order of minutes, as in sounding rocket flights) affects fast cellular processes such as phagocytosis at the single-cell level. Most existing studies focus on long-term exposure (hours to days) and gene expression changes, whereas early mechanosensitive responses occurring within seconds to minutes remain poorly characterized. This limits our ability to disentangle immediate physical effects from longer-term biological adaptation.

Phagocytosis is a fundamental and highly conserved cellular process by which eukaryotic cells recognize, internalize, and degrade particulate targets. It relies on actin-driven membrane remodeling, receptor-mediated capture, and tightly regulated intracellular trafficking leading to phagosome maturation and acidification. Because these steps depend on cytoskeletal dynamics and vesicle transport, phagocytosis is considered a sensitive functional readout of how altered gravity affects cell mechanics and intracellular organization.

The phagocytic process can be divided into several successive stages. First, the cell encounters a bacterial target through diffusion and active migration. Surface receptors then recognize the particle, triggering local actin polymerization and the formation of a phagocytic cup. The plasma membrane progressively extends around the bacterium until membrane closure results in the formation of an intracellular phagosome. The newly formed phagosome subsequently undergoes maturation through membrane fusion, intracellular trafficking and progressive acidification before degradation of the engulfed target. Because each of these steps depends on cytoskeletal dynamics and intracellular transport, phagocytosis provides an integrated functional readout of cellular behavior under altered gravity. Figure 01 illustrates the successive stages of phagosome maturation following bacterial internalization. (A) After engulfment, the newly formed phagosome first acquires early endosomal markers such as Rab5 and EEA1 while V-ATPases begin acidifying the lumen. (B) The compartment subsequently matures into a late phagosome characterized by Rab7 and LAMP proteins, accompanied by progressive

acidification and membrane remodeling through fusion and recycling events. (C) Finally, fusion with lysosomes generates a phagolysosome enriched in hydrolytic enzymes and antimicrobial components, where bacterial degradation takes place. This highly regulated maturation process involves coordinated membrane trafficking, cytoskeletal dynamics, vesicle recycling and intracellular acidification, making phagocytosis a sensitive functional readout of cellular responses to altered gravity.

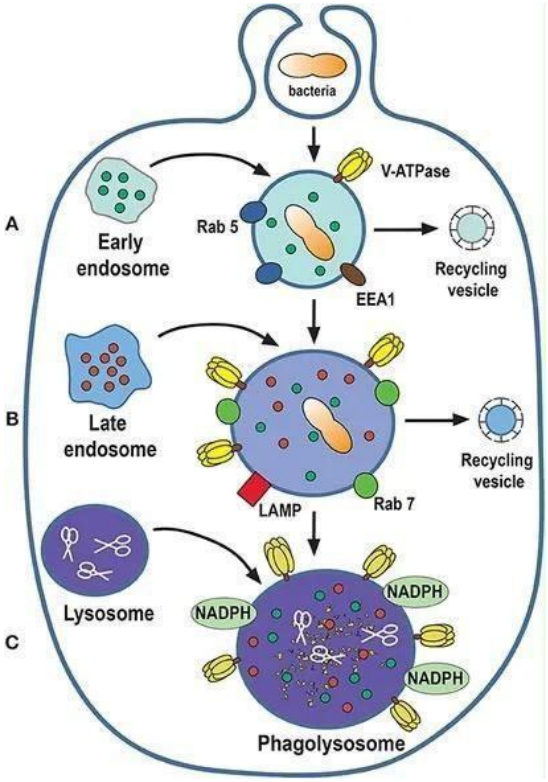


Figure 1: Schematic Representation of Phagosome Maturation in *Dictyostelium discoideum*

Previous studies have shown that cytoskeletal organization and cell motility are rapidly affected by changes in the gravitational vector, with transient increases or decreases in motility depending on cell density and experimental conditions. These findings suggest that gravity-sensitive processes such as actin polymerization and membrane deformation could directly impact phagocytosis efficiency. However, quantitative measurements of phagocytosis under real microgravity conditions remain scarce, especially in simplified and controlled model systems.

To study this process in a robust and flight-compatible manner, the STAR-D experiment employs *Dictyostelium discoideum* (strain AX2), a non-pathogenic

social amoeba widely used as a model system for phagocytosis and innate immune-like behavior. *Dictyostelium* shares key cellular mechanisms with mammalian macrophages, including chemotactic navigation, actin-driven engulfment, and phagosomal acidification, while offering several practical advantages for suborbital experimentation: genetic homogeneity, high resilience to environmental stress, and compatibility with simple culture conditions. The organism has a long history of use in space biology and microgravity-related studies, including investigations of development, morphology, and cytoskeletal organization, providing operational heritage for flight experiments. While *Dictyostelium* has been previously used in space biology experiments to study development and cytoskeletal dynamics, its use as a quantitative model for phagocytosis under microgravity remains limited. This represents a key scientific opportunity, as *Dictyostelium* provides a simplified and highly controllable system to isolate physical effects of gravity from complex immune signaling pathways present in mammalian cells.

A defining physical feature of microgravity is the suppression of sedimentation and buoyancy-driven convection. Under these conditions, cell–particle encounters occur primarily through diffusion and active cellular motility rather than gravity-assisted settling. This altered transport regime may significantly influence the frequency of encounters between cells and targets, the mechanics of capture, and the subsequent internalization process. The STAR-D experiment specifically addresses how these gravity-dependent physical constraints affect phagocytosis dynamics, rather than focusing on long-term gene expression or adaptation. In addition, fluid dynamics in microgravity differ significantly from ground conditions. The absence of buoyancy-driven convection leads to diffusion-dominated transport, which can alter particle distribution, encounter rates, and local concentration gradients. Experimental studies have shown that even small differences in fluid shear stress or mixing behavior can significantly influence biological responses. This highlights the importance of carefully controlling fluid injection and mixing processes in microgravity experiments

Phagocytic activity is monitored using FITC-labelled *Escherichia coli* K12 cells. FITC (fluorescein isothiocyanate) is a widely used fluorescent dye that absorbs blue light around 495 nm and emits green fluorescence around 519 nm. Fluorescent antibodies are first conjugated with FITC and subsequently used to label bacterial targets. This approach allows direct visualization of bacteria throughout the experiment, enabling observation of cell–particle encounters, engulfment and intracellular localization in real time while remaining fully compatible with the short duration of the microgravity phase. The experiment initially considered a pHrodo-based fluorescence strategy; however, based on the results of the biological validation campaign, FITC was selected as the final fluorescence readout. The rationale for this transition is presented in the Biology section.

The biological assay is integrated into an autonomous microfluidic lab-on-chip system combining live-cell culture chambers, a controlled fluid injection, and onboard optical microscopy. Microfluidic approaches are increasingly used in space biology due to their low reagent volume, containment safety, repeatability, and compatibility with automation. Similar architectures have been successfully demonstrated in previous suborbital and orbital payloads, including projects such as FlightScope, which combine microfluidics and microscopy to observe dynamic biological processes during short microgravity windows. STAR-D builds on this technical lineage while targeting a distinct biological endpoint: quantitative phagocytosis kinetics. However, the implementation of microfluidic systems in sounding rocket environments introduces several challenges that are not fully addressed in current state-of-the-art platforms. These include transient high acceleration phases during launch, potential bubble formation, and variability in fluid mixing under reduced gravity. While previous missions such as FlightScope have demonstrated the feasibility of combining microfluidics and microscopy in microgravity, quantitative biological assays involving active fluid injection and time-resolved imaging remain limited. STAR-D aims to address this technological gap by integrating controlled fluid delivery, stable cell adhesion, and synchronized imaging within a fully autonomous system.

The experiment is designed under the assumption that FITC-labelled bacterial targets can be continuously visualized throughout the experiment, allowing phagocytic events to be identified through cell-particle interactions, engulfment, and intracellular localization, that fluid transport during the microgravity phase is diffusion-dominated and reproducible, and that matched ground-reference experiments allow gravity-related effects to be distinguished from operational variables. Within these constraints, STAR-D provides a compact, safe (BSL-1), and technically mature approach to studying immune-like cellular behavior in a suborbital microgravity environment. In addition, the experiment assumes that transient hypergravity during launch does not irreversibly affect cell viability but may induce short-term cytoskeletal perturbations that are reversible before the start of the measurement phase. This aspect is further addressed through experimental design choices and ground testing.

The present project is grounded in prior experimental work in space biology and supported by a structured review of the relevant literature, which is further detailed in the following sections to substantiate both the scientific rationale and the technological design of the STAR-D experiment.

1.2 Mission Statement

The STAR-D experiment aims to improve understanding of how microgravity affects phagocytosis, a fundamental immune-like cellular process that underpins pathogen clearance and cellular homeostasis. While altered immune function has been repeatedly observed during spaceflight, the physical mechanisms driving these changes remain insufficiently characterized,

particularly at the level of single-cell interactions. In microgravity, the absence of sedimentation and buoyancy-driven convection fundamentally alters how cells encounter and internalize particulate targets, yet the direct impact of these altered transport conditions on phagocytosis dynamics is still poorly constrained.

To address this gap, STAR-D implements a compact, autonomous, and quantitative biological assay to observe phagocytosis under real microgravity conditions during a suborbital rocket flight. Using *Dictyostelium discoideum* as a robust analogue of professional phagocytes and a fluorescence-based, self-discriminating readout of particle internalization, the experiment will capture real-time phagocytic activity within a microfluidic lab-on-chip system. By comparing in-flight measurements with matched ground-reference data, STAR-D will identify qualitative differences in phagocytosis behavior attributable to altered gravity. The experiment thereby establishes a technically mature and biologically relevant platform for probing gravity-dependent cell–particle interactions, forming a clear foundation for the definition of experiment objectives and requirements in subsequent sections.

1.3 Experiment Objectives

Table 1.1: Experiment Objectives

O.1	Observe Phagocytosis of *Dictyostelium* in microgravity and compare it to Earth-based conditions (primary objective)
O.2	Design a cost-effective microscope platform for sounding rocket use (secondary objective)

1.4 Experiment Concept

The STAR-D experiment relies on a biological method that is widely employed in phagocytosis assays. It consists of mixing the cell to be studied with some sort of target to be “ingested” and observing the interaction with a microscope.

Before launch, *Dictyostelium discoideum* AX2 cells are cultured in nutrient-rich HL5 medium, harvested during exponential growth, washed, and transferred into a phosphate buffer before being loaded into the microfluidic observation chamber, where they adhere to the transparent bottom surface. In parallel, *E. coli* K12 cells are fluorescently labelled using FITC-conjugated antibodies and stored in a separate fluidic reservoir. During the microgravity phase, the bacterial suspension is injected into the observation chamber through the microfluidic circuit, allowing the amoebae to encounter, engulf and internalize the fluorescent bacterial targets. These interactions are continuously monitored

by onboard fluorescence microscopy, providing time-resolved images of the phagocytic process.

The first two key components of the experimental system are therefore the Open Flexure epifluorescence microscope and microfluidic system. Both are assembled in a single structure called the Microfluidic Assembly (MA), which also contains the control electronics of the experiment. The MA is housed inside the Pressure Vessel (PV) which shields it from vacuum conditions. The PV is the component added inside the rocket during the early access window. It is stored inside a structure mounted on a vibration absorber, itself bolted to the bottom bulkhead of the module. The bottom bulkhead also hosts the electronic interface box (EIB), which contains power electronics and communication electronics interfacing with the rocket. The EIB is connected to the PV to provide electrical power and data transmission;

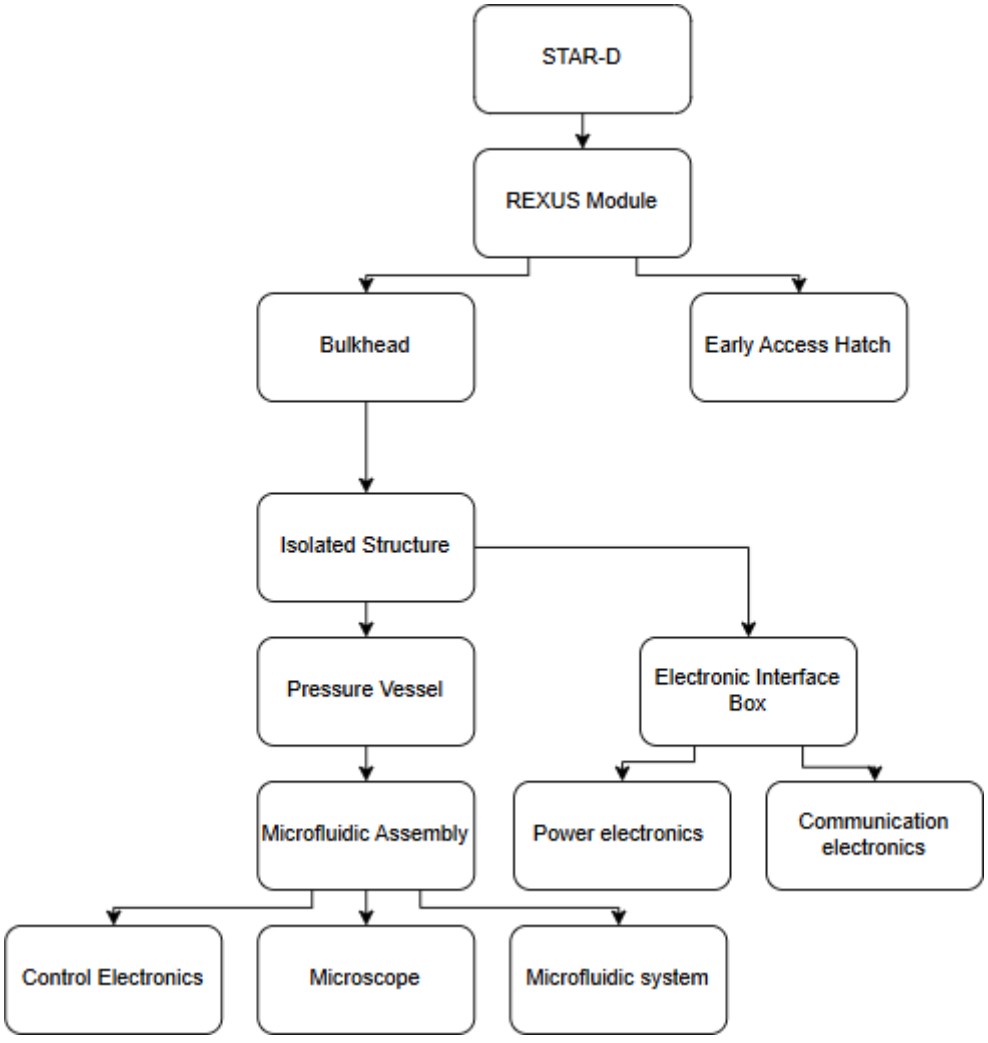


Figure 2: Breakdown Structure

1.5 Team Details

1.5.1 Contact Point

Scope	Person	Contact
General Topics / Team	Léopold REHLINGER	leopold.rehlinger@edu.devinci.fr +33 6 75 95 00 53
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1.5.2 Team Members

Name	Function	Course of study	University	ECTS
Léopold Rehlinger	Team Lead	Mechanical engineering	ESILV	5
Eden Elfassy	Electronics Lead	Electrical engineering	ESILV	5
Alexandre Rollet	Software Lead	Aerospace engineering	ESILV	5
Enzo Bertaut	Biology Lead	Life Sciences engineering	AgroParisTech	0
Manon Gaudillière	Mechanical Lead	Mechanical engineering	ESILV	5
Méliane Eydalene	Outreach Lead	Mechanical engineering	ESILV	5
Dominic Alderson	Mechanical Team	PhD Bio-Physics	University of Newcastle	0
Tomas Ducai	Biology Team	Biology degree	University of Vienna	0

Julien Finkler	Electronics Team	Electrical engineering	ESILV	0
Cléo El Gueziri	Mechanical Team	Mechanical engineering	Université de Bordeaux	0
Joshua Chen	Mechanical Team	PhD Chemistry	Université de Bordeaux	0
Pierre-Alexis Quillon	Software Team	Cybersecurity engineering	ESILV	5
Marco Acúrcio	Electronics Team	Space Science and Technology	Aalto University	0
Maxime Tardieu	Mechanical Team	Mechanical engineering	ESILV	5
Sarah Rosier	Biology Team	Biology	AgroParisTech	0
Mia Dubuis	Biology Team	Biology	AgroParisTech	0
Caroline Jourdain	Software Team	UX/UI design	IIM	0

2 EXPERIMENT REQUIREMENTS AND CONSTRAINTS

2.1 Functional Requirements

Table 2.1: Functional Requirements

ID	Description
F.1	The experiment shall have a sample container holding the cells
F.2	The experiment shall mix the cells with the phagocytosis target
F.3	The experiment shall have a video recording of the sample cells
F.4	The experiment shall have a bright field illumination for the sample container
F.5	The experiment shall have an epi fluorescent illumination for the sample container
F.6	The experiment shall measure the flow of the phagocytic target
F.7	The experiment shall control the flow of the phagocytic target
F.8	The experiment shall measure the pressure of the phagocytic target
F.9	The experiment shall control the pressure of the phagocytic target
F.10	The experiment shall measure the temperature of the sample container
F.11	The experiment shall control the temperature of the sample container
F.12	The experiment shall keep the sample container environment at a constant pressure
F.13	The experiment must measure the ambient pressure outside the sample container
F.14	The experiment must measure the vibration of the experiment module

2.2 Performance Requirements

Table 2.2: Performance Requirements

ID	Description
P.1	The experiment sample holder shall hold 500µL of cells
P.2	The experiment sample holder shall hold the cells aligned with the camera for 180s
P.3	The experiment shall be able to mix 2mL of the phagocytic target
P.4	The experiment shall record video for at least 400s
P.5	The experiment shall record video with a quality of 1080p
P.6	The experiment shall record video at 3 fps
P.7	The experiment shall record video with a zoom of x40
P.8	The experiment shall illuminate the sample container with a range of 0.1 W/mm^2 to 1 W/mm^2 with the bright field system
P.9	The experiment shall illuminate the sample container with a light frequency of 560nm ±5 nm with the epi fluorescent system
P.10	The experiment shall measure the pressure of the phagocytic target from 0.001 to 0.1 bar
P.11	The experiment shall measure the pressure of the phagocytic target with a precision of 1mbar
P.12	The experiment shall measure the pressure of the phagocytic target every 0.5s
P.13	The experiment shall mix the phagocytic target with the cells at a pressure of 0 to 500mbar (performance)
P.14	The experiment shall measure the flow of the phagocytic target from 10µL to 500µL
P.15	The experiment shall measure the flow of the phagocytic target with a precision of 50µL
P.16	The experiment shall measure the flow of the phagocytic target every 0.5s

P.17	The experiment shall mix the cells with up to 500µL of phagocytic target (a non-zero amount must be injected of at least 100 µL)
P.18	The experiment shall measure the temperature of the sample container from -30°C to 30°C
P.19	The experiment shall measure the temperature of the sample container with a precision of 0.5°C
P.20	The experiment shall measure the temperature of the sample container ever 0.5s
P.21	The experiment shall maintain the temperature of the sample container in a range of 18°C to 25°C
P.22	The experiment should maintain the sample container at a constant pressure of 1bar ±0.1 bar
P.23	The experiment must measure the ambient pressure outside of the sample container in a range of 0.001bar to 1bar
P.24	The experiment must measure the ambient pressure outside of the sample container with a precision of 0.1bar
P.25	The experiment must measure the ambient pressure outside of the sample container every 0.5s
P.26	The experiment must measure the vibration of the experiment module in a range of 0.01G to 1G:
P.26.1	On the x-axis
P.26.2	On the y-axis
P.26.3	On the z-axis
P.27	The experiment must measure the vibration of the experiment module with a precision of 0.1G
P.28	The experiment must measure the vibration of the experiment module every 0.1s

2.3 Design Requirements

Table 2.3: Design Requirements

ID	Description
D.1	The experiment shall have a standard EuroLaunch REXUS D-SUB bracket placed 20mm under the upper radax joint
D.2	A section of the bulkhead shall be left open to accommodate the feed through of up to 4 D-SUB 15 connectors
D.3	The hatch shall be flush with the module skin
D.4	The hatch shall follow the shape of the module
D.5	The hatch shall not protrude outside of the module into the air flow
D.6	The hatch may be offset by a maximum of 0.5mm inwards only
D.7	The maximum hatch cut-out shall be 156mm (horizontal) x 156mm (vertical)
D.8	The hatch shall have the same thickness as the module wall minus the tolerance to make it flush (4mm)
D.9	The hatch shall have a surrounding supporting frame
D.10	The frame shall be at least of the same thickness as the module (standard 4mm)
D.11	The gap between the module's cut-out and the hatch should be 0.1mm to 0.5mm maximum
D.12	For frame attachment, pop rivets with a diameter of 4mm or 5mm shall be used
D.13	The overlap frame to module shall be 2.5 times the thickness of the frame
D.14	The overlap frame to hatch shall be 2.5 times the thickness of the frame
D.16	Experiments with fluids quantity greater than 2mL shall use absorbent material at the top and bottom of the module The absorbent material shall have the capacity to absorb twenty times the quantity of fluid used

D.17	Pressure systems shall have a 2:1 safety factor
D.18	The experiment shall be dimensioned to withstand 20Gs of acceleration in all three axes
D.19	The experiment shall withstand the applicable REXUS User Manual vibration qualification profile
D.20	The center of gravity of the module shall be as close as possible to the longitudinal axis
D.21	The experiment shall have a BNC connector
D.22	The module's internal thermal dissipation must not heat up the outer structure more than 10°C ±5°C above the ambient temperature
D.23	The module's internal thermal dissipation must not heat up the parts close to or in contact with the feed-through cable to more than 70°C
D.24	The module's internal thermal dissipation must not heat up parts facing other modules to more than 50°C
D.25	The heat transport by convection must be limited in such a way that the air temperature at the module interfaces does not exceed the ambient temperature by more than 10 °C
D.26	The experiment interface connector should be a D-SUB 15 male type
D.27	The downlink baud rate must not exceed 80% (30kbit/s) of the maximum data throughput
D.28	The downlink format shall be 8 bits, 1 start and stop bit, no parity
D.29	The uplink baud rate must not exceed 80% (30kbit/s) of the maximum data throughput
D.30	The uplink format shall be 8 bits, 1 start and stop bit, no parity
D.31	The experiment should be able to deal with voltage steps (from 0 V to 32 V, and in reverse from 32 V to 0 V)
D.32	The peak power consumption should not exceed 3A per experiment line during switching
D.33	The mean value should not exceed 1A (approx. 30W)
D.34	The experiment must take precautions to limit voltage ripple feedback to the RXSM over the power line to a maximum of 100mVpp

D.35	Optocoupler shall be used on control interface lines
D.36	The experiment shall have a radio silence where no current flow may take place in the experiment
D.37	All power supply cables shall be twisted
D.38	Data cables shall be twisted
D.39	The radiation levels of the circuitry shall be kept under -115 dBm within the sensitive GPS frequency range (1573 MHz to 1578 MHz)
D.40	The telecommand frequencies of 448 MHz and 449.95 MHz shall remain free of spurious emissions, i.e. below -100 dBm measured at the telecommand reception antenna
D.41	Power emissions shall be kept under the required 100mVss measured on the 28 V lines

2.4 Operational Requirements

Table 2.4: Operational Requirements

ID	Description
OP.1	The experiment shall safely contain biological samples throughout all flight phases
OP.2	The experiment shall be capable of autonomous operation without ground intervention
OP.3	Biological samples withstand and survive hypergravity
OP.4	The PV shall be able to be inserted into the EM without mechanical interference
OP.5	The PV shall not show any displacement once fixed and secured into the EM
OP.6	The PV shall be installed and removable during early access procedure without interference with the electrical cables

2.5 Constraints

Table 2.5: Constraints

ID	Description
C.1	The home university is not equipped with a wet lab
C.2	The home university is not equipped with proper environment testing facilities
C.3	The home university is closed from the 17th of July to the 17th of August 2026
C.4	Limited access to BSL-1 certified laboratories

3 PROJECT PLANNING

3.1 Work Breakdown Structure (WBS)

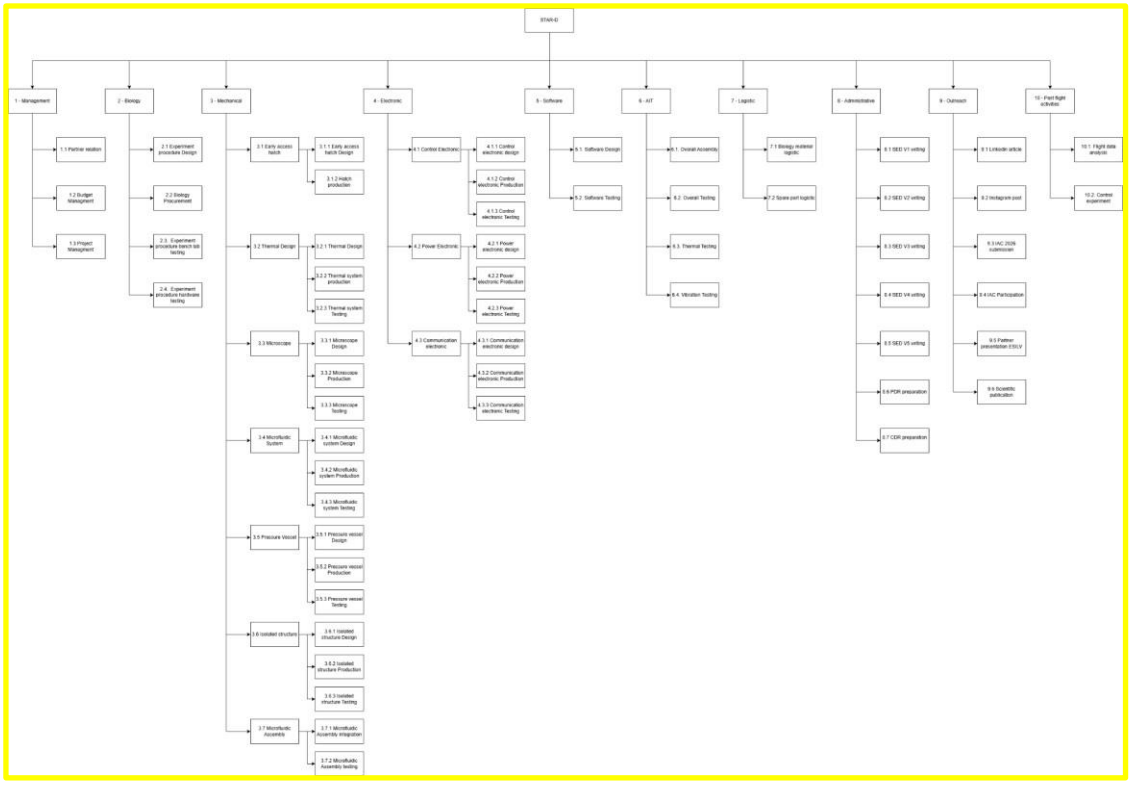
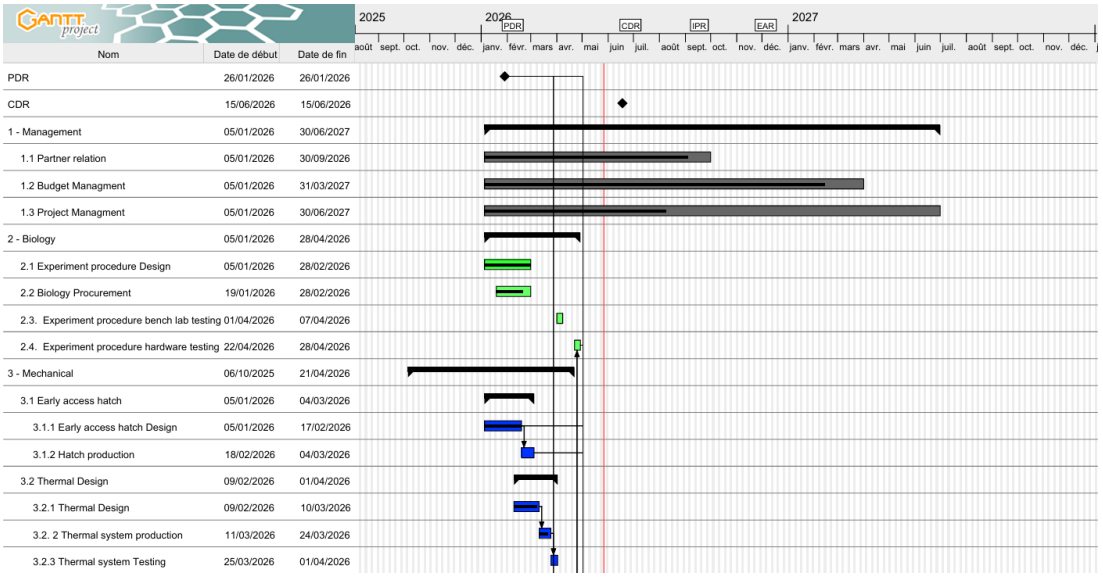


Figure 3: WBS

3.2 Schedule

The following GANTT chart represents the tasks that must be done throughout the development timeline.



Nom	Date de début	Date de fin
PDR	26/01/2026	26/01/2026
CDR	15/06/2026	15/06/2026
1 - Management	05/01/2026	30/06/2027
1.1 Partner relation	05/01/2026	30/09/2026
1.2 Budget Managment	05/01/2026	31/03/2027
1.3 Project Managment	05/01/2026	30/06/2027
2 - Biology	05/01/2026	28/04/2026
2.1 Experiment procedure Design	05/01/2026	28/02/2026
2.2 Biology Procurement	19/01/2026	28/02/2026
2.3. Experiment procedure bench lab testing	01/04/2026	07/04/2026
2.4. Experiment procedure hardware testing	22/04/2026	28/04/2026
3 - Mechanical	06/10/2025	21/04/2026
3.1 Early access hatch	05/01/2026	04/03/2026
3.1.1 Early access hatch Design	05/01/2026	17/02/2026
3.1.2 Hatch production	18/02/2026	04/03/2026
3.2 Thermal Design	09/02/2026	01/04/2026
3.2.1 Thermal Design	09/02/2026	10/03/2026
3.2. 2 Thermal system production	11/03/2026	24/03/2026
3.2.3 Thermal system Testing	25/03/2026	01/04/2026
3.3 Microscope	06/10/2025	05/04/2026
3.3.1 Microscope Design	06/10/2025	14/03/2026
3.3.2 Microscope Production	15/03/2026	28/03/2026
3.3.3 Microscope Testing	29/03/2026	05/04/2026
3.4 Microfluidic System	06/10/2025	06/04/2026
3.4.1 Microfluidic system Design	06/10/2025	14/03/2026
3.4.2 Microfluidic system Production	15/03/2026	29/03/2026
3.4.3 Microfluidic system Testing	30/03/2026	06/04/2026
3.5 Pressure Vessel	10/11/2025	01/04/2026
3.5.1 Pressure vessel Design	10/11/2025	09/03/2026
3.5.2 Pressure vessel Production	10/03/2026	24/03/2026
3.5.3 Pressure vessel Testing	25/03/2026	01/04/2026
3.6 Isolated structure	09/02/2026	01/04/2026
3.6.1 Isolated structure Design	09/02/2026	09/03/2026
3.6.2 Isolated structure Production	10/03/2026	24/03/2026
3.6.3 Isolated structure Testing	25/03/2026	01/04/2026
3.7 Microfluidic Assembly	07/04/2026	21/04/2026
3.7.1 Microfluidic Assembly integration	07/04/2026	14/04/2026
3.7.2 Microfluidic Assembly testing	15/04/2026	21/04/2026

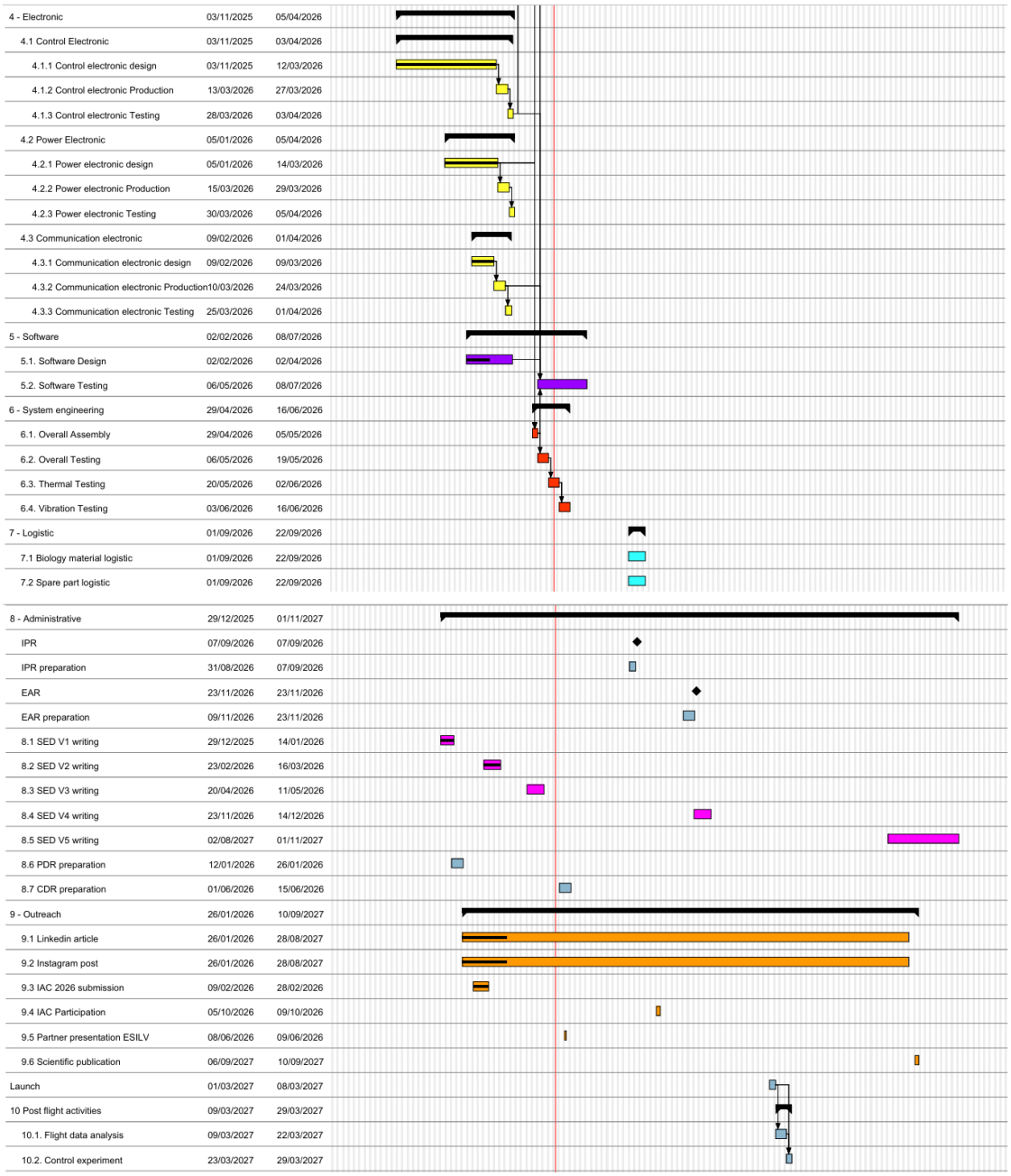


Figure 4: Gantt Chart

Table 3.1: Schedule/overview of work in progress until EAR

Workstream	15 – 21 Sep	22 – 30 Sep	1 – 7 Oct	8 – 15 Oct	16 – 31 Oct	1 – 9 Nov
Mechanical	Parts ordered – awaiting delivery	**Assembly of received parts complete by 30 Sep**	Mechanical / structural testing	**Testing complete 15 Oct**	Retest margin, close-out	
Pressure vessel	Pressure test re-run with new seals	**Re-test passed by 27 Sep**	Integration into structure		Margin	
Electronics	Cables received – integration testing underway	Integration testing	Integration testing	**All electronics tested by 15 Oct**	Fixes and retests	
Software – flight (UART → SPI)	Switch to SPI; tests on simulation files	Simulation-file testing	Testing with real electronics	**Link validated by 15 Oct**	Fixes and retests	
Software – ground station	**First prototype 21 Sep**	Development	Development	**Tested with simulated downlink 15 Oct**	Refinement	
Biology	Ordered – awaiting remaining consumables	**Testing resumes; transport research by 30 Sep**	Testing: nominal + backup experiment	**Biology testing complete 15 Oct**	Transport booking (airline / courier)	
Outreach	IPR post published	**Module + electronics test post by 30 Sep**				
Documentation & review			Test results logged in verification plan	Test results logged in verification plan	SED v4 writing	**SED v4 delivered 2 Nov EAR 9 Nov**

Work ongoing Deadline / milestone Done

3.2.1 Workforce

The team is divided into five operational teams: Mechanical, Electronic & Electrical, Biological, Software and Outreach. The distribution is as follows in the diagram:

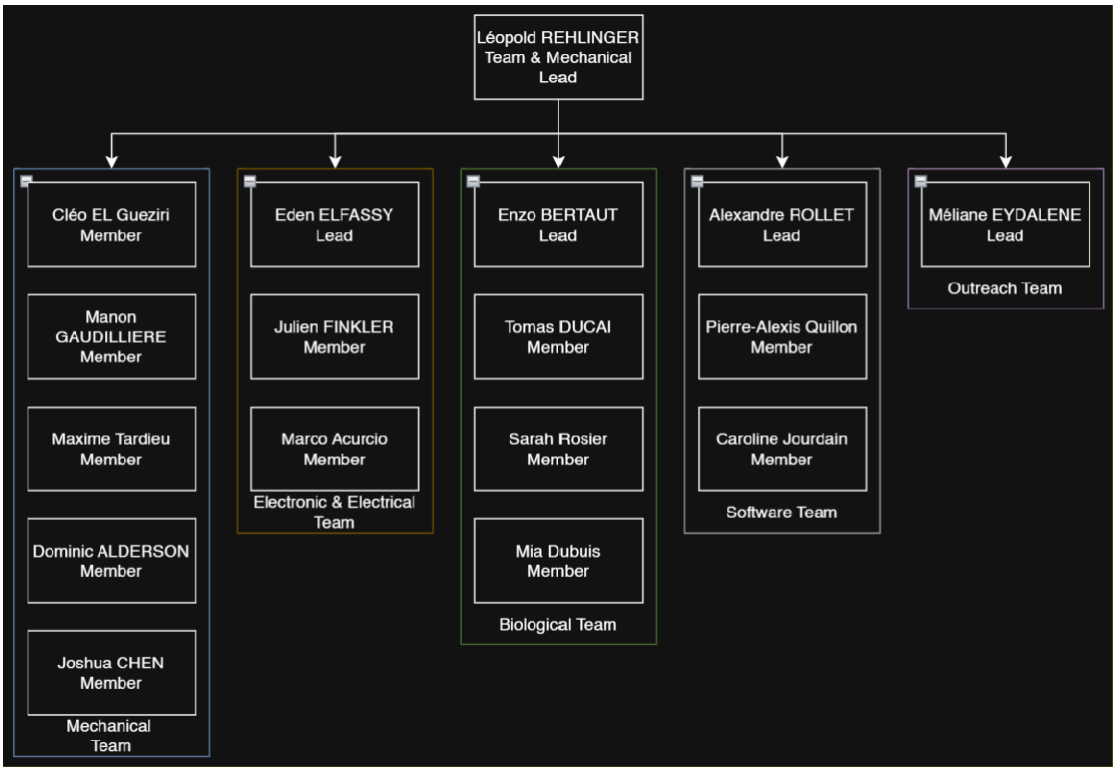
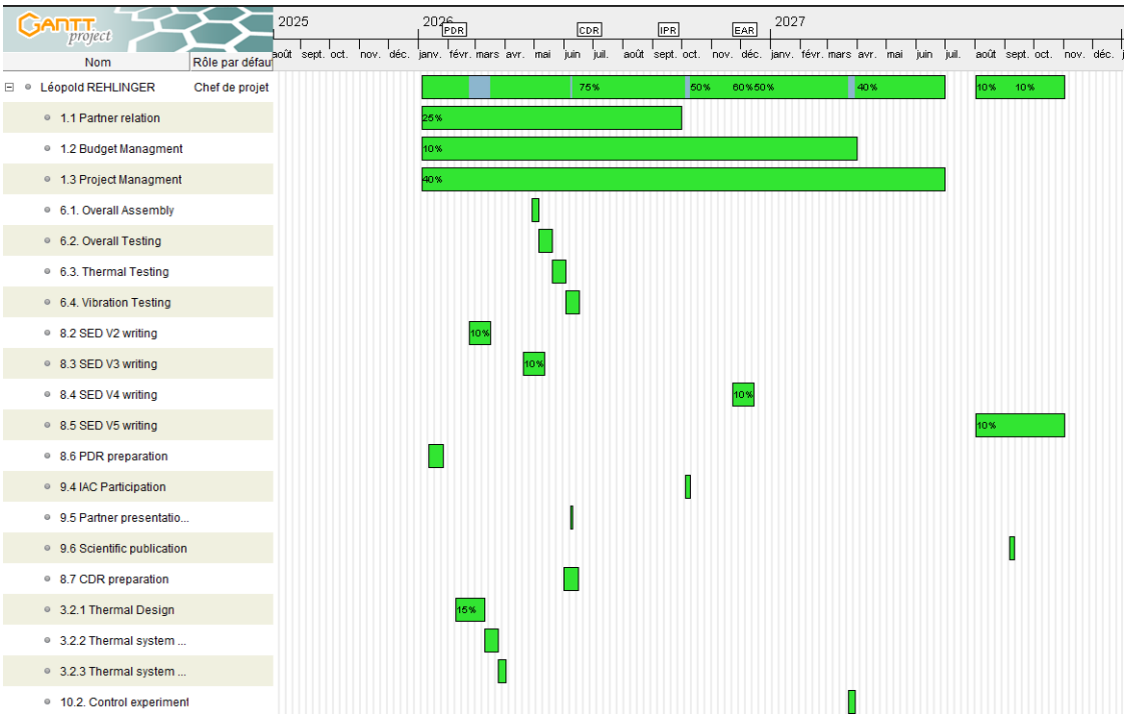
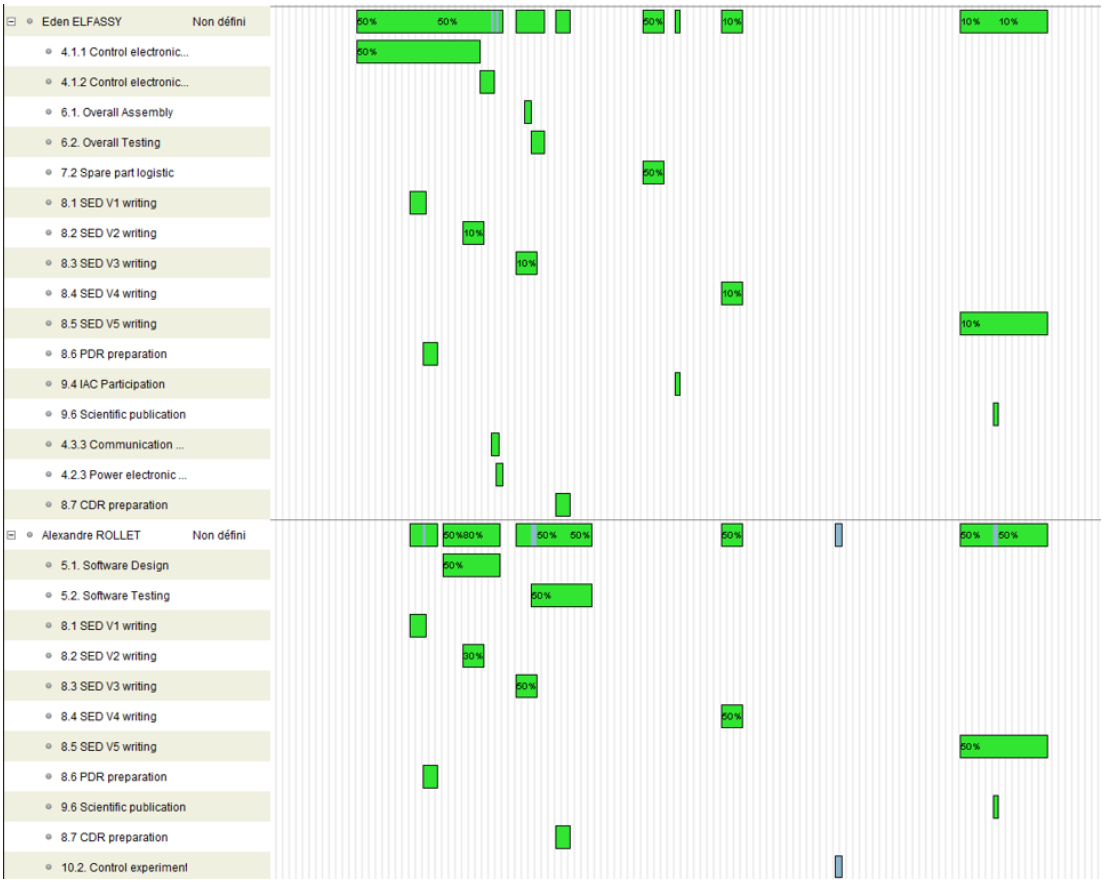


Figure 5: Team Structure

Each member was added to our project management software and assigned tasks within their teams:


GANTT project
Nom
Rôle par défau
2025
2026
2027
PDR
CDR
IPR
EAR
août sept. oct. nov. déc. janv. févr. mars avr. mai juin juil. août sept. oct. nov. déc. janv. févr. mars avr. mai juin juil. août sept. oct. nov. déc.
Léopold REHLINGER Chef de projet
1.1 Partner relation
1.2 Budget Managment
1.3 Project Managment
6.1. Overall Assembly
6.2. Overall Testing
6.3. Thermal Testing
6.4. Vibration Testing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
8.6 PDR preparation
9.4 IAC Participation
9.5 Partner presentatio...
9.6 Scientific publication
8.7 CDR preparation
3.2.1 Thermal Design
3.2.2 Thermal system ...
3.2.3 Thermal system ...
10.2. Control experiment


Eden ELFASSY Non défini
4.1.1 Control electronic...
4.1.2 Control electronic...
6.1. Overall Assembly
6.2. Overall Testing
7.2 Spare part logistic
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
8.6 PDR preparation
9.4 IAC Participation
9.6 Scientific publication
4.3.3 Communication ...
4.2.3 Power electronic ...
8.7 CDR preparation
Alexandre ROLLET Non défini
5.1. Software Design
5.2. Software Testing
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
8.6 PDR preparation
9.6 Scientific publication
8.7 CDR preparation
10.2. Control experiment


Enzo BERTAUT Non défini
2.1 Experiment proced...
2.3. Experiment proce...
2.4. Experiment proce...
6.2. Overall Testing
7.1 Biology material lo...
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
9.6 Scientific publication
8.7 CDR preparation
10.1. Flight data analys...
10.2. Control experiment
Tomas DUCAI Non défini
2.2 Biology Procurement
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
9.6 Scientific publication


Dominic ALDERSON Non défini
3.3.1 Microscope Desi...
3.3.2 Microscope Prod...
3.3.3 Microscope Testi...
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
9.6 Scientific publication
Julien FINKLER Non défini
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
9.6 Scientific publication
4.1.2 Control electronic...
4.1.3 Control electronic...
4.1.1 Control electronic...
Manon GAUDILLIERE Non défini
6.1. Overall Assembly
8.1 SED V1 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
9.6 Scientific publication
3.5.1 Pressure vessel ...
3.5.3 Pressure vessel ...
3.5.2 Pressure vessel ...
8.7 CDR preparation


Maxime Tardieu
Non défini
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
9.6 Scientific publication
3.7.1 Microfluidic Asse...
3.4.1 Microfluidic syste...
3.7.2 Microfluidic Asse...
3.4.3 Microfluidic syste...
3.2.1 Thermal Design
3.2.2 Thermal system ...
3.2.3 Thermal system ...
Kacem EL GEZIRI
Non défini
3.6.1 Isolated structure...
3.6.2 Isolated structure...
3.6.3 Isolated structure...
6.1. Overall Assembly
6.2. Overall Testing
6.4. Vibration Testing
7.2 Spare part logistic
8.1 SED V1 writing
8.2 SED V2 writing
8.3 SED V3 writing
8.4 SED V4 writing
8.5 SED V5 writing
8.6 PDR preparation
9.6 Scientific publication



Figure 6: Project Schedule and Task Overvie

3.2.2 Budget

Table 3.2: Budget

Items	Price (€)	Qty	Total (€)
Mechanical			
Aluminum Experiment Support Frame	300	1	300
Screws and bolts	100	1	100
PV structure	500	1	500
Frame Bracket	100	2	200
(Total for category)			**1100€**
Electrical			
ESP32 module	6.28	1	6.28
Raspberry Pi 5	106.99	1	106.99
NVMe SSD M.2 2230	111	1	111
BME280 (internal)	19.74	1	19.74
BME280 (external)	19.74	1	19.74
DS18B20	7.59	1	7.59
LSM6DSMIMU	3.09	1	3.09
A4988 driver - logic	9.95	1	9.95
NEMA-8 stepper motor (bipolar)	26.40	1	26.40

28BYJ-48 stepper motor	6	1	6
ULN2003 board	1.85	1	1.85
LTC7001	8.72	1	8.72
Microscope LEDs / PCB LEDs	30.96	3	92.88
(Total for category)			**420.23**
Biology			
HSP60 Monoclonal Antibody (LK2), FITC	476.71	1	476.71
Escherichia coli K12 strain	25.02	1	25.02
Nutrient Agar	19.26	100 g	19.26
Dictyostelium Culture Kit	90	2	180
Microfluidic & other	450	1	450
(Total for category)			**1739.3**
Events	**Price per person**	**Number of members**	**Total price per event**
Pre-selection	661	4	2644
STW & PDR	1330	4	5320
CDR	470	4	1880
ITW	450	3	1350
Bench Test	470	4	1880
Launch Campaign	2660	4	10640
(Total for category)			**23 714.00**
Total Budget			**27 453.53 €**

Funding:

Table 3.3: Funding Sources

Partner	Subject of partnership
Thermo fisher scientific	pHrodo™ bioparticles
Croda	Phase changing material
Bordeaux University	Raw materials (frame and CNC parts)
A proctor group	Thermal insulation
European Space Agency	Event cost coverage & technical guidance

3.2.3 External Support

We can conte on a mix of industrial and researcher to develop our experiment:

Adam Wollman, Associate professor at the university of Newcastle and with experience in developing a microscope and microfluidic system for use in microgravity

Hannah Burgess, researcher at ISSET and King's College, working on a similar experiment with *Dictyostelium* that will be sent on the ISS

Marine Kirat, mechanical engineering lead at Interstellar lab

Zac Bimson, mechanical engineer at interstellar lab

Sacha Yzerman, embedded system lead at interstellar lab

Openflexure Foundation, a group of researchers behind the Openflexure microscope on which our system is based.

Socitec: company specialized in vibration mitigation, helping us with vibration modeling

AgroParisTech: Access to a biolab

3.3 Outreach Approach

The outreach approach is based on three main lines, the two first are through our LinkedIn and Instagram accompts, the first one to reach French students and encourage them to participate in the REXUS/BEXUS program (French absence in this program is quite an anomaly) while the LinkedIn is aimed at a broader audience of students from all across Europe, as well as companies in Europe to show the value of the REXUS/BEXUS program. Finally, the third line is around scientific papers, the first one would be a technical development paper to be presented at IAC 2026, the second would be more focused on the scientific result post flight, to be published in a specialized review such as NPJ microgravity. To write those papers, we would strongly rely on the existing SED to reduce the workload it would imply, and use the summertime, where experiment development is at a slower pace.

Table 3.4: Team social media links

Team	Facebook	Instagram	LinkedIn	Website
STAR-D	https://www.facebook.com/LeoFlyPULV/	https://www.instagram.com/leoflypulv/?hl=en	https://www.linkedin.com/company/stardrexus	https://www.leofly.fr/stard

3.4 Risk Register

Risk ID
TC – technical/implementation
MS – mission (operational performance)
SF – safety
VE – vehicle
PE – personnel
EN – environmental

Table 3.5: Risk Register

ID	Risk & Consequence	P	S	P x S	Prevention/Mitigation	Acceptance
TC-10	Failure of power conversion leading to loss of one or more voltage rails,	B	4	Low	Conservative power budget, margin on converters, ground testing	

	causing partial or total experiment shutdown				under nominal and peak loads	
TC-11	Loss-of-rail consequence unchanged (no rail redundancy); likelihood reduced by converter margin and verified peak-load testing.	A	4	Low	As TC-10, plus rail-level voltage/current telemetry for early detection. No further hardware change proposed.	Accepted for flight, contingent on successful nominal + peak-load qualification test.
TC-20	Overcurrent on 5 V due to simultaneous activation of motors, heater and Raspberry pi	B	4	Low	Conservative power budget, margin on converters, ground testing under nominal and peak loads	
TC-21	Draw bounded by budget; consequence remains a possible 5 V brown-out.	A	4	Low	As TC-20, plus proposed firmware load-sequencing (stagger heater / motor / Pi peaks) and resettable current limiting. If adopted, residual severity falls to 3 (protection trip, not brown-out) → Very Low.	Accepted; recommend adopting load-sequencing before design freeze.
TC-30	RasPi overheating or instability leading to loss of microscope imaging	C	3	Low	Downclocking, reduced workload, thermal monitoring and controlled operating profile	We accept this risk

TC-40	Stepper motor or driver malfunction causing loss of mechanical actuation	B	3	Low	Multiple sensors (internal/external), plausibility checks and logging of raw data	We accept this risk
TC-50	Sensor malfunction (temperature, pressure, IMU) leading to degraded environmental monitoring	B	2	Very Low	Local NVMe storage, buffered writes, post-flight data retrieval independent of telemetry	We accept this risk
TC-60	*Dictyostelium discoideum* cells do not adhere properly to the microfluidic chamber surface, leading to cell drifting during microgravity conditions and reduced effective interaction with fluorescent E. coli bioparticles. This may result in poor image quality, reduced phagocytosis probability, and loss of biological data	C	3	Low	Optimize surface coating; validate adhesion time on ground; allow settling period after loading; flush non-adherent cells before experiment start	We accept this risk
TC-70	Fluorescent E. coli bioparticles may aggregate during storage or injection, leading	C	2	Low	Use BSA to reduce adhesion to tubing and reservoirs;	We accept this risk

	to non-uniform particle distribution, altered encounter rates with *Dictyostelium* cells, reduced phagocytosis efficiency, or partial clogging of microfluidic channels				optional low-concentration methylcellulose to stabilize suspension and limit sedimentation; gentle resuspension before loading (avoiding harsh sonication if incompatible with bioparticles); ground validation of bioparticle homogeneity and flow behavior prior to flight	
TC-80	Air bubbles form in microfluidic lines or chambers, disrupting flow, detaching cells, or blocking optical observation, resulting in total loss of biological data	B	4	Low	Degas all buffers and suspensions; pre-fill all lines carefully; ground validation of bubble-free operation; conservative flow rates; strict filling procedure	We accept this risk
TC-81	Residual re-evaluation of TC-80 (air bubbles in microfluidic lines). Occurrence strongly reduced by procedure; if a bubble still	A	4	Low	As TC-80, plus a documented bubble-free fill procedure verified at integration and a pre-activation optical bubble check.	Residual Low; consequence is loss of science data only, no vehicle/safety impact. Accepted for flight.

	forms, biological-data loss is unchanged (science-only, no safety impact).					
TC-100	PV not fully locked after early access causes motion of the PV	B	2	Very Low	Optimal locking mechanism	We accept this risk
TC-110	PV structural support insufficient for flight acceleration	B	3	Low	Stiffness-oriented frame, modal check and computer vibration/load testing	We accept this risk
TC-120	Rail misalignment or jamming during insertion	B	4	Low	Repeated dry runs, assembly with tolerance	
TC-121	Likelihood reduced by toleranced assembly and dry runs; consequence if it occurs is delayed or aborted early-access integration.	A	4	Low	As TC-120, plus a defined recovery/abort procedure and a go/no-go criterion at the early-access dry run.	Residual Low. Accepted, contingent on a successful early-access dry run (links to CDR Action 40).
TC-130	Tests fail after the CDR with major re-design required	C	3	Low	Take high margin in design, test at the earliest	We accept this risk

MS-10	Loss or corruption of experiment data during flight	C	3	Low	Local NVMe storage, buffered writes, post-flight data retrieval independent of telemetry	We accept this risk
MS-20	Fluorescence signal decreases due to photobleaching or suboptimal illumination, leading to reduced signal-to-noise ratio and limited detection of phagocytosis (ingestion) events	B	3	Low	Pre-flight calibration of illumination intensity and exposure time; use photostable pH-sensitive fluorescent dyes (e.g. FITC) conjugated to E. coli cells; limit illumination duration and duty cycle; validate fluorescence signal stability and activation kinetics during ground tests	We accept this risk
MS-30	Extended pre-launch holding time induces cellular stress (nutrient deprivation, oxygen limitation, temperature drift), reducing motility and phagocytosis at experiment activation	C	3	Low	Define and validate maximum acceptable standby duration; maintain temperature within biological tolerance; ground tests with representative	We accept this risk

					standby times; use physiologically compatible buffer (KK2)	
MS-40	In microgravity, the absence of sedimentation reduces passive encounters between *Dictyostelium* cells and E. coli bioparticles, potentially leading to fewer observable phagocytosis events during the experiment	B	2	Very Low	Increase E. coli bioparticle concentration if needed; rely on active cell motility and chemotactic behavior; ensure homogeneous particle distribution at injection; systematically compare with ground control experiments performed under identical conditions	We accept this risk
SF-10	Overheating of the heating element causing unsafe internal temperatures	B	4	Low	Thermostat control, temperature feedback loop, power limitation of heating pad	
SF-11	Power limitation physically caps the maximum attainable temperature, so both likelihood and the severity of an unsafe over-temperature condition are reduced.	A	3	Very Low	As SF-10, plus a proposed independent hardware over-temperature cutoff (thermal fuse), independent of the control firmware, as defence-in-depth.	Residual Very Low with independent cutoff. Accepted; given this is a safety-class risk, recommend the hardware cutoff be included before freeze.

VE-10	Electrical short or ground fault impacting the REXUS vehicle interface	A	5	Low	Single-point grounding, insulation, compliance with REXUS electrical interface requirements	
VE-11	Worst-case consequence to the vehicle remains catastrophic by definition (S = 5); mitigation drives the likelihood of a vehicle-affecting fault to the matrix floor by containing faults to the experiment side.	A	5	Low	As VE-10, plus inline fusing on the experiment supply and verification of isolation resistance and fuse/breaker coordination by test (links to DV-T04 / DV-T05).	Likelihood at matrix floor and fault contained to the experiment side. Accepted, contingent on measured isolation resistance and fuse coordination meeting the REXUS electrical-interface requirement.

4 EXPERIMENT DESCRIPTION

4.1 Experiment Setup

The experiment is based on an autonomous microfluidic lab-on-chip system integrating a sealed microfluidic cartridge, controlled fluid injection, and optical microscopy. The microfluidic subsystem enables controlled interaction between *Dictyostelium* cells and fluorescent E. coli bioparticles during the microgravity phase. The chip is housed within a sealed microfluidic cartridge and integrated inside the experiment pressure vessel.

4.2 Experiment Interfaces

4.2.1 Mechanical

The STAR-D experiment is mechanically interfaced with the REXUS payload through the experiment bulkhead. The bulkhead is provided by the REXUS/BEXUS programme and constitutes the primary mechanical interface between the experiment and the launch vehicle.

The STAR-D mechanical assembly is mounted to the bulkhead through two rigid aluminium support feet. Each support foot is attached to the bulkhead using three M6 bolted connections, resulting in a total of six M6 attachment points between the experiment structure and the bulkhead.

The upper experiment structure is rigidly connected to the two support feet. Each foot provides two M5 threaded interfaces. The M5 screws pass through clearance holes in the upper structural elements and are fastened directly into threaded holes in the support feet, resulting in four M5 connections between the support feet and the upper structure.

The previously considered wire-rope isolation system has been removed from the flight design following the CDR. Consequently, the STAR-D structure is now rigidly mounted to the bulkhead, with no mechanical isolation elements between the experiment structure and the REXUS mechanical interface.

Cable routing and connector accessibility

The REXUS electrical harness is routed behind the Experiment Support Frame before being guided towards the upper electronics compartment. This routing keeps the cables away from the moving parts of the suspended assembly and minimizes the risk of mechanical interference during integration and operation.

The interface PCB is mounted on the upper structural platform. External electrical connectors are directly accessible from outside the experiment through dedicated openings in the electronics enclosure.

A dedicated internal cable runs from the interface PCB to the front side of the pressure vessel, facing the early-access hatch. This routing provides a direct electrical connection between the electronics and the pressure vessel while keeping the harness protected and avoiding interference with the mechanical support structure.

Cable routing has been designed to avoid excessive bending, tension or contact with the suspended assembly throughout the operational range of motion. All electrical connections remain accessible during integration and maintenance operations.

Table 4.1: Part of the mechanical Interfaces

Num of Part	Name.	Use
1	REXUS Bulkhead	Bottom Mounted Surface
2	Aluminium support feet	Lift the PV structure
1	PV Structure	Aluminum structure containing the PV
1	Pressure Vessel	Containing the Experiment
2	End-stop	Prevent the slide of the PV
4	Rail	
2	PV lock	The handle fix the PV to those lock
1	Handle	Screwed to the PV and to the PV lock
1	E-Box	Mechanical protection and EMI shielding of the electronics

4.2.2 Electrical

Table 4.2: REXUS connector pin out usage.

Pin No.	Name.	Use
1	+28 V	Used
2	Charging (28 V/1 A)	Not used
3	SODS	Used
4	SOE	Used
5	LO	Used
6	EXP out+	Used
7	EXP out-	Used
8	28 V Ground	Used

9	+28 V	Used
10	n.c.	Not used
11	n.c.	Not used

Pin No.	Name.	Use
1	+28 V	Used
2	Charging (28 V/1 A)	Not used
3	SODS	Used
4	SOE	Used
5	LO	Used
6	EXP out+	Used
7	EXP out-	Used
8	28 V Ground	Used
12	Charging Return	Not used
13	EXP in +	Used
14	EXP in -	Used
15	28 V Ground	Used

The STAR-D experiment interfaces electrically with the REXUS service module through the standard 15-pin D-SUB experiment connector. The experiment is powered exclusively by the +28 V DC supply provided by REXUS (Pins 1 and 9, with ground on Pins 8 and 15). This voltage constitutes the sole external power source and is internally converted to the different voltage rails required by the experiment electronics. No current is drawn from signal or communication pins, and the charging lines available on the connector are not used.

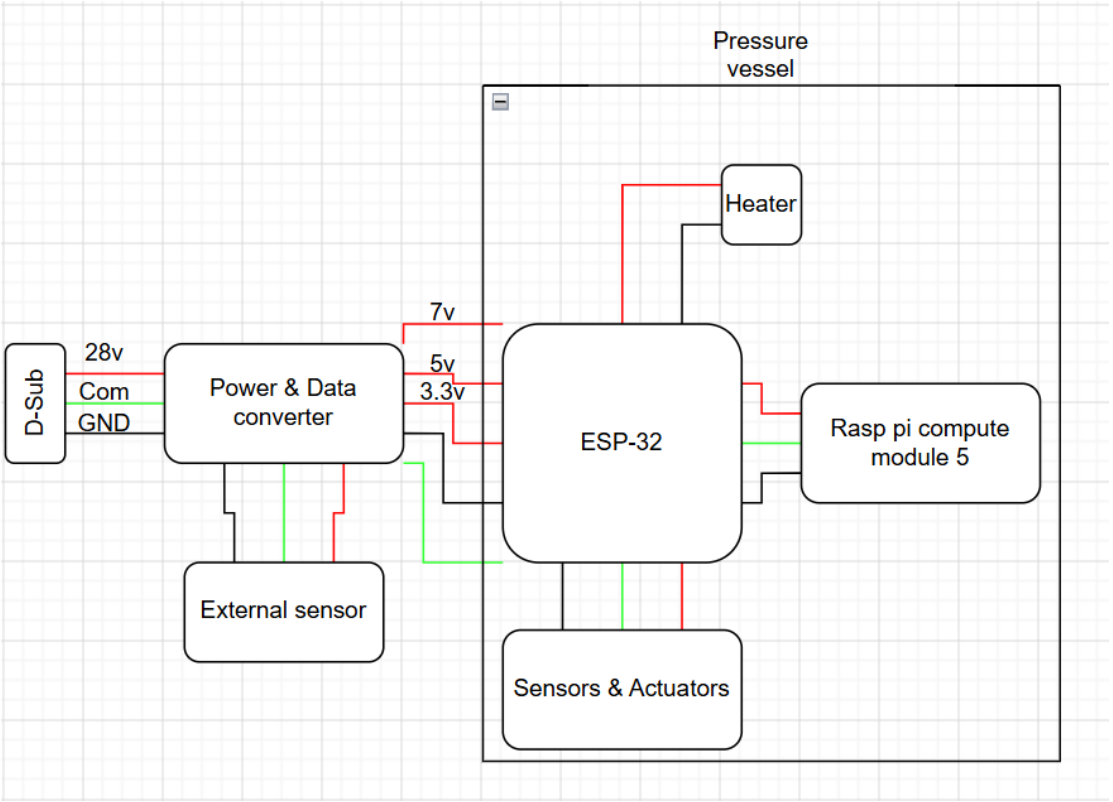


Figure 7: Simplified Internal Electronics Architecture

Following the electrical diagram, the internal electronics of STAR-D are organized into clearly defined subsystems.

The incoming 28 V supply is routed to an internal voltage unit converter, which generates the regulated voltage rails required by the experiment electronics. These rails supply both low-power logic circuits and higher-current loads, with a clear separation between sensitive electronics and power-intensive subsystems.

An ESP32 module acts as the central control unit of the experiment. It interfaces with sensors, actuator drivers, heating system, and illumination electronics, and manages the real-time sequencing of the experiment following the SOE signal.

High-level processing and image acquisition are performed by a Raspberry Pi Compute Module 5 dedicated to the microscopy subsystem. The Raspberry Pi interfaces with the camera and local storage and operates in a reduced-performance configuration to limit power consumption. Synchronization with the control electronics is achieved through a dedicated digital interface.

Environmental and inertial sensors monitor the conditions inside and outside the experiment volume, as well as the flight dynamics. Actuators and their drivers are grouped in a dedicated section of electronics to simplify routing and limit interference. Thermal regulation and microscope illumination are controlled through dedicated power stages and are operated only when required to reduce power consumption.

4.3 Experiment Components

Table 4.3: Experiment summary table

Experiment mass (in kg):	Estimated 12.2kg (including REXUS module)
Experiment dimensions (for BEXUS) or module size (for REXUS) (in m):	0.220m
Experiment expected COG (center of gravity) position:	X: 90.457 Y: 0 Z: -5.019

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Table 4.4: Weight per Mechanical Part (module included)

Component	Subtotal (g)
Module (+hatch)	5080
Bulkhead	1240

Pressure vessel	2036
Rail (x4)	50
PV structure	2011
PV lock (x2)	8
Handle	85
End stop (x2)	40
Support feet (x2)	524
PCB Box	439
TOTAL	**12235 g (12.2 kg)**

4.4 Mechanical Design

4.4.1 Mechanical Structure Design

Overall mechanical architecture

The STAR-D experiment is integrated as a bottom-mounted assembly inside the REXUS module. The mechanical architecture provides a rigid and direct load path between the experiment assembly and the REXUS bottom bulkhead, while allowing early-access insertion and removal of the Pressure Vessel (PV).

Following the CDR, the previously considered wire-rope vibration isolation system was removed from the design. The final architecture therefore consists of two rigid aluminum support feet extending from the bottom bulkhead to the Experiment Support Frame. The support feet carry the main experiment structure and transfer the mechanical loads directly to the REXUS bulkhead.

The Experiment Support Frame supports and retains the Pressure Vessel and provides the mechanical interfaces for the associated external experiment hardware. The internal Microfluidic Assembly (MA) is contained within the Pressure Vessel and is described separately.

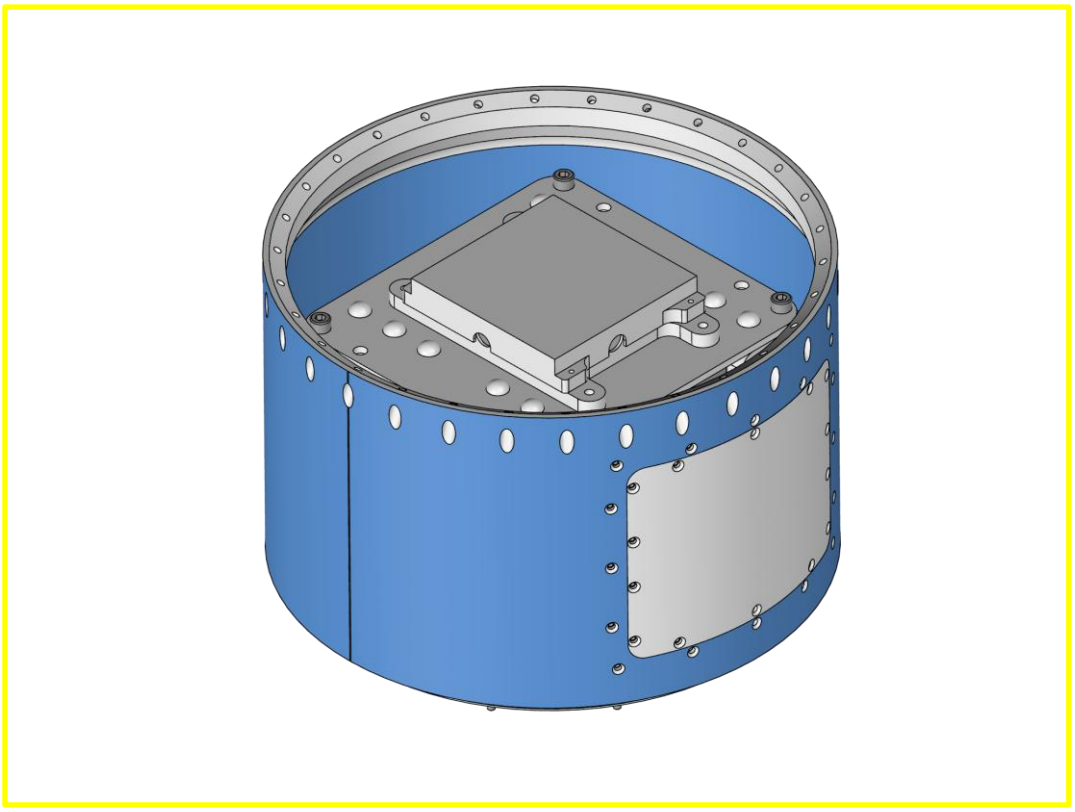


Figure 8: Module Perspective View

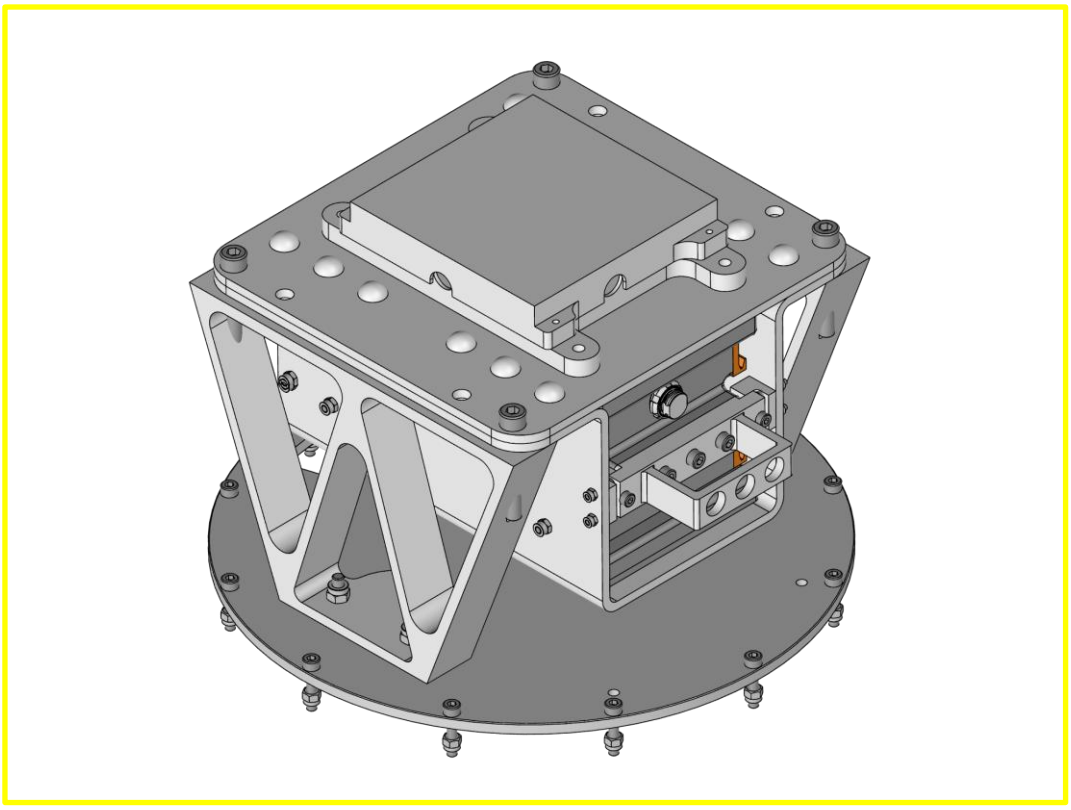


Figure 9: Module Perspective View without REXUS module

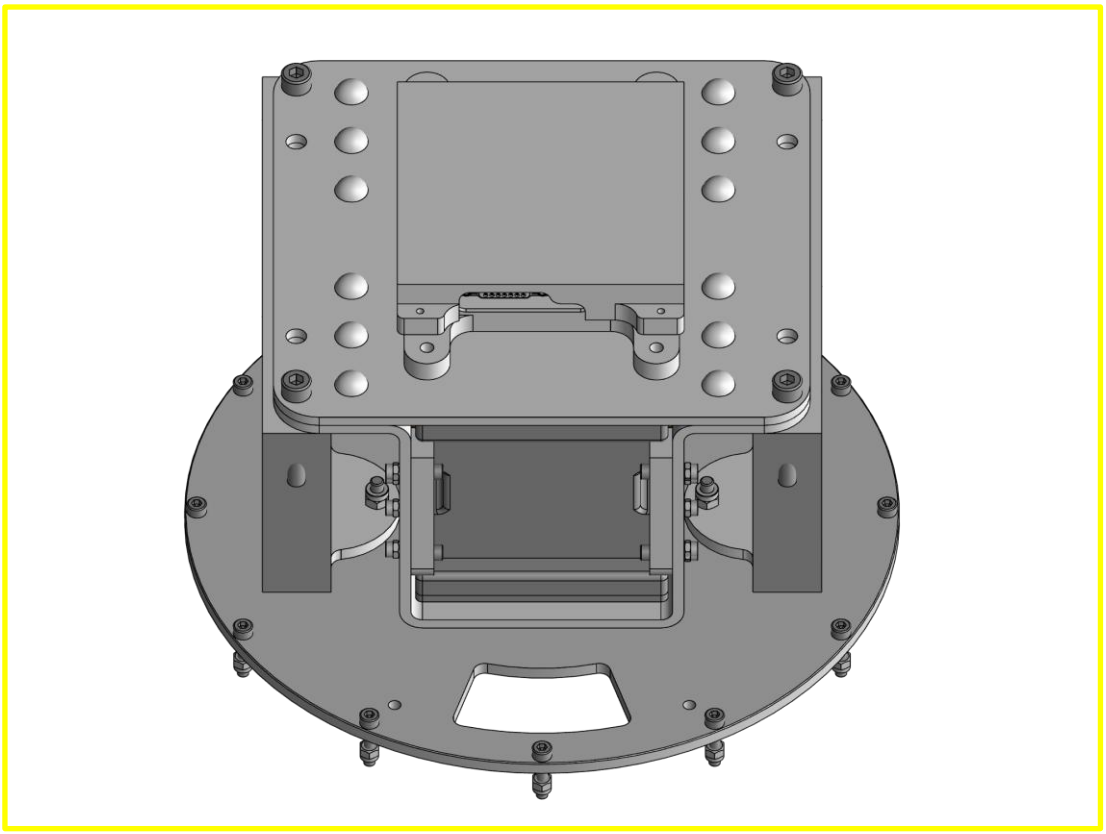


Figure 10: Experiment High Rear View

Early access hatch and integration concept

The THRIVE module and its early-access hatch will be reused for this flight. The existing hatch is compatible with the current STAR-D mechanical configuration and does not require modification. The Pressure Vessel is inserted into the Experiment Support Frame through the early-access hatch and mechanically secured in its flight position before vehicle arming.

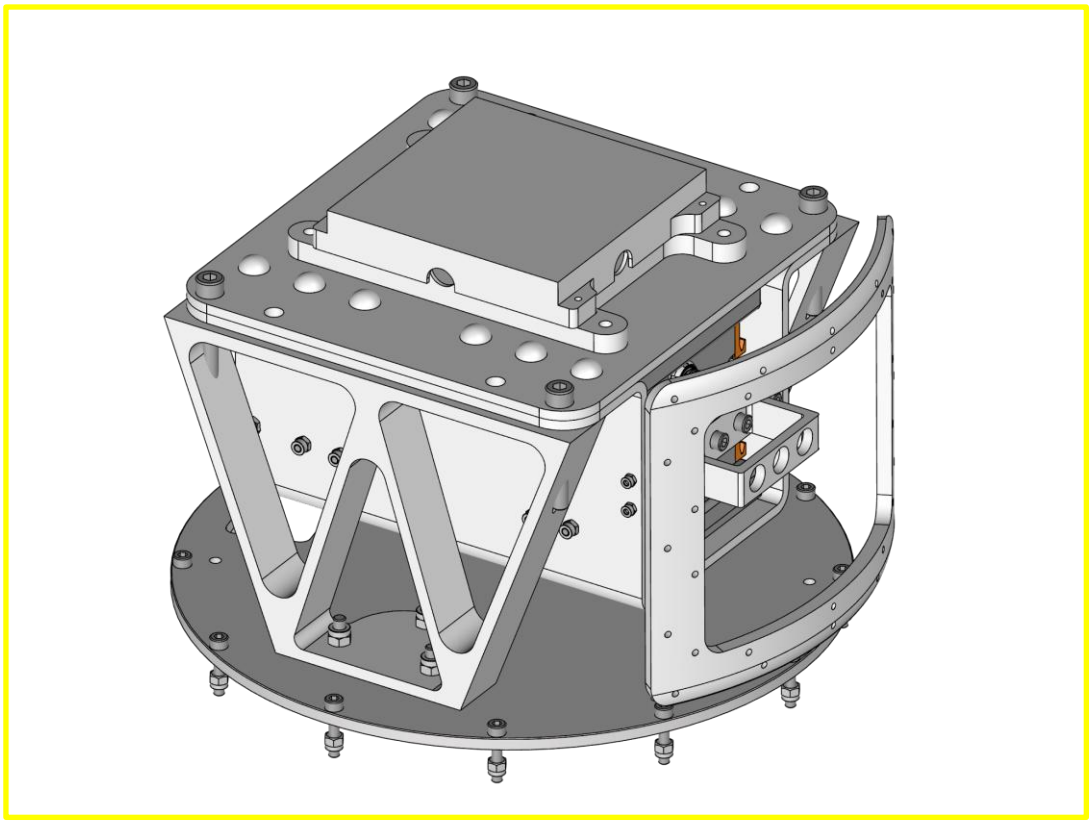


Figure 11: Experiment Assembly In Front of the Early Access Hatch

Bottom bulkhead arrangement

The bottom bulkhead of the REXUS module is used as the primary mechanical interface between STAR-D and the launch vehicle. All loads generated by the experiment assembly are transferred to the rocket structure through this bulkhead.

Two machined aluminum support feet are mounted directly onto the bulkhead. Each support foot is attached using three M6 bolted connections, resulting in six M6 attachment points in total. The connections pass through the bulkhead and are secured using nuts on the opposite side.

At their upper end, the support feet are connected to the Experiment Support Frame using two M5 screws per foot. The screws pass through clearance holes in the structural plates and are fastened into threaded holes machined directly into the support feet.

A dedicated opening in the bulkhead allows the D-SUB electrical connection to pass through the bulkhead.

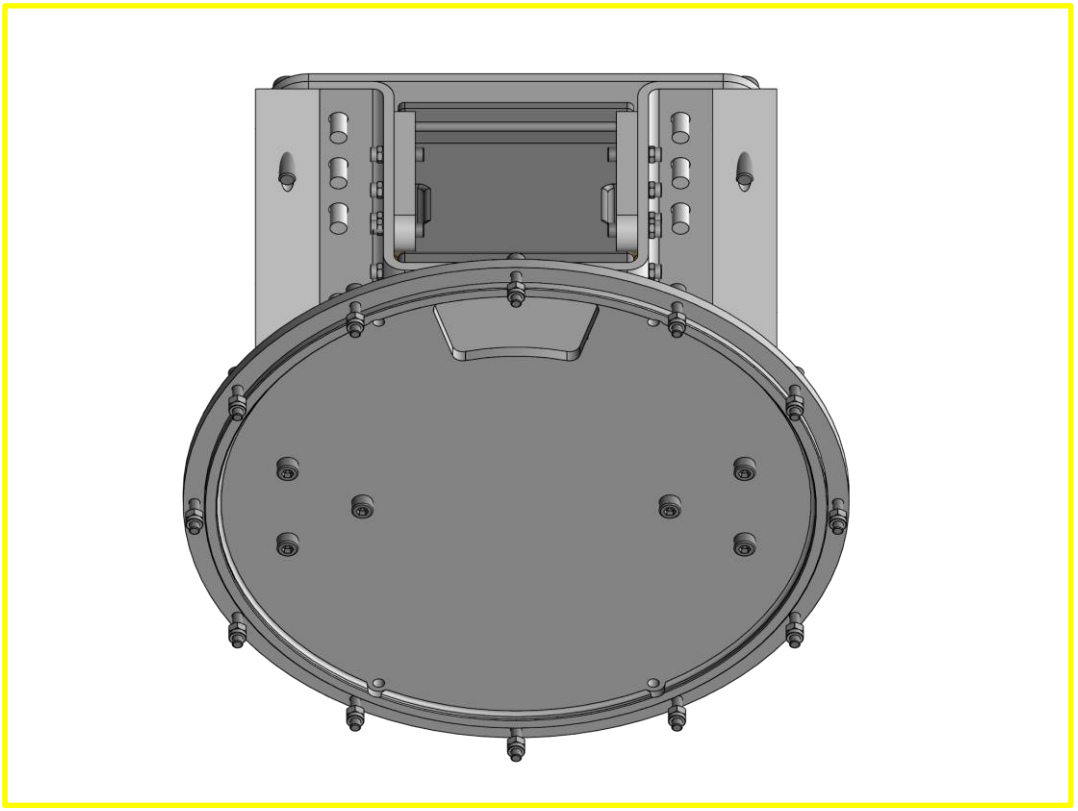


Figure 12: Bottom View of the BulkHead

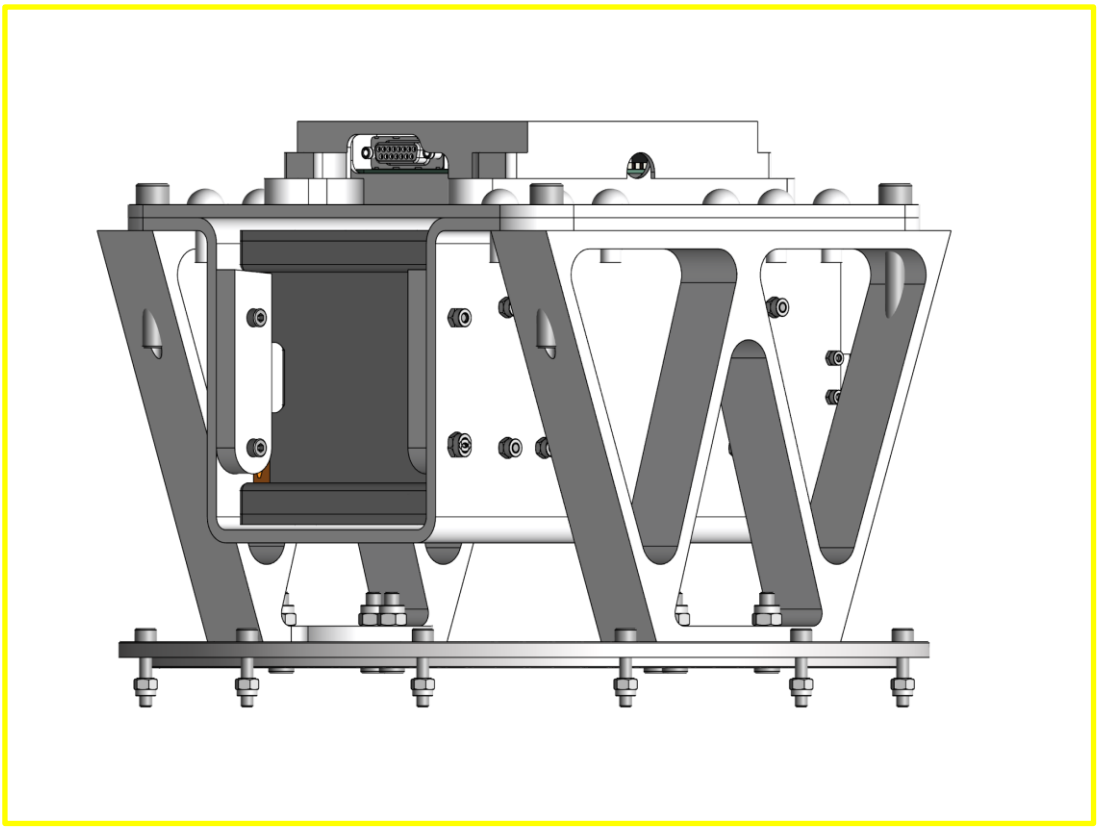


Figure 13: Back View of the Experiment Assembly

Rigid support structure

The wire-rope vibration isolation system presented in the previous design iteration was removed following the CDR. No dedicated vibration isolation element is therefore included in the final mechanical architecture.

The Experiment Support Frame is rigidly connected to the REXUS bottom bulkhead through the two machined aluminum support feet. Increasing the height of the support feet allows them to directly bridge the distance between the bulkhead and the upper structure, replacing the former support and wire-rope assembly.

This configuration simplifies the mechanical load path and removes the relative displacement previously associated with the compliant wire-rope isolators. Consequently, the dedicated hard stops associated with the wire-rope system are no longer required and have also been removed from the design.

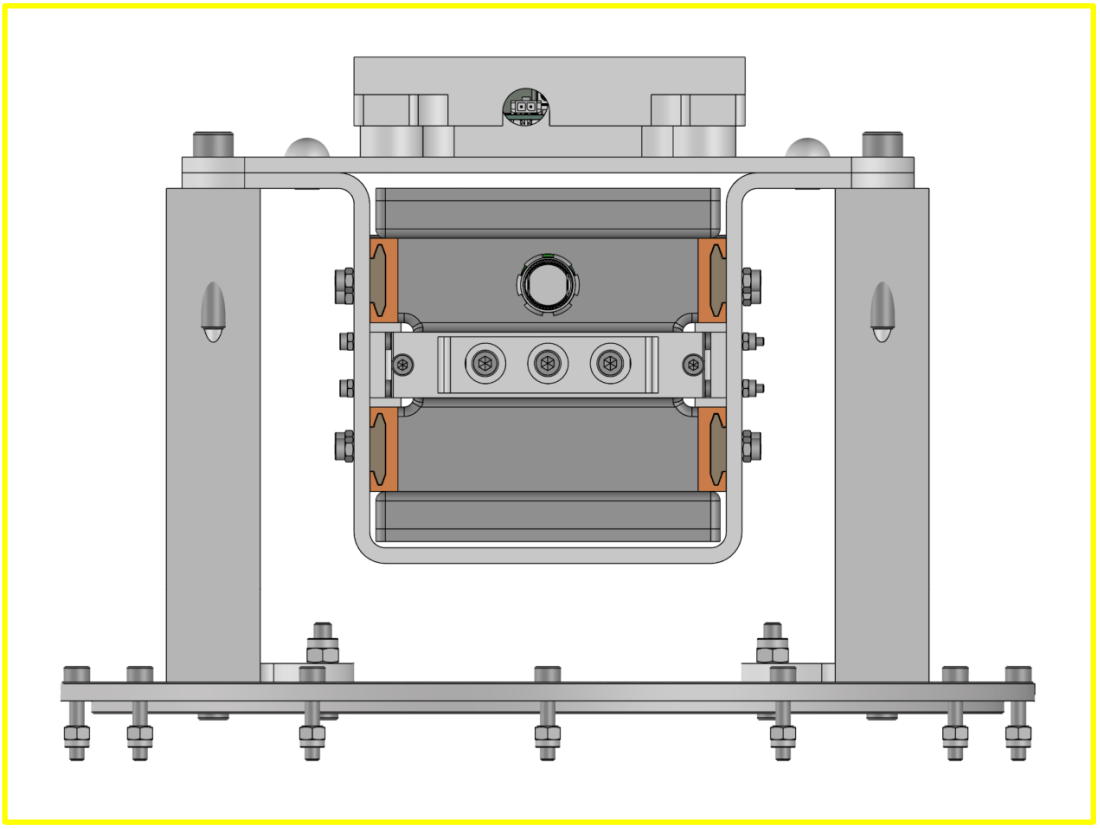


Figure 14: Front View of the Experiment Assembly

Load-carrying structure (Experiment Support Frame)

The Experiment Support Frame is mounted directly onto the upper interfaces of the two rigid support feet. It provides the primary mechanical support and retention interface for the Pressure Vessel.

The upper structure consists of two aluminum structural elements: an upper plate and a folded U-shaped sheet-metal component. The components are not welded together. Instead, they are mechanically clamped to the support feet using four M5 screws, two per foot and with bolt. Each screw passes through clearance holes in both structural elements and engages directly with a threaded hole in the corresponding support foot.

The structure is designed to remain mechanically independent from the REXUS module walls. Loads are therefore transferred through the support feet to the bottom bulkhead rather than through contact with the module casing.

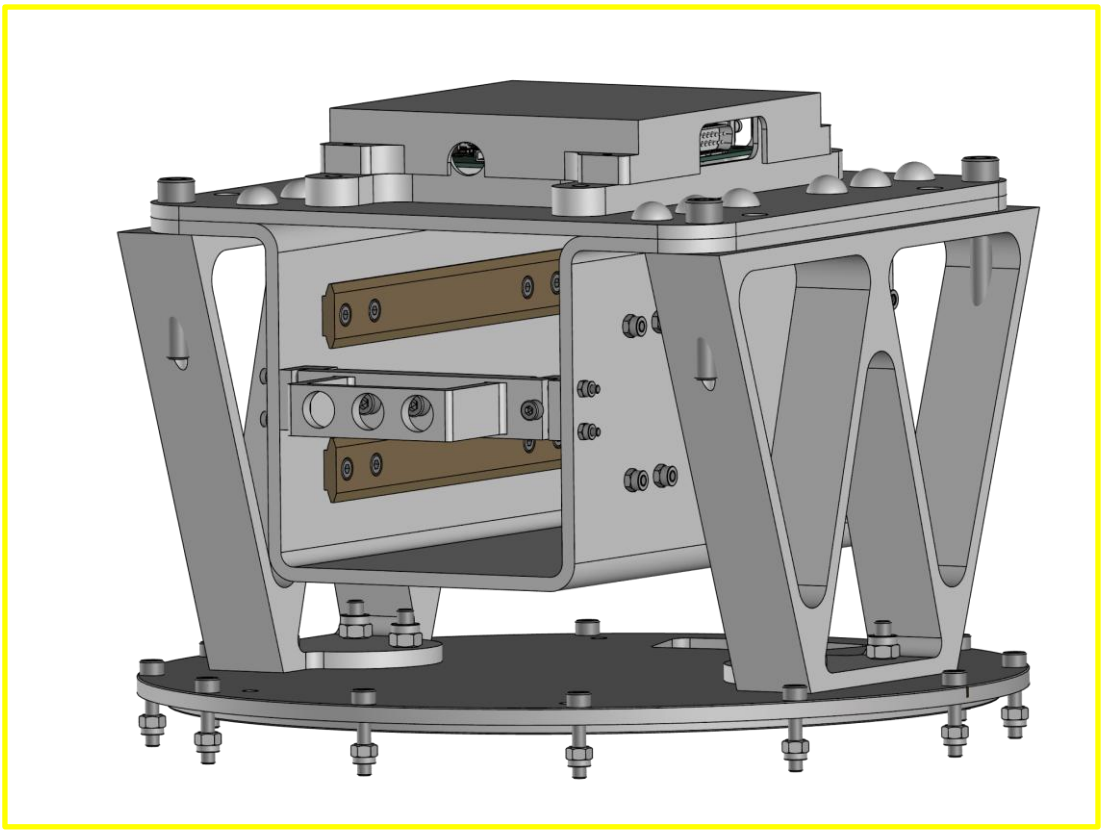


Figure 15: Perspective View of the Experiment Support Frame

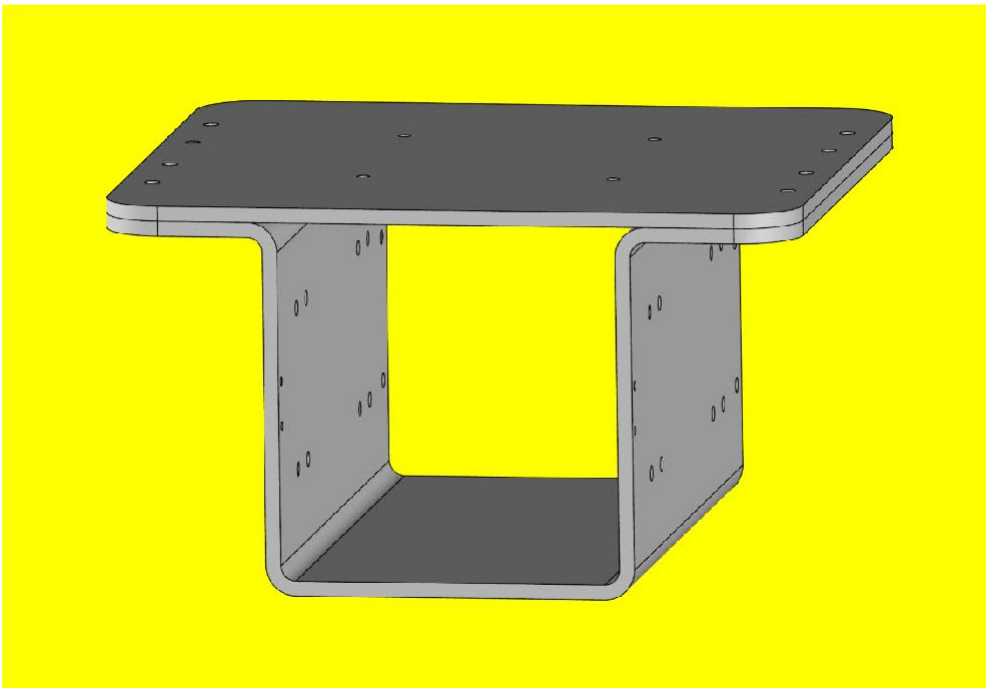


Figure 16: Perspective View of the Experiment Support Frame

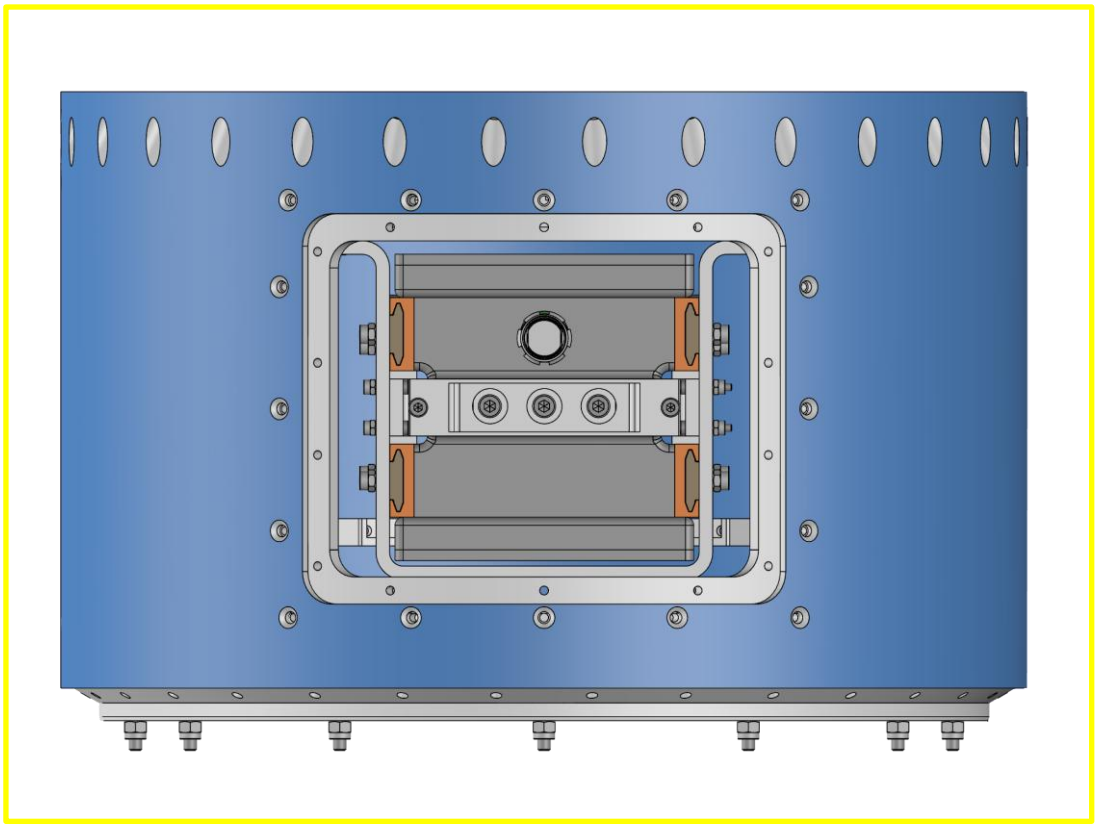


Figure 17: Front View of the Experiment Assembly

General Mechanical Assembly

The experiment has been designed as a fully modular mechanical assembly. Each major subsystem is mechanically fixed to the Experiment Support Frame using dedicated interfaces, allowing straightforward manufacturing, assembly and maintenance.

The Experiment Support Frame is directly attached to the REXUS bulkhead and constitutes the primary load-carrying structure.

The pressure vessel is inserted into the support frame using four PEEK guide rails and secured by front and rear retention features together with a rear locking mechanism.

The electronics enclosure is mounted directly to the upper structural platform using bolted joints. All remaining structural components are attached using removable mechanical fasteners, allowing complete disassembly of the experiment without permanent modification of the structure Loctite will be used for the flight.

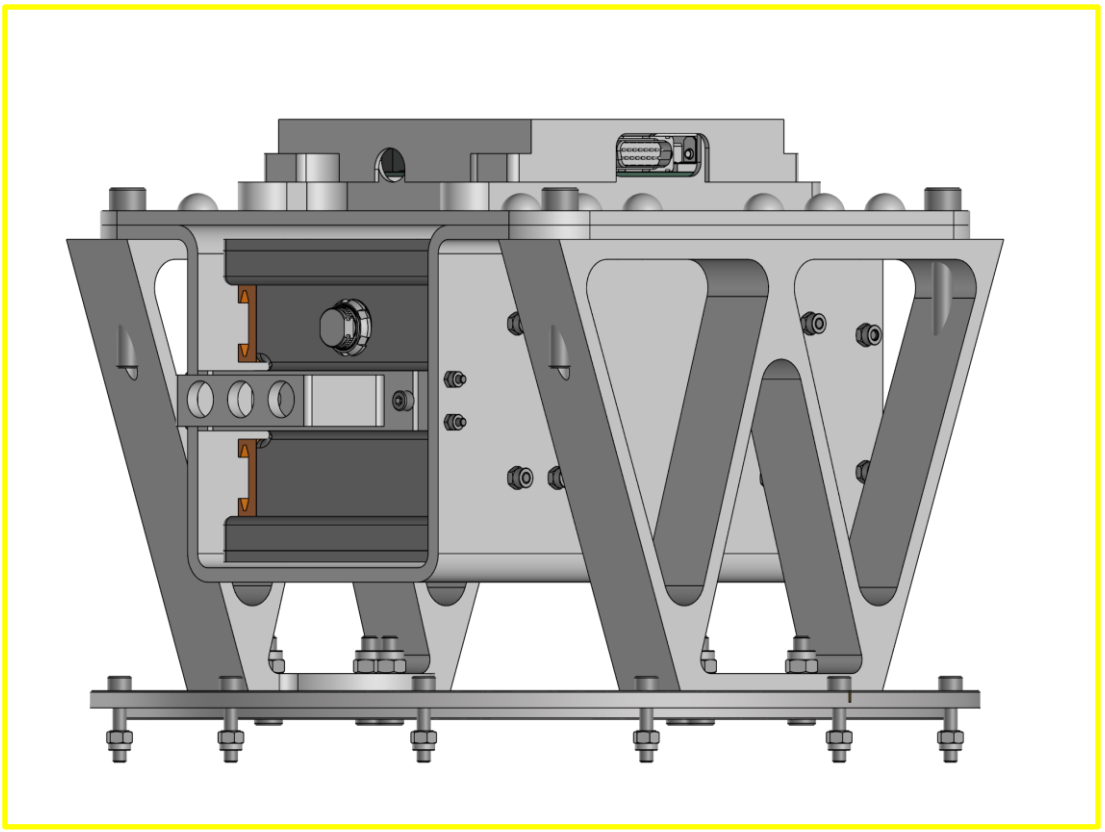


Figure 17.1 Perspective vision

Pressure vessel locking brackets

Two aluminum locking brackets are used to secure the pressure vessel in its flight position after insertion into the Experiment Support Frame. Each bracket is **machined from aluminum** with overall dimensions of 18 × 13 × 21 mm.

Each locking bracket is attached to the Experiment Support Frame using two structural bolts, providing a rigid attachment to the primary structure. A third bolt passes through the pressure vessel handle, securing the pressure vessel in its nominal flight position and preventing unintended withdrawal during launch, flight and recovery.

The locking brackets are fully removable, allowing straightforward installation and removal of the pressure vessel during early-access operations while ensuring reliable mechanical retention during flight.

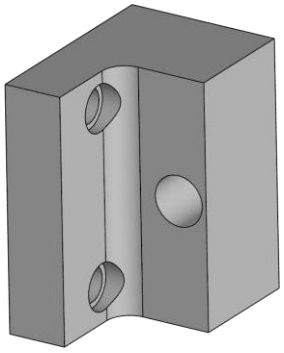


Figure 17.2 Locking mechanism component

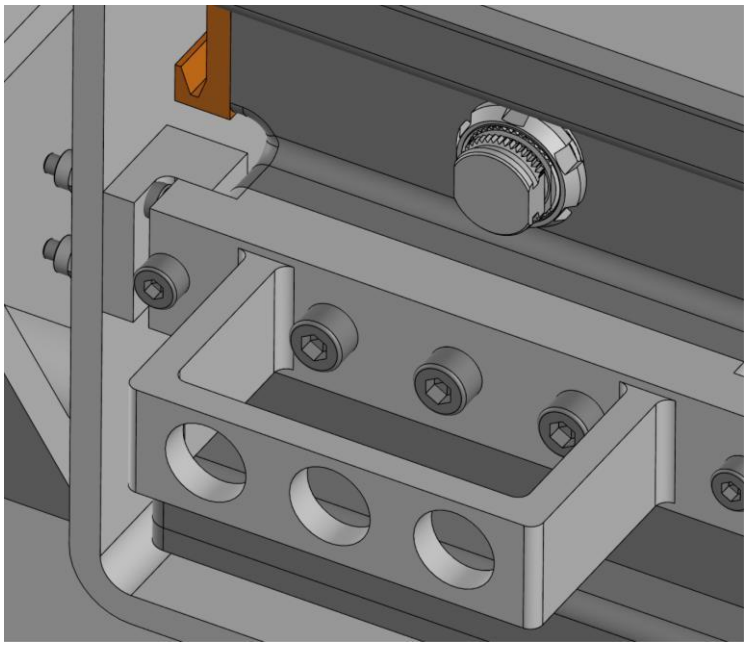


Figure 17.3: Locking mechanism

PCB shielding enclosure

The interface PCB is enclosed in a **2 mm thick aluminum housing** mounted on the upper structural platform. The enclosure provides mechanical protection and electromagnetic shielding while allowing the required electrical connections.

Dedicated cut-outs are integrated into the enclosure walls for the cables and connectors. These openings are limited to the dimensions required for cable passage and connector access.

Small gaps may remain along the folded corner regions of the metal housing. Additional copper tape will therefore be added locally over these regions to reduce the open areas and improve the electrical continuity of the enclosure.

Inside the enclosure, the PCB is mounted on insulating standoffs. These standoffs maintain a defined clearance between the PCB, the supporting structure and the aluminum housing, preventing unintended electrical contact and reducing the risk of short circuits.

A removable aluminum cover closes the upper side of the enclosure while preserving access for assembly and maintenance.

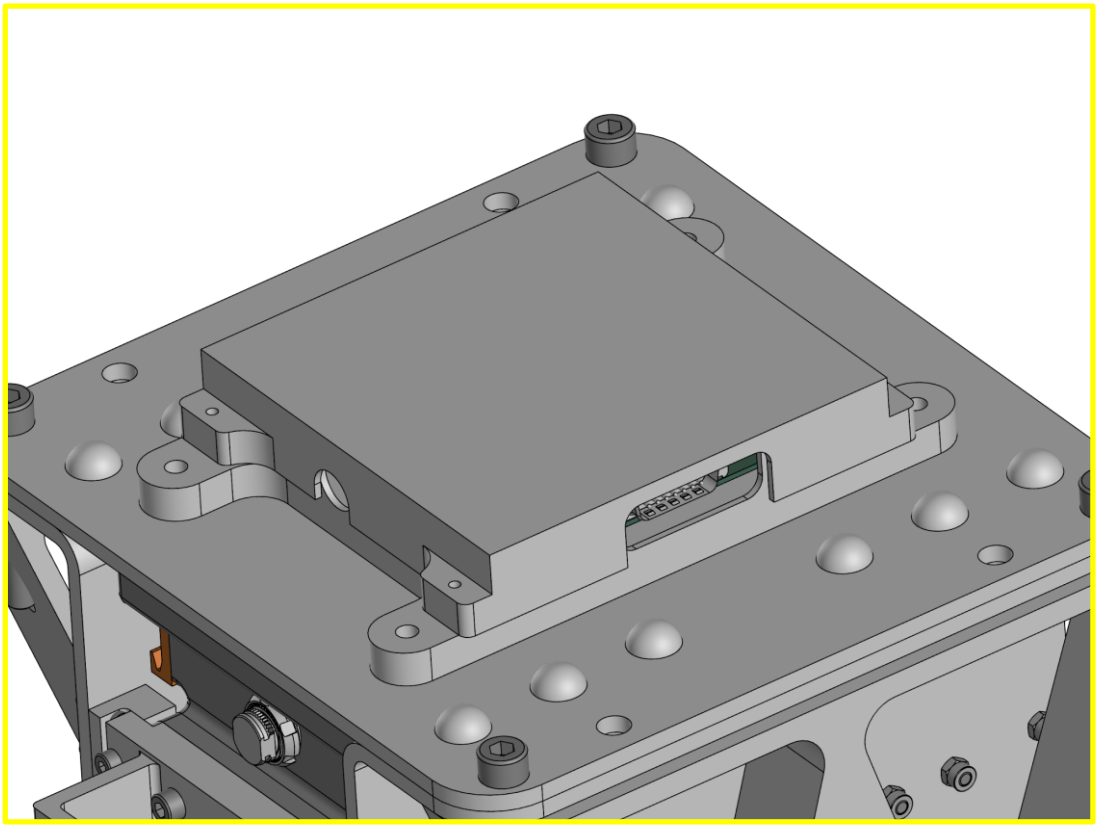


Figure 17.4: PCB Box

4.4.2 Pressure Vessel

Pressure Vessel Design

The Pressure Vessel (PV) provides a sealed environment for the Microfluidic Assembly during the rocket flight. Its role is to keep the internal pressure close

to atmospheric conditions despite the very low external pressure reached at high altitude.

The PV is made of a machined Aluminum 6061-T6 enclosure closed by a removable cover plate. Sealing between the body and the cover plate is achieved using a static O-ring (2mm diameter) placed in a rectangular groove machined in the plate.

The groove dimensions were selected to obtain an O-ring compression of about 25%, which is within the usual range for static sealing and ensures sufficient contact pressure. The cover plate is secured to the enclosure using a distributed pattern of M4 bolts that provide the clamping force required to maintain the seal during the mission.

Preliminary analytical calculations indicate that the loads generated by the expected pressure differential remain well within the structural limits of the enclosure and fastening system.

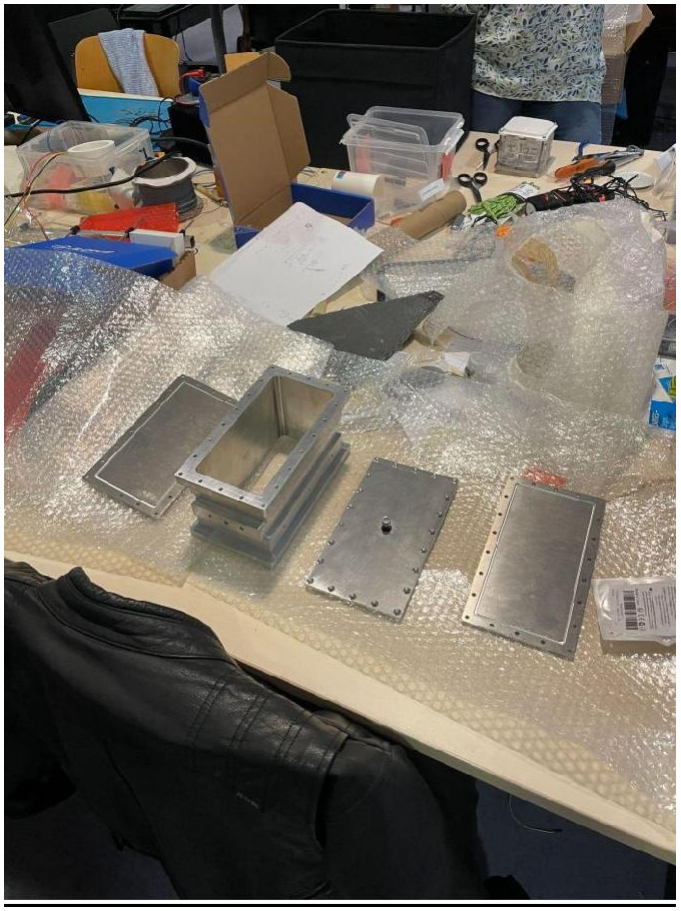


Figure 18: Pressure vessel parts

An electrical hermetic connector is integrated on the front of the pressure vessel in order to route power and data between the internal experiment electronics and the external system. The connector is mounted through a threaded

interface in the vessel wall, ensuring a sealed mechanical interface compatible with the pressurized environment.

A compact push-pull connector qualified for vibration environments is used in order to ensure reliable electrical connection during launch and flight conditions.

Pressure Vessel guidance and locking

The Pressure Vessel is inserted through the early access hatch and guided into the Experiment Support Frame using rail guides. These rails ensure controlled insertion, precise lateral positioning, and repeatability during integration.

Once inserted, the PV is mechanically constrained in all six degrees of freedom using a combination of rail guidance and end-stop. This prevents axial sliding and relative motion of the PV during launch, flight, and recovery.

The end-stop is screwed to the Experiment Support Frame from the back.

The handle is used as an end-stop and is screwed to the pressure vessel when installed to the two supports.

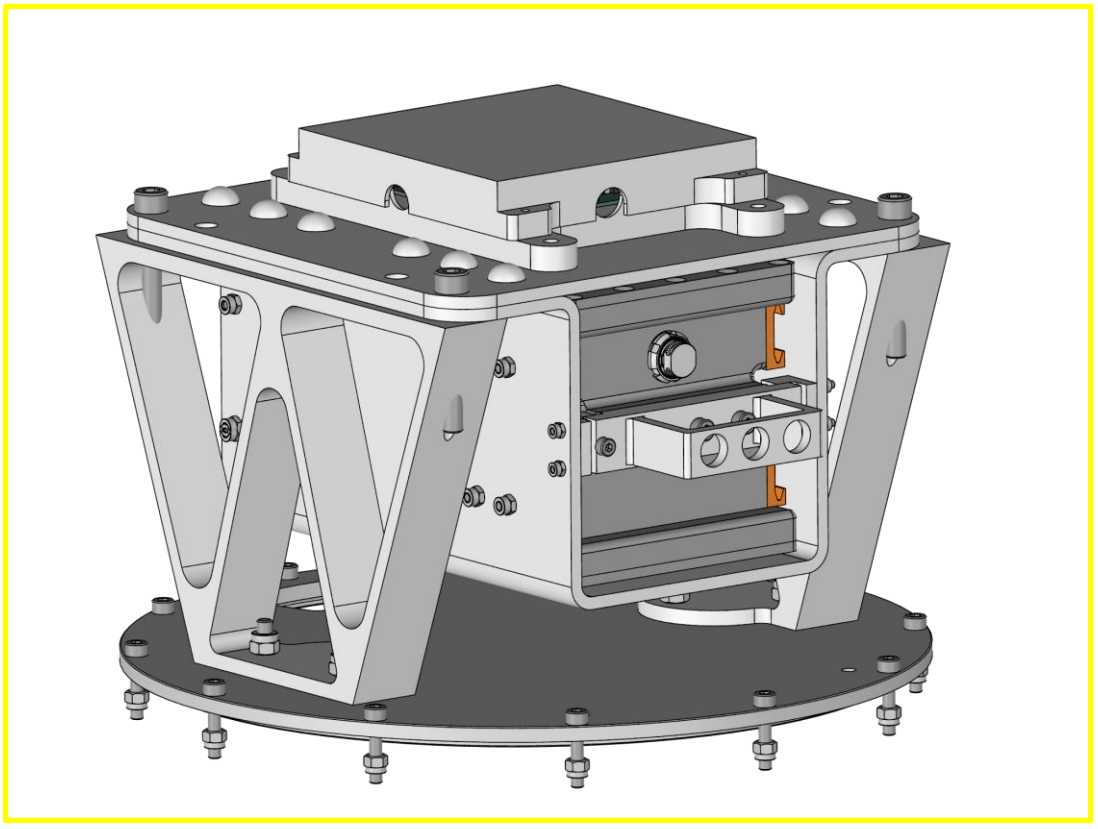


Figure 19: Front Perspective View of the Pressure Vessel inside the Experiment Support Frame

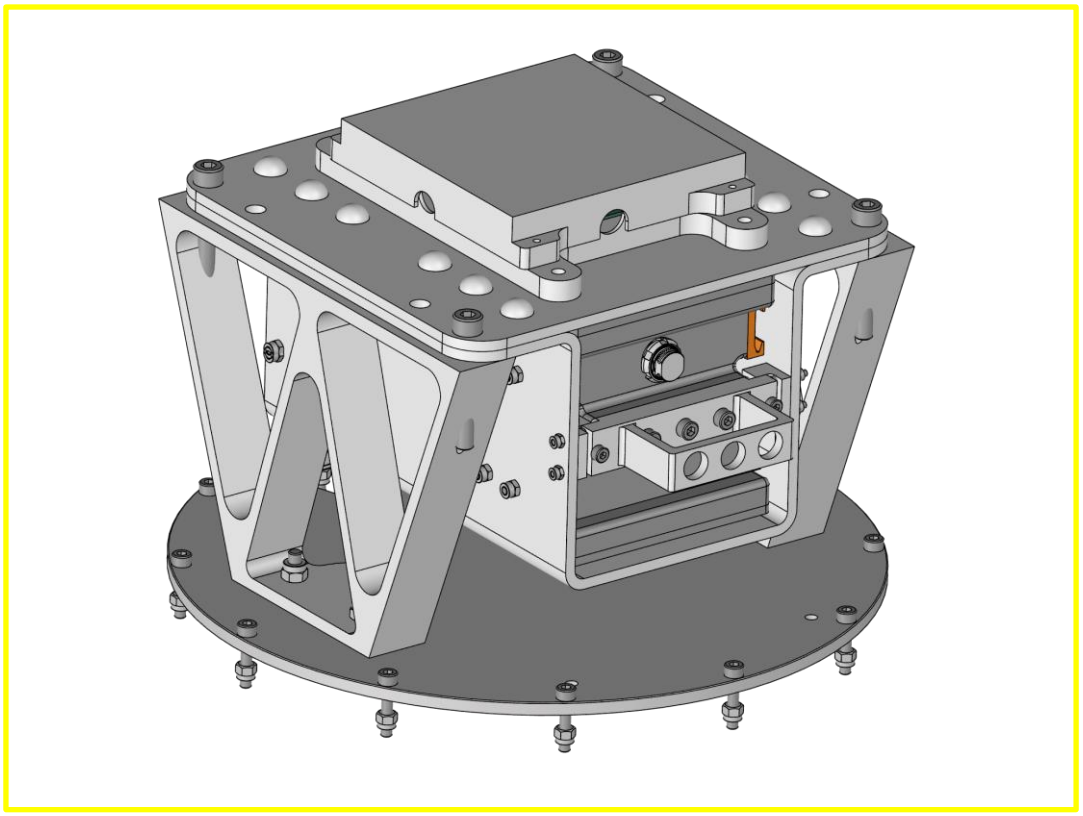


Figure 20: Front Perspective View of the Experiment Assembly

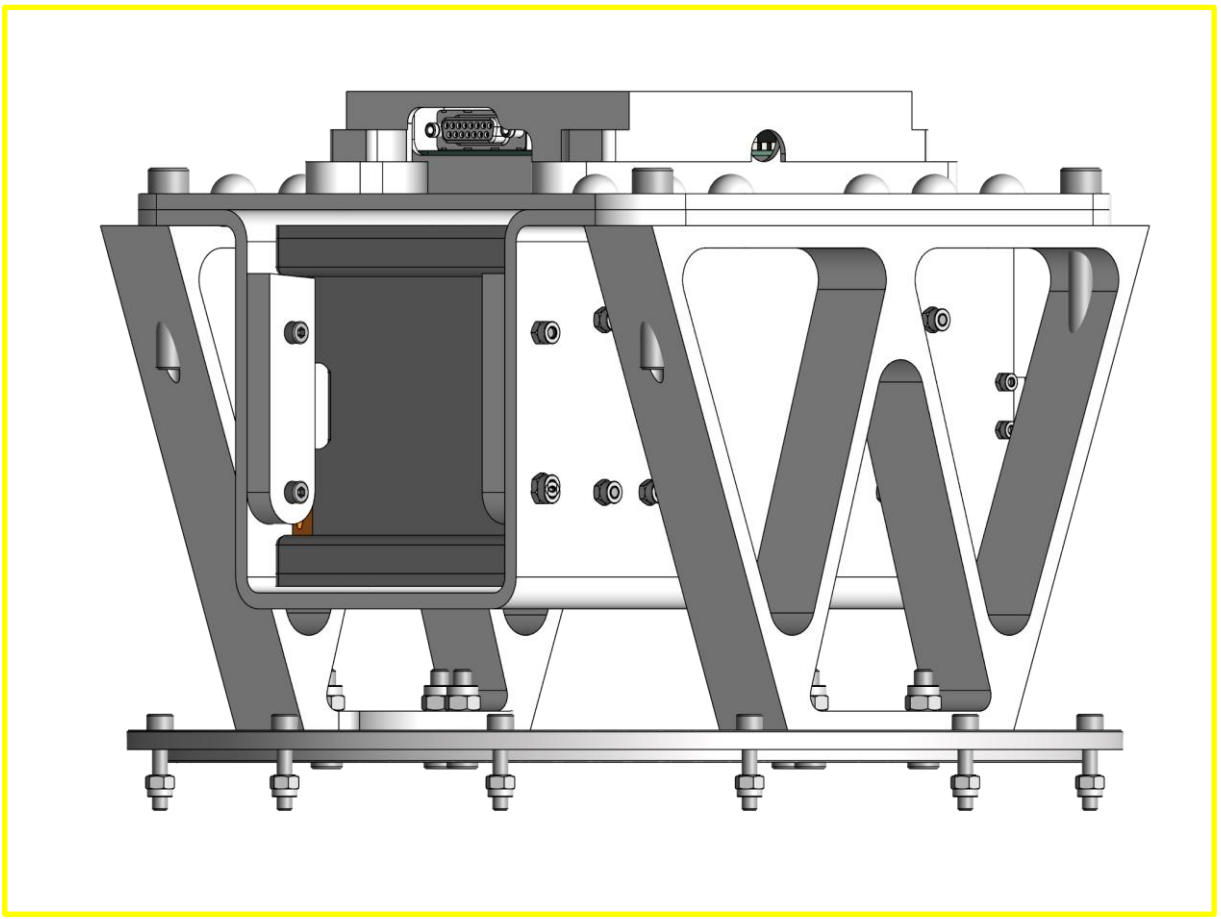


Figure 21: Back View of the Pressure Vessel

Load path definition

The primary structural load path of the STAR-D experiment extends from the Experiment Support Frame through the two rigid aluminum support feet to the REXUS bottom bulkhead.

Loads generated by the Pressure Vessel and the equipment supported by the Experiment Support Frame are transferred from the frame to the two support feet through four M5 bolted connections, with two screws per foot. These screws pass through clearance holes in the upper structural elements and engage with threaded holes machined directly into the support feet.

The loads are subsequently transferred from the support feet to the REXUS bottom bulkhead through six M6 bolted connections, with three attachment points per foot. The M6 fasteners pass through the support feet and bulkhead and are secured using nuts.

This configuration provides a continuous rigid load path between the experiment assembly and the REXUS structure. No load-bearing connection is made between the Experiment Support Frame and the cylindrical module wall.

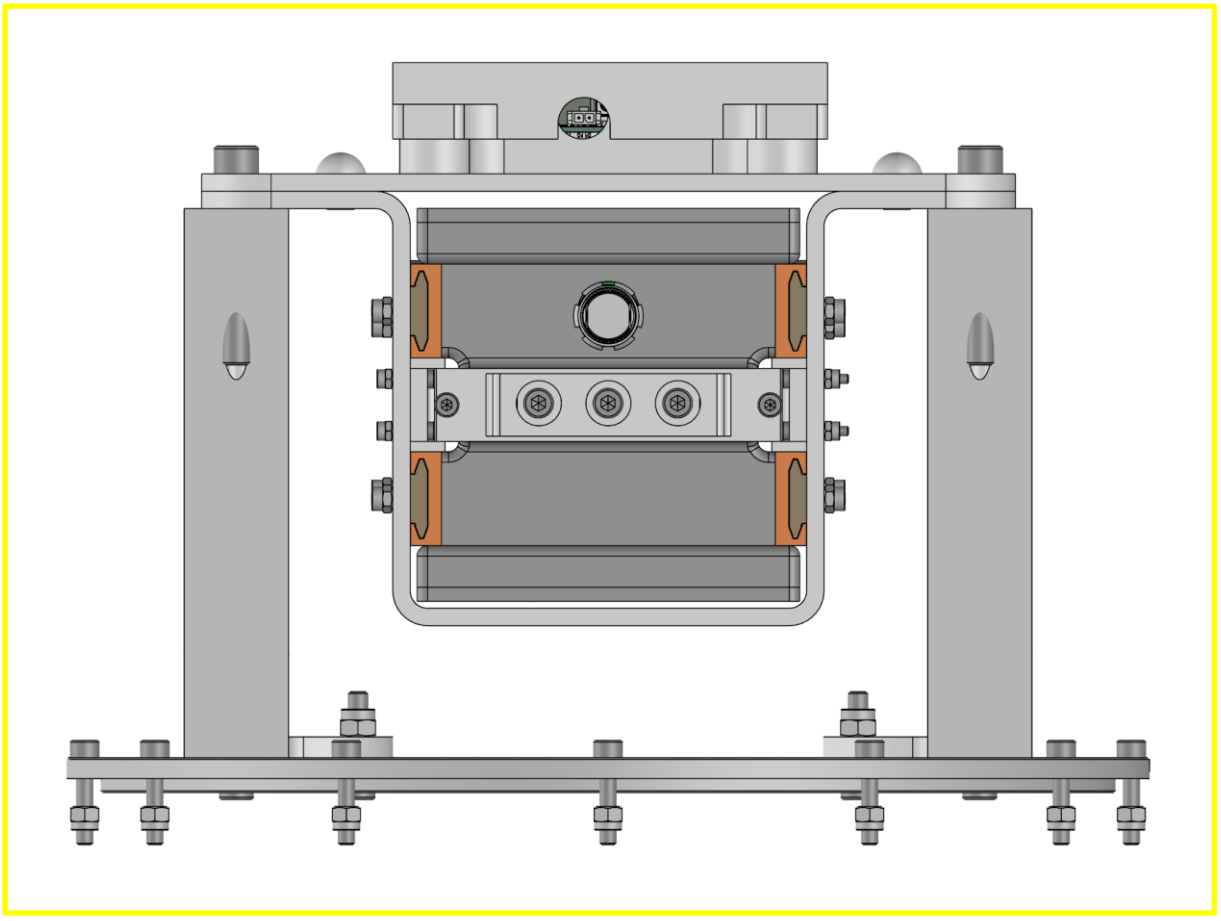


Figure 22: Front View of the Experiment Assembly

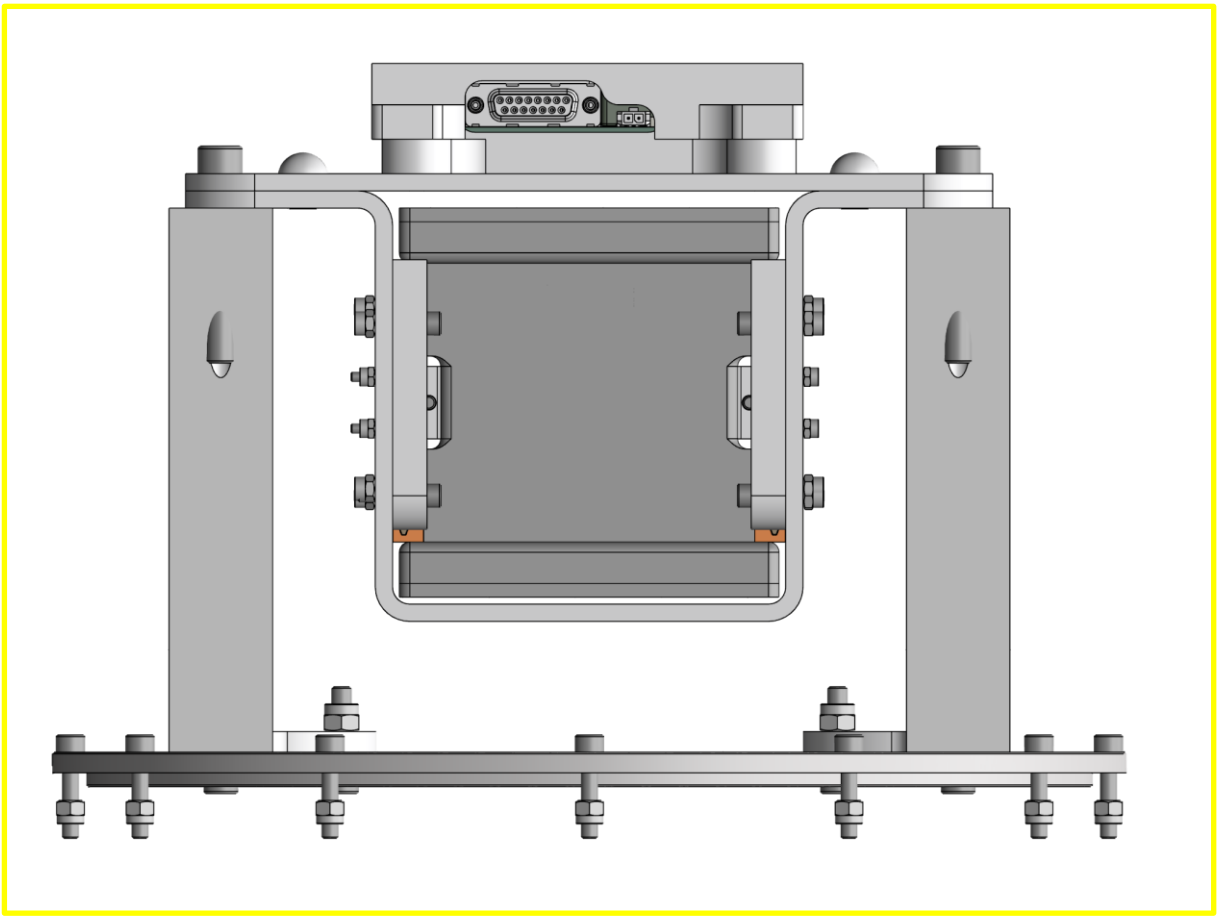


Figure 23: Back View of the Experiment Assembly

Pressure vessel guidance and retention

The pressure vessel is inserted into the Experiment Support Frame along four aluminium rail guides. These rails ensure repeatable positioning during early-access integration and provide the main structural support for the pressure vessel once installed.

In the flight configuration, the pressure vessel bears against the four rails. The acceleration loads acting in the corresponding support directions are therefore transferred through the rail contact surfaces into the Experiment Support Frame. The use of four rails distributes the contact loads, limits local deformation and improves the alignment of the pressure vessel within the frame.

Longitudinal displacement and withdrawal are prevented by the front and rear retention features. The rear handle is used for manual handling and is secured to the support frame by two screws, one on each side, after insertion. These screws maintain the pressure vessel in its installed position and prevent unintended withdrawal. They are not considered the sole structural load path for the pressure vessel.

Depending on the direction of the applied acceleration, the loads are transferred through the four rail contact surfaces and the corresponding front or rear retention feature into the Experiment Support Frame.

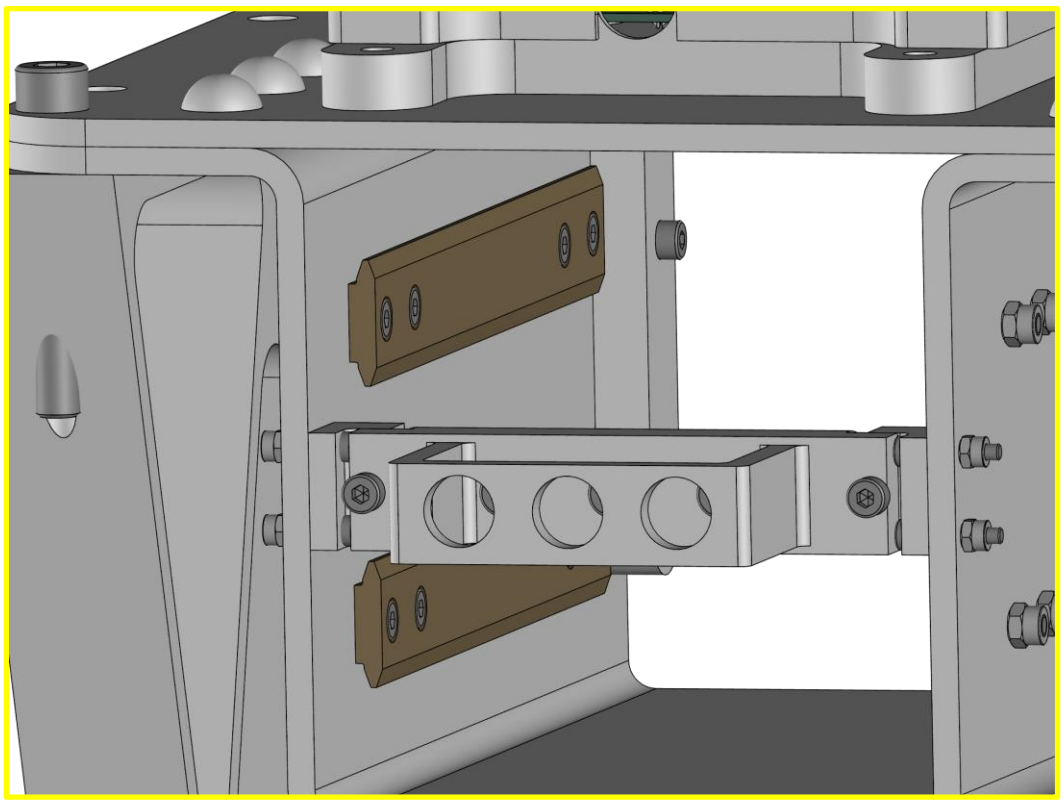


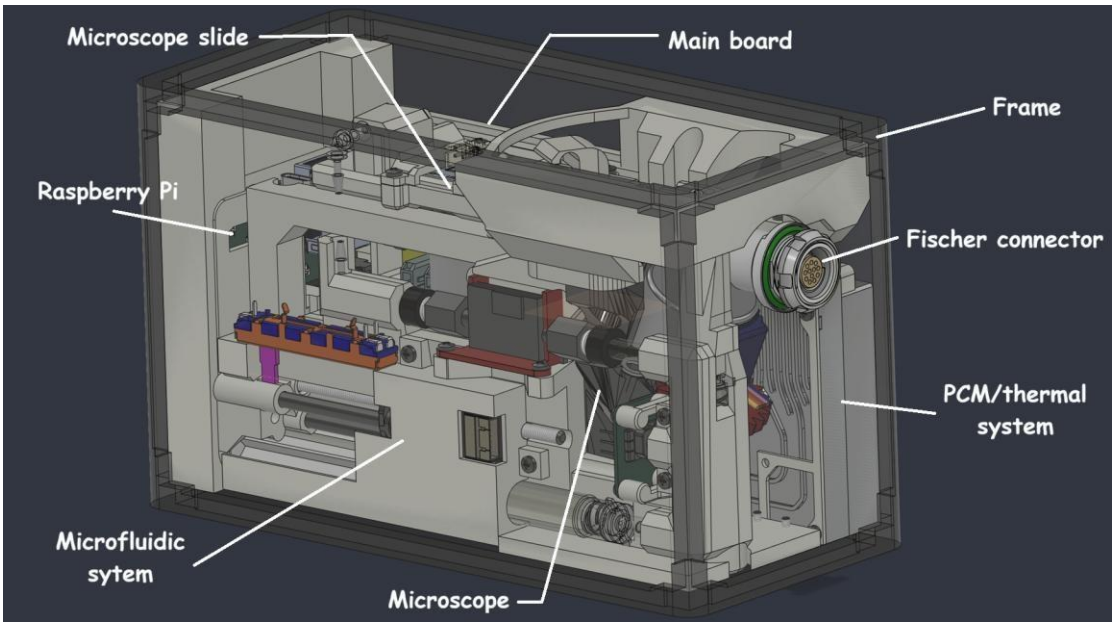
Figure 24: Rails

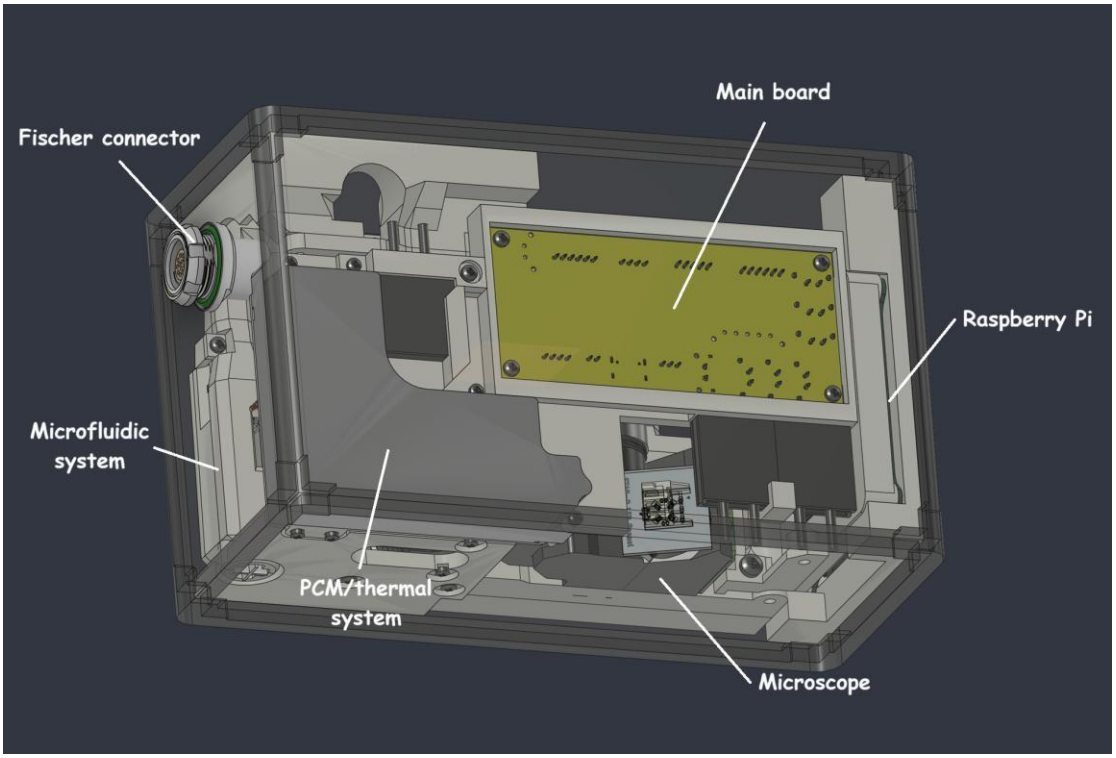
4.4.4 Microfluidic Assembly (MA)

The MA represents all the parts contained inside the pressure vessel of the experiment. It contains the following systems:

- **Microscope**
- **Microfluidic system**
- **Sample observation/Microscope slide**
- **Electronics integration**
- **Thermal integration/PCM**

The global design is built around a single structure built by SLS nylon 3D printing. Modules or parts are inserted into their recessed mounting location into the structure. Caps containing recessed location for the top of several components are then screwed into threaded inserts.





Figures 25: whole microfluidic assembly - two view angles

Microscope design

The microscope is built around the OpenFlexure microscope platform. The optics was modified to make the setup an epifluorescence microscope. The original CAD files are given ready-to-print. Functional surfaces of the components are kept unmodified. It is 3D printed with classic PLA filament to

keep its intended properties; it is not considered structural but a component to fix to the structure.

The microscope focuses by vertically translating a unified optics module via a stepper-driven flexure stage. To maintain strict optical alignment and prevent aberrations, the objective, a 50 mm achromatic correction lens, and the digital sensor are secured at fixed distances within a rigid 3D-printed housing. Moving this entire optical assembly as a single unit maintains a constant mechanical tube length whilst aligning the focal plane with the stationary sample.

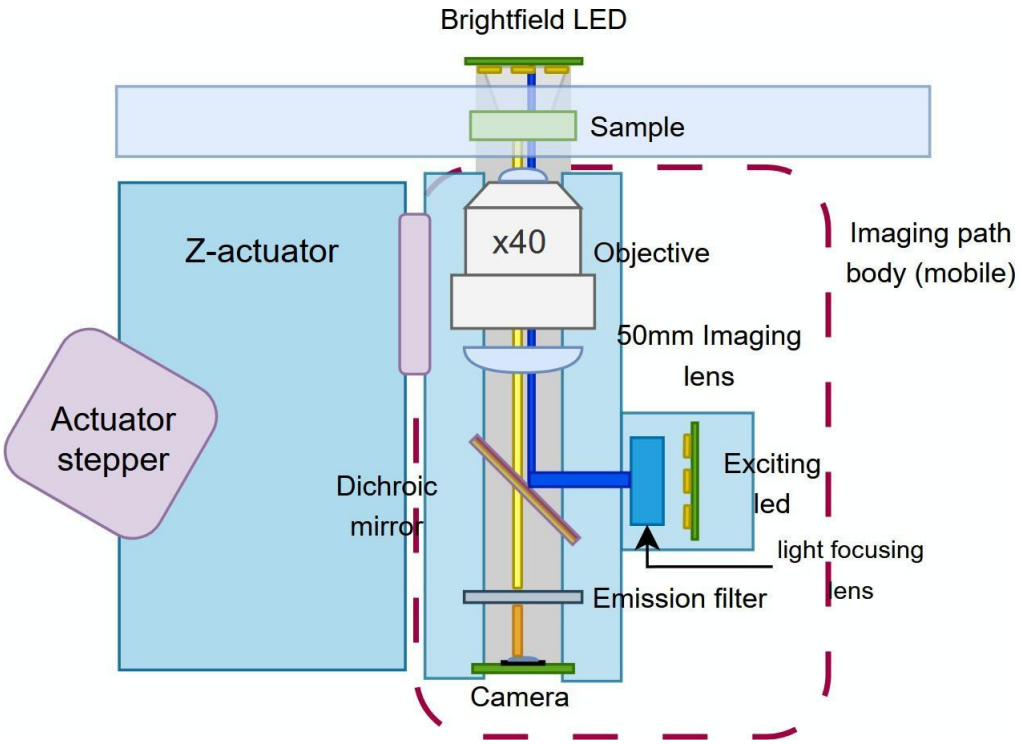


Figure 26: Diagram of the microscope setup(before camera modification)

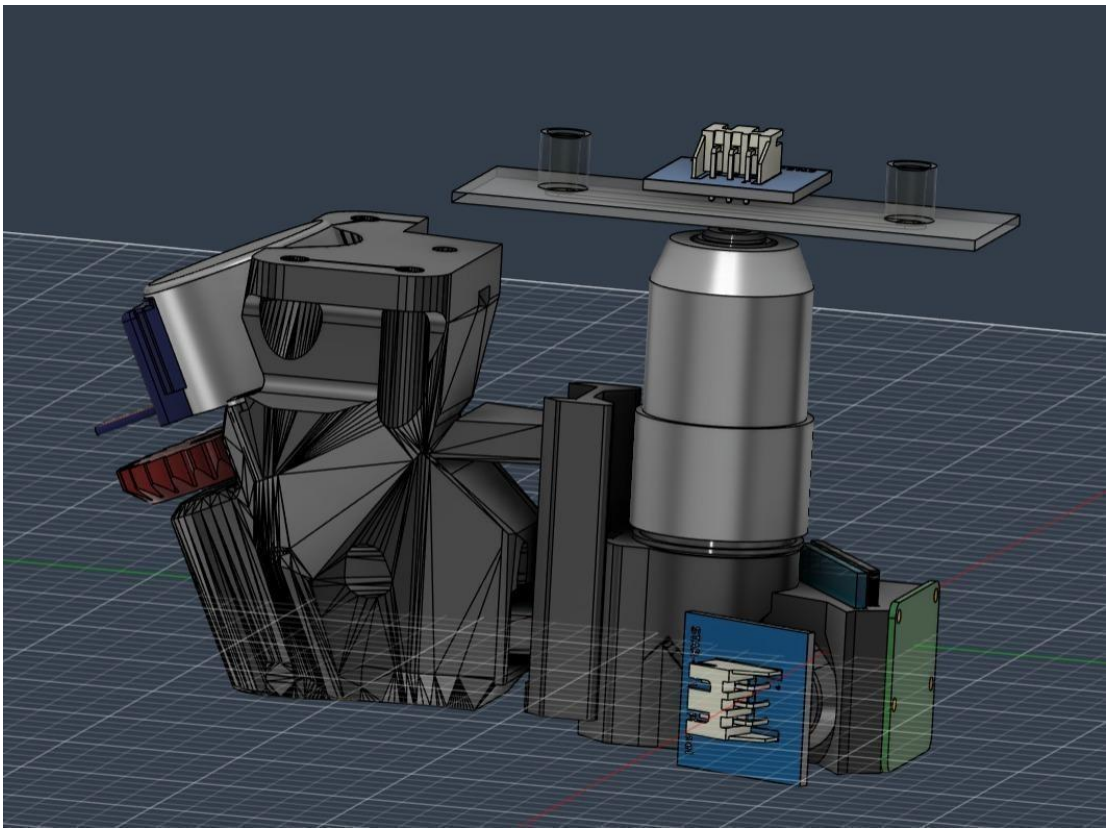


Figure 27: Representation of the microscope system

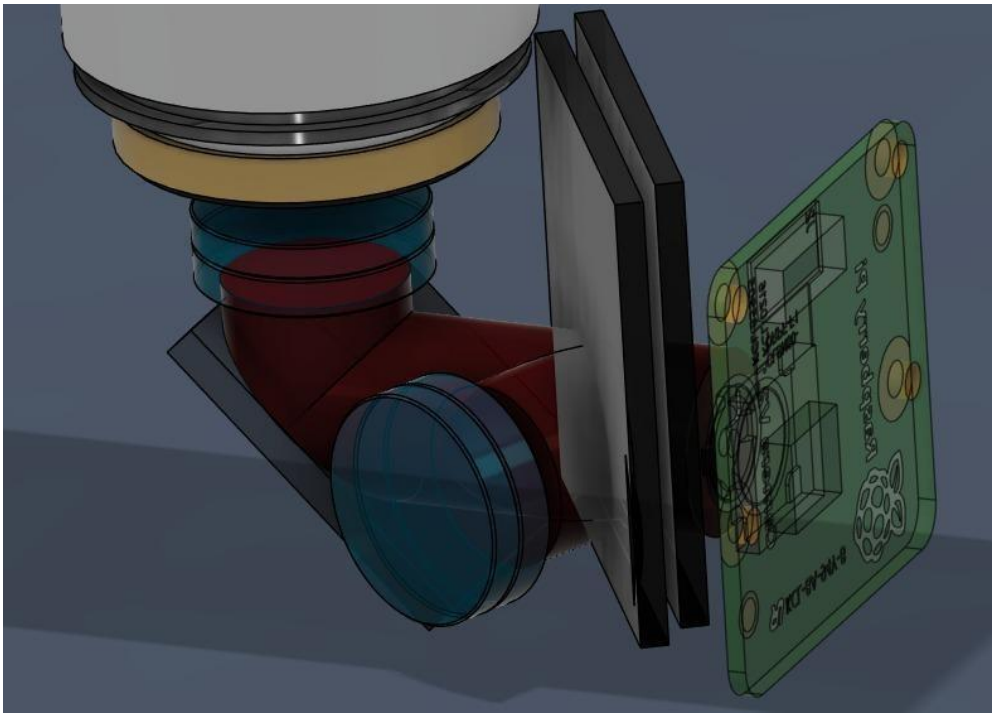


Figure 28: Light path inside of the microscope focus

Z-actuator

The Z-actuator uses a 28BYJ-48 stepper motor maintained to the frame with screws. The reduction setup is made by two reducting gears connected to a continuous plastic part that uses elastic deformation to achieve the reduction. The reduction setup has a position resolution of less than 0.1µm in its original design. Screw holes are added at the bottom of the actuator frame for rigidity.

Optics system

The electronics and optics are mounted on the moving part of the microscope. They are both enclosed in the moving part by closing together the halves-frames with screws. During launch the moving frame rests on an external support to diminish the length of recalibration step and ensure no damage occurs to it.

Camera

The camera is a RasPi V2 Camera module used for imaging; it is screwed at the bottom of the light tube in threaded inserts. The geometry of the end of the light tube enables direct contact between the tube and the board to minimize vibrations.

The ribbon to connect the cam to the Raspberry 5 card is secured with caps that follow curves on the global structure made for bending the ribbon with correct radius.

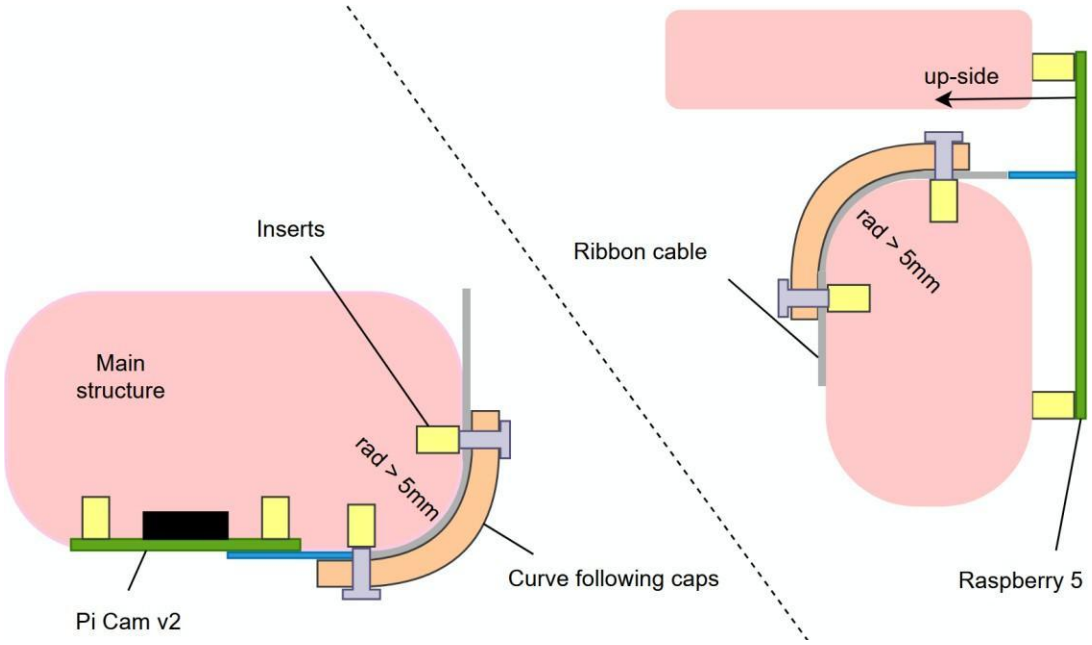


Figure 29: Camera mounting and ribbon securer

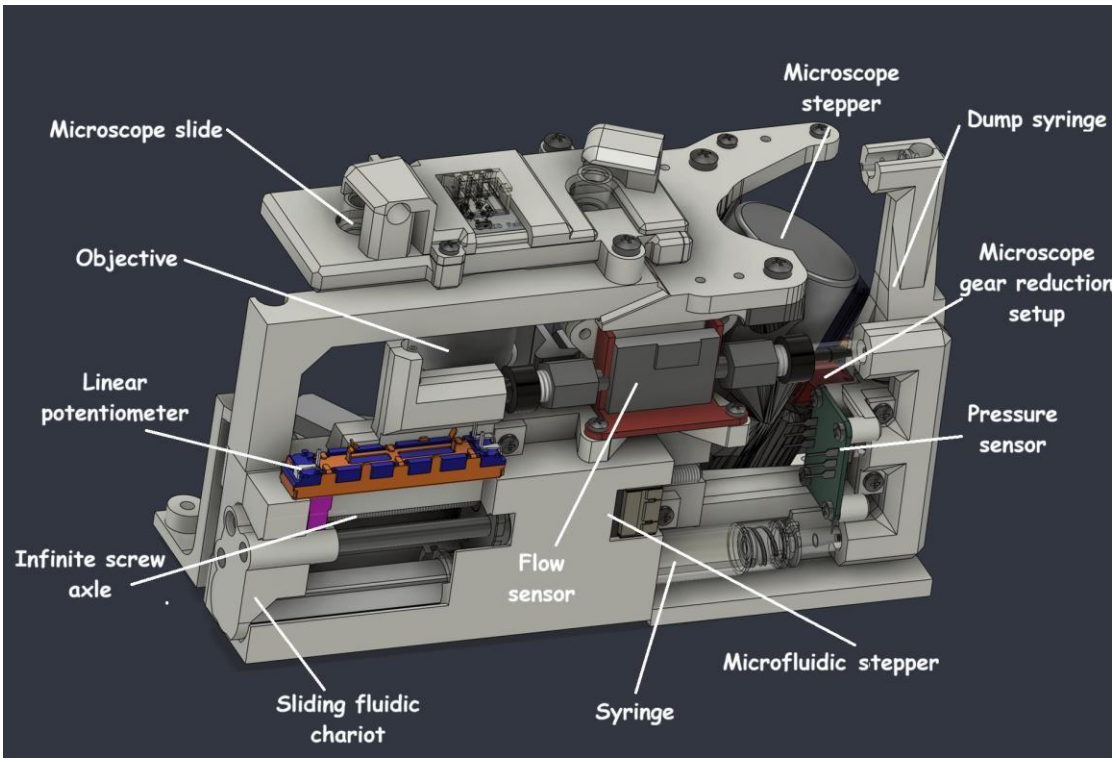


Figure 30: Microscope and microfluidic assembly

Microfluidic System Design

The microfluidic system comprises:

- A **Chip/Slide with a channel** to observe the sample cells interactions with the microscope
- A **3mL syringe** that contains and delivers cells and particles to the slide
- **Pressure and Flow sensors** to have feedback for the PID of the control of the flow delivered by the syringe
- The **Pump system** to control and push the plunger of the syringe
- A **Dump tank** paired with an **Anti-backflow valve** to retain used fluid
- **Tubes and fluidic fittings/adapters** to connect the components

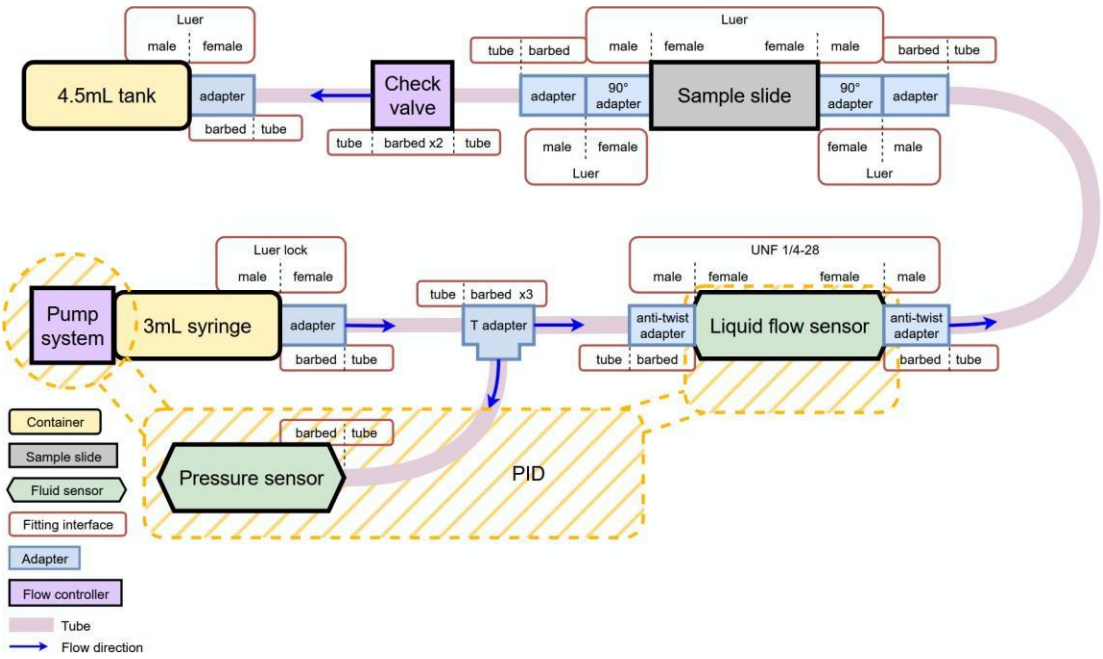


Figure 31: Microfluidic system diagram

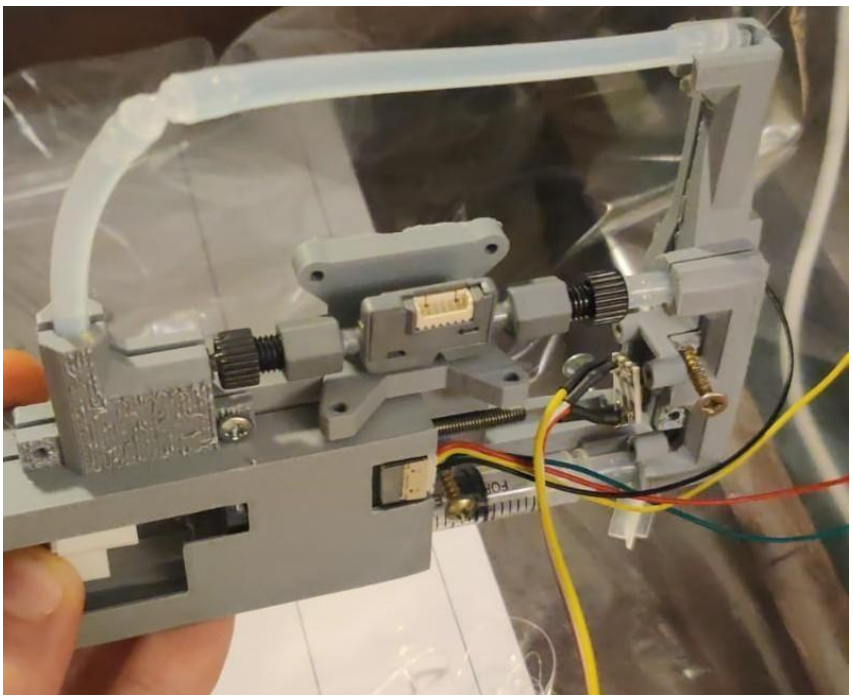


Figure 32: Prototype of the microfluidic system

Pump system

This system is made around a stepper motor that controls an infinite screw axle on which slides a cart guided with a smooth axle. This cart is fixed to the end of the syringe plunger to control its position. The alignment between motor axle, guide axle and syringe tube is made with 2 aluminum alignment plates at the

end of the axles. The system is maintained by a single cap maintained by threaded inserts

To fit inside the pressure vessel, syringe capacity is limited to 2.5mL. The infinite screw threads are 0.6mm long and the angular resolution of the stepper is 1.8°. With an internal diameter of 10mm inside the syringe, we have a fluidic control resolution of 0.24µL(mm³).

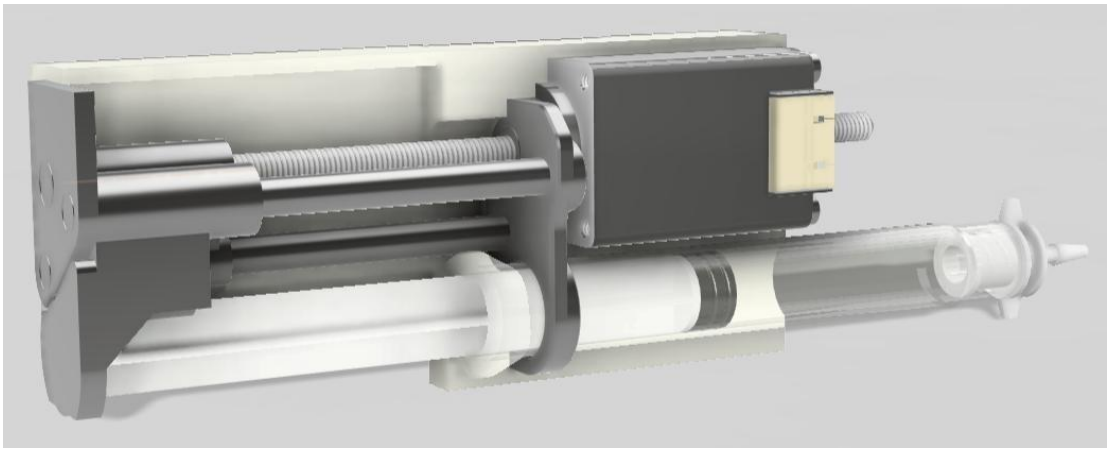


Figure 33: Microfluidic pump setup

Sensors

Both pressure and flow sensors are made with screw holes in a plate, so they are screwed into inserts into the main frame.

Containers

The syringe is inserted fully loaded and floods completely the system before fixation. The dump tank is a same size syringe that is preceded by a one-way valve so air and slushing cannot interfere with the cell observation system. Using another syringe for the dump tank allows it to be fully air and liquid tight because of the pusher that adapts to the quantity of liquid inside.

Adapters and components

Every adapter or component is contained into the closed structure. Tubes are kept free for the minimal length possible between sections of fixation.

Pressure vessel assembly

Tolerances

3D printed parts are designed for clearance fittings between each other. The exact dimensions on the CAD are adapted due to the 3D printer having a tendency to reduce internal hole dimensions and increase external dimensions.

Tolerances achieved on critical parts such as the objective thread supporter are measured to be accurate at 0.1mm. Non calibrated parts have an accuracy of around 0.3mm.

CNC parts have a 0.05-0.1mm tolerance.

Objective assembly

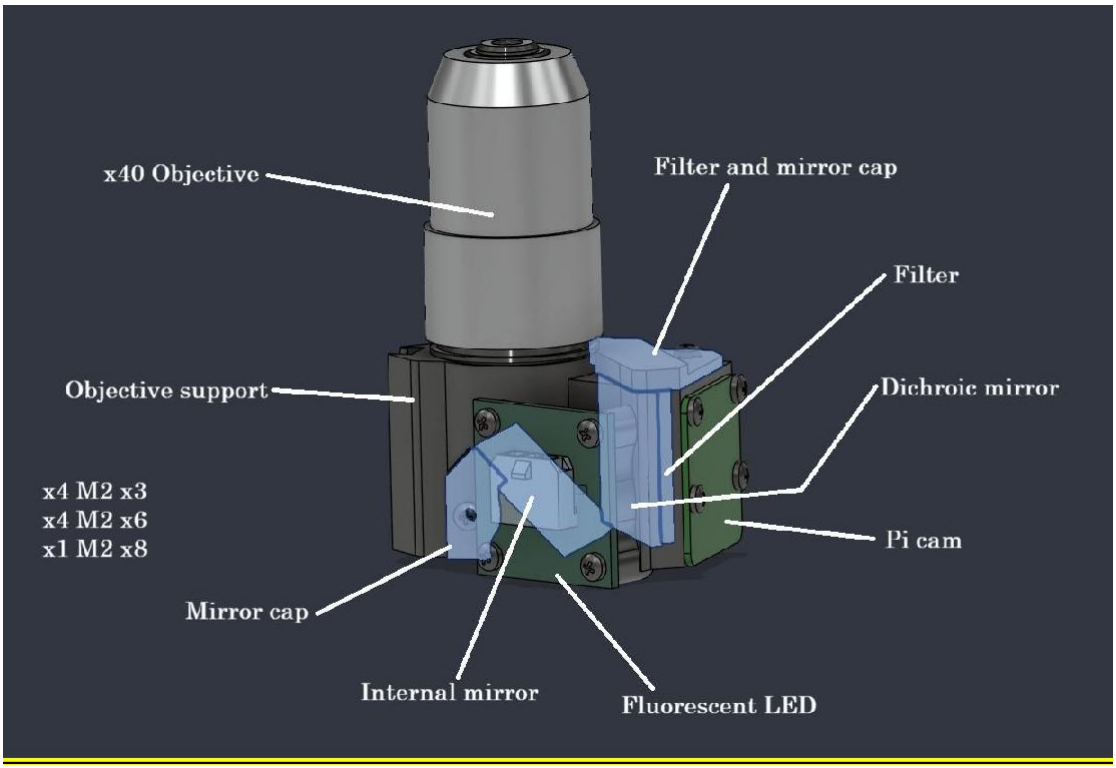
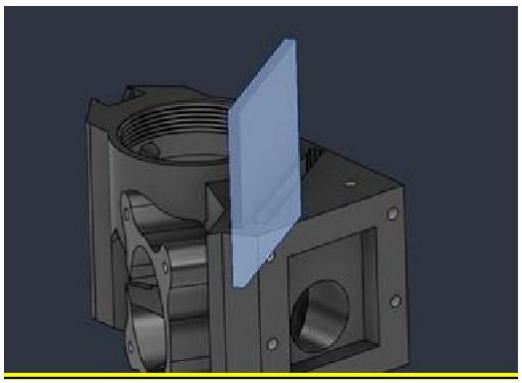
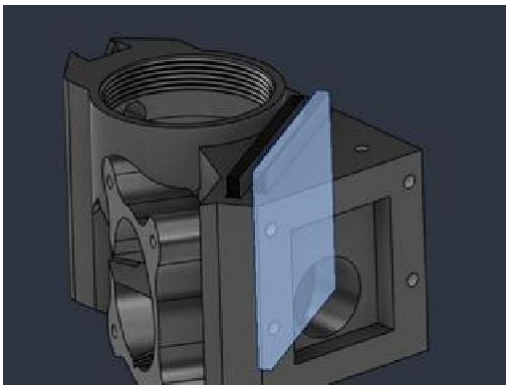


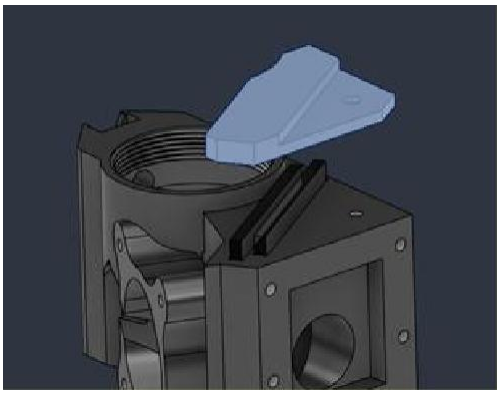
Figure 34: Objective assembly



Figures 34.1: Insert dichroic mirror into the internal filter placement hole



Figures 34.2: Insert filter into the external placement hole



Figures 34.3: Add the filter cap

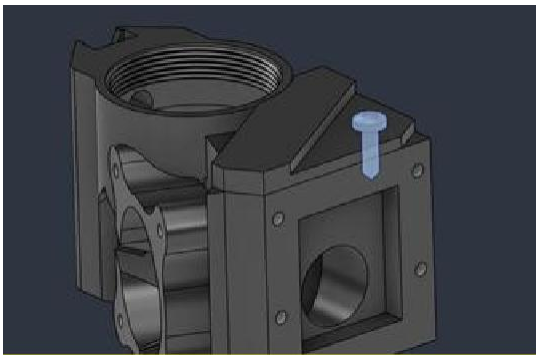


Figure 34.4: using a M2x6 screw tighten filter cap

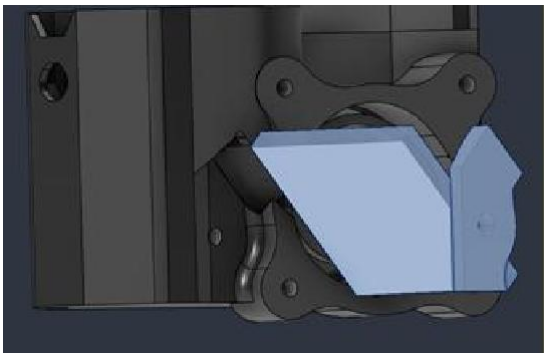


Figure 34.5: In the side of the assembly insert the mirror and align its cap

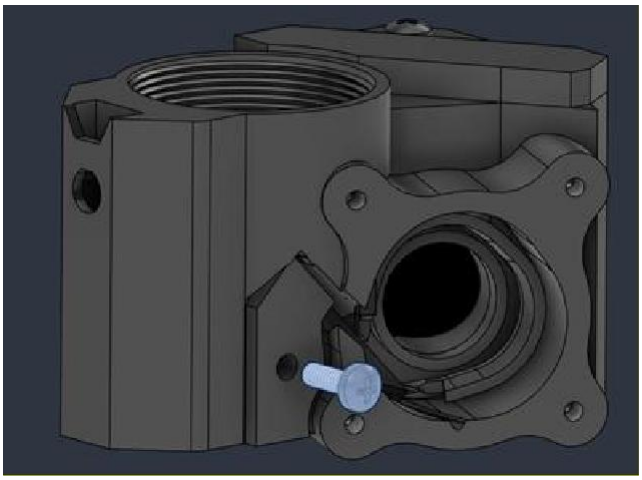
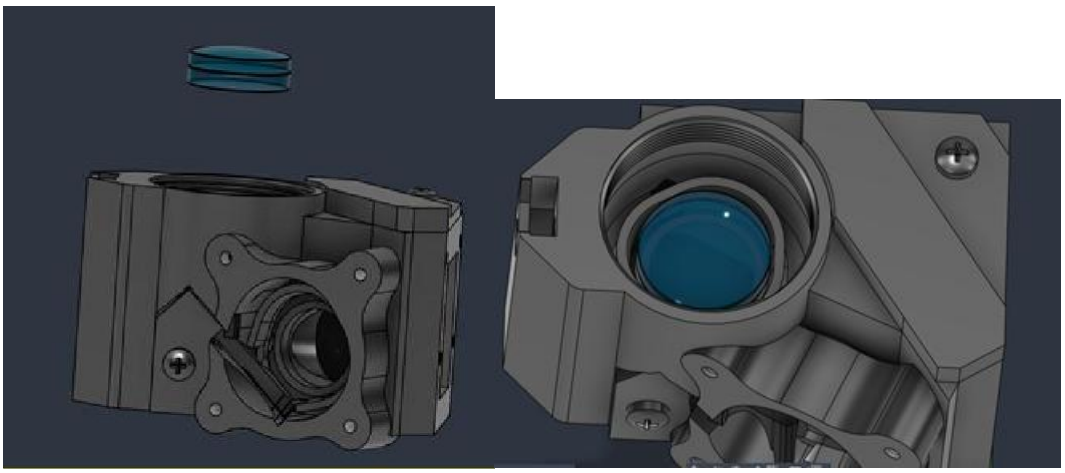
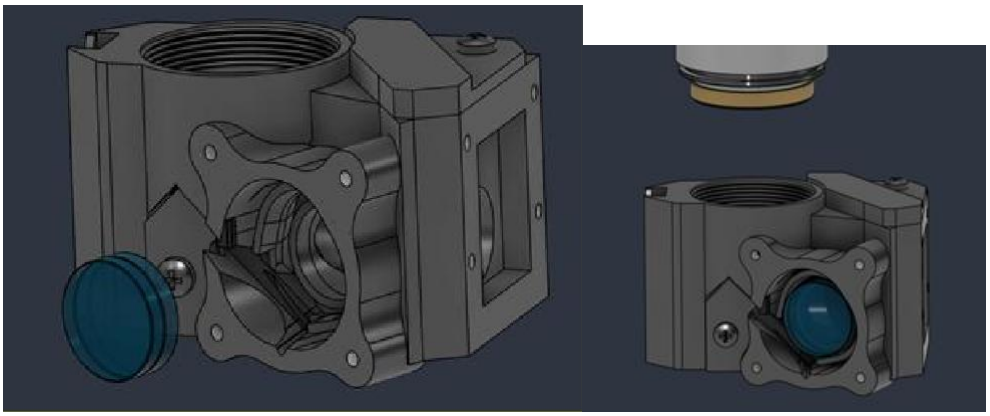


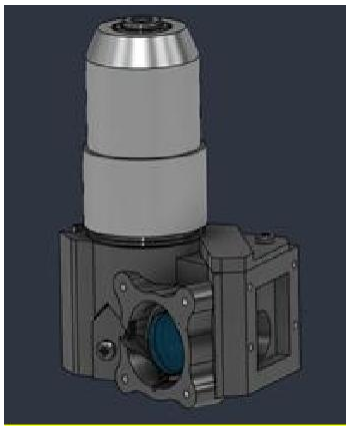
Figure 34.6: Using a M2x8 tighten the mirror cap



Figures 34.7, 34.8: push the 12.5mm lens until its blocked



Figures 34.9, 34.10: push the second 12.5mm lens (identical) until its blocked



Figures 34.11: screw the 40x objective onto the thread

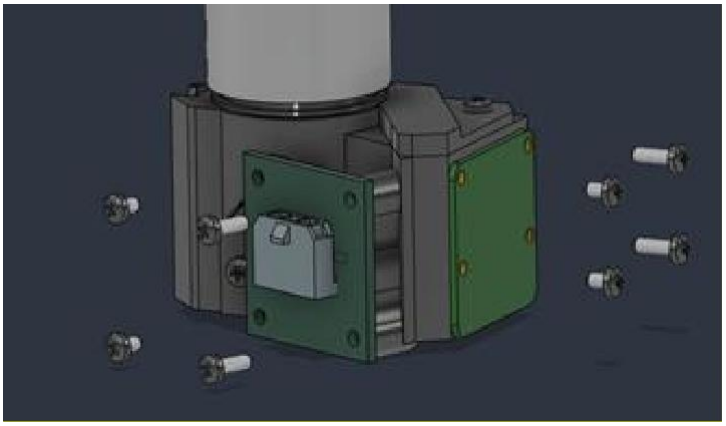
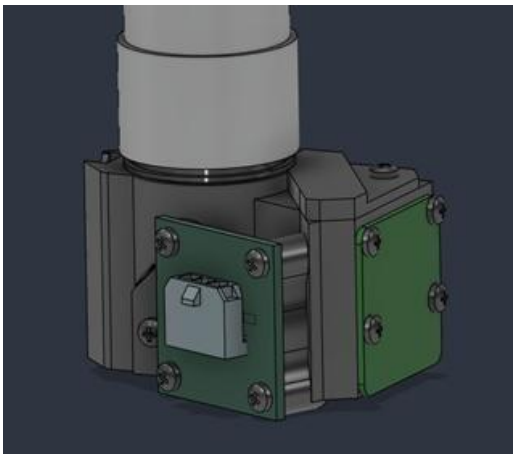


Figure 34.12: align the LED pcb over the lens and Pi cam and prepare 4 M2x3 and 4M2x6



Figures 34.13: tighten the screws

Electronics integration

The compute module 5 and the ESP32 control card are both integrated vertically using their built-in screw holes.

Connectors are maintained during boost phase through an attachment at the beginning of the cables that is screwed into place onto the structure that holds the card. Which itself is screwed into the main structure.

Cables between components and ESP32 PCB do follow branches on the structure and are zip-tied to it.

4.5 Electronics Design

The electronics design of STAR-D is based on a modular architecture separating power conversion, control, processing, sensing, actuation, and illumination functions. The overall electrical system is documented through a set of schematics providing both a system-level overview and detailed representations of custom electronic subsystems.

A global electrical schematic describes the complete experiment architecture, including the REXUS electrical interface, internal power conversion, control and processing units, sensors, actuators, and illumination system. This schematic serves as the reference for all internal electrical connections and signal flows.

Power conversion from the REXUS 28 V supply is handled by a dedicated voltage unit converter, which generates the internal voltage rails (3.3 V, 5 V and 7 V) required by the experiment electronics. The corresponding schematic documents show the distribution of these rails to the various subsystems. This

board includes three almost identical buck converters, as well as several protections against noise and emissions, and some debugging features (LEDs and test points).

Custom electronics developed specifically for the experiment include a dedicated LED PCB used for microscope illumination. The schematic of this board details the LED configuration, current control elements, and electrical interface with the rest of the system.

Internal communication between subsystems is implemented using standard digital interfaces. The ESP32 control unit communicates with sensors and actuator drivers through low-level digital buses, while the Raspberry Pi processing unit interfaces with the camera and local storage using dedicated high-speed interfaces. These interfaces are defined consistently across the schematics to ensure correct integration and operation.

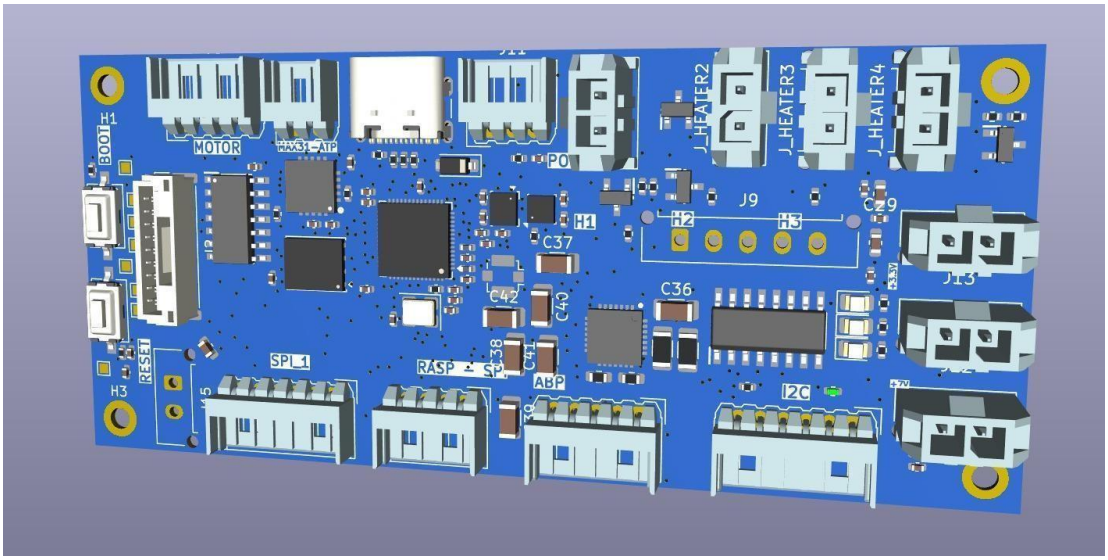


Figure 35: Updated STAR-D Main Board layout integrating the TMC2209 stepper motor driver and locking connectors

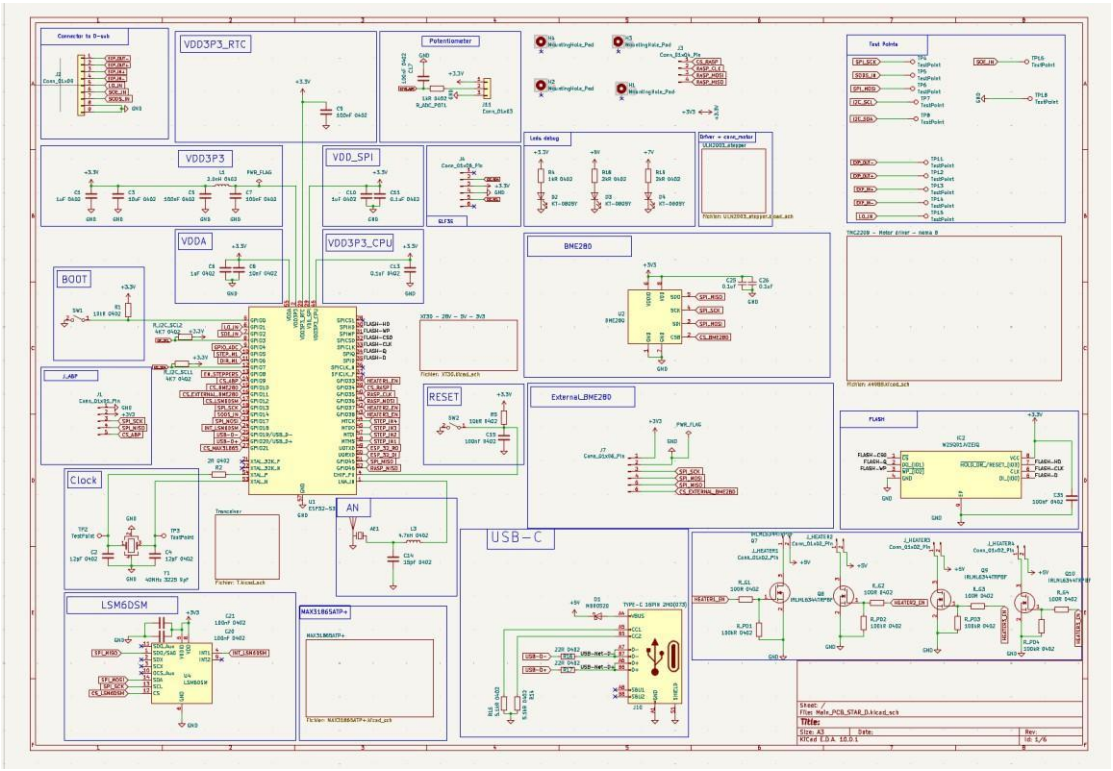


Figure 36: Simplified Schematic of ESP32 PCB

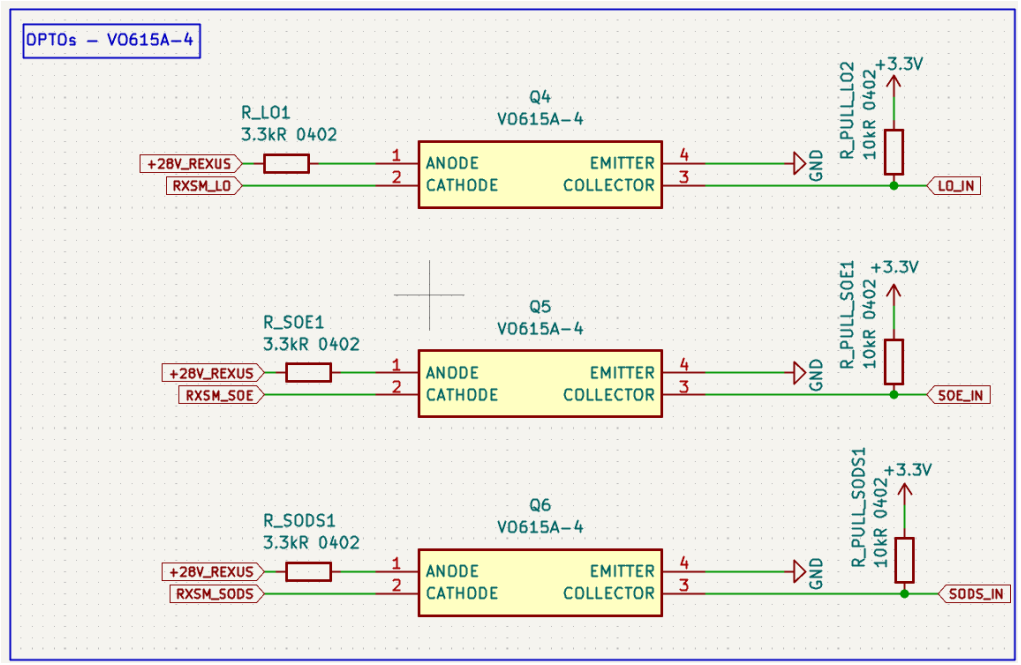


Figure 37: Optocoupler interface between REXUS service signals and onboard control electronics

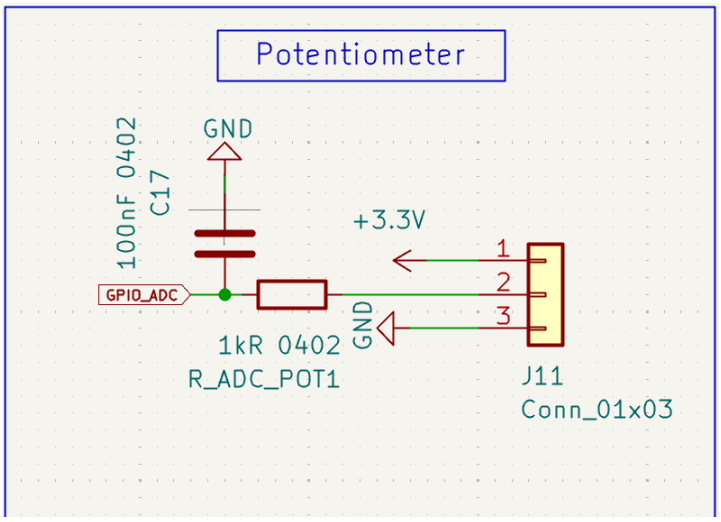


Figure 38: Potentiometer interface with analog input (ADC)

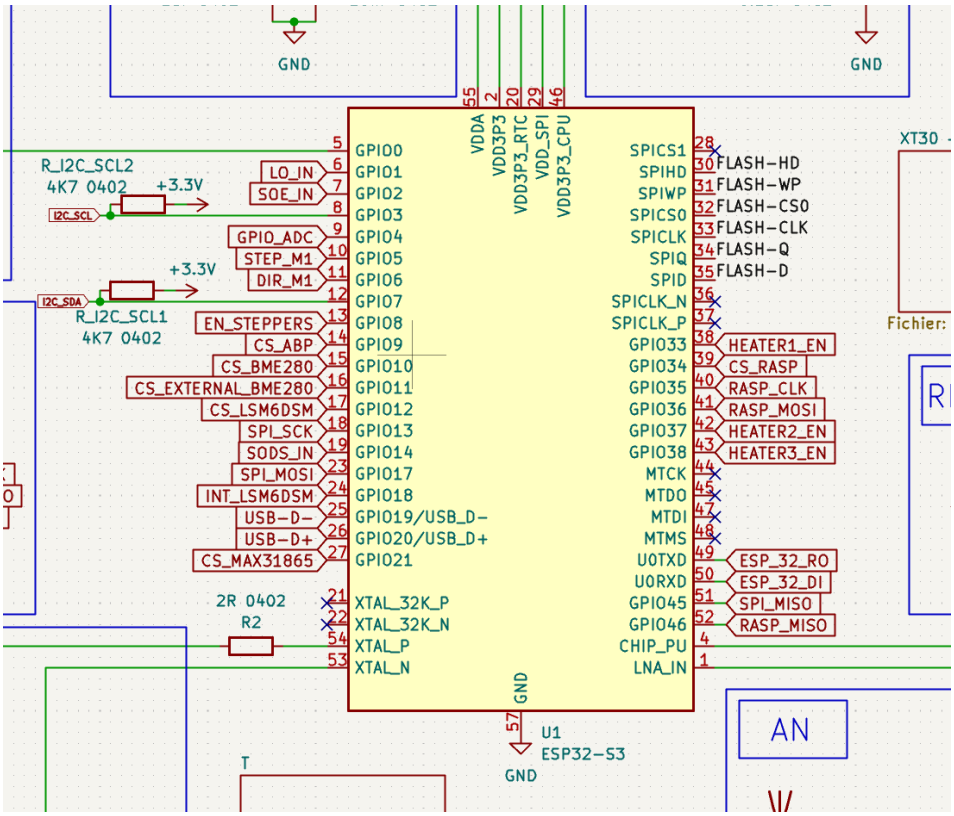


Figure 39: ESP32-S3 microcontroller integration and peripheral interfaces

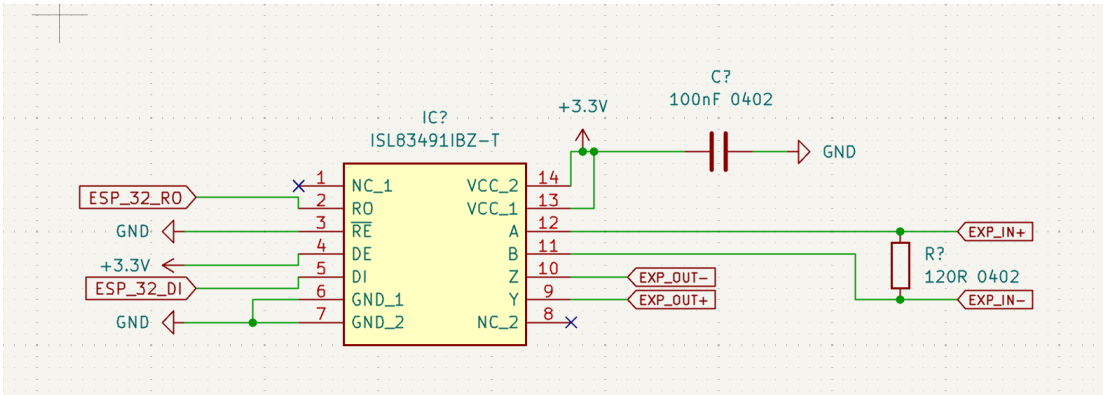


Figure 40: RS-422 transceiver interface for experiment telemetry communication

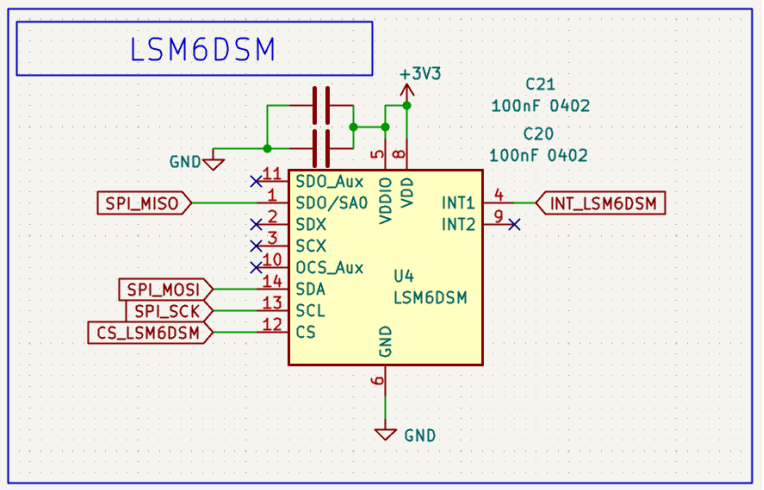


Figure 41: LSM6DSM IMU integration and SPI interface

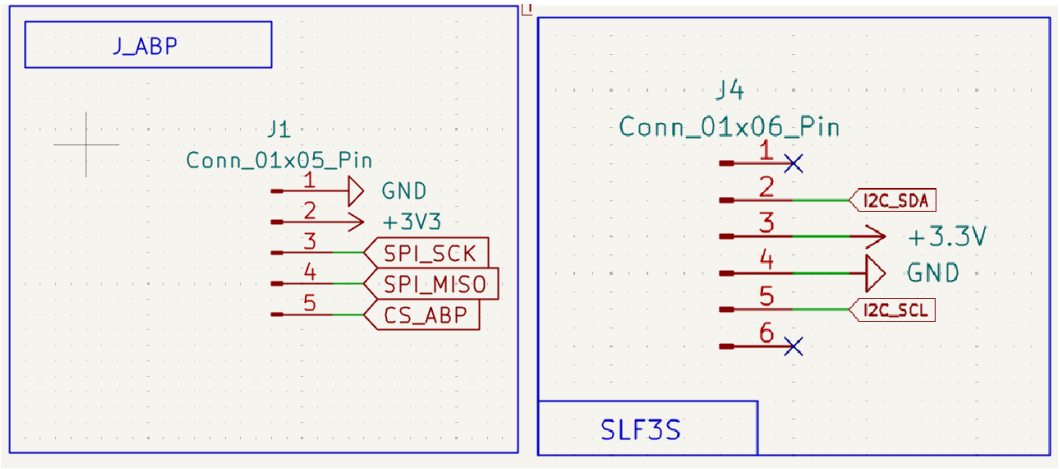


Figure 42: Sensor interface connectors for ABP pressure sensor (SPI) and SLF3S flow sensor (I²C)

Table 4.5: Electronic components description

Component	Function	Communication Protocol	Role in System
Raspberry Pi	Onboard processing & imaging	SPI	Handles image acquisition, storage, and high-level processing
ESP32	Main control unit	SPI / I²C / UART	Central controller: manages sensors, actuators, telemetry and sequencing
LSM6DSM	IMU sensor	SPI	Provides acceleration and angular velocity measurements
BME280	Environmental sensor	I²C / SPI	Measures temperature, pressure and humidity
ABP pressure sensor	Pressur e sensing	SPI	Measures pressure in the microfluidic system
SLF3S flow sensor	Flow sensing	I²C	Measures fluid flow rate
MAX31865 + PT100	Temperature sensing	SPI	Measures precise temperature near biological sample

RS-422 transceiver	Telemetry interface	UART (via ESP32)	Converts UART to differential RS-422 for communication with REXUS
***TMC2209 stepper motor driver**	**Stepper motor driver**	**GPIO (STEP/DIR/EN)**	**Controls syringe pump motor**
ULN2003ADR	Stepper motor driver	GPIO-based 4-phase stepper motor	Controls precise movement for microscopy system

*The TMC2209 provides improved current regulation and smoother microstepping operation, resulting in reduced motor noise and vibration. It remains compatible with the STEP/DIR control principle used by the ESP32, with the addition of an enable signal. The updated driver is therefore controlled through the STEP, DIR and EN signals.

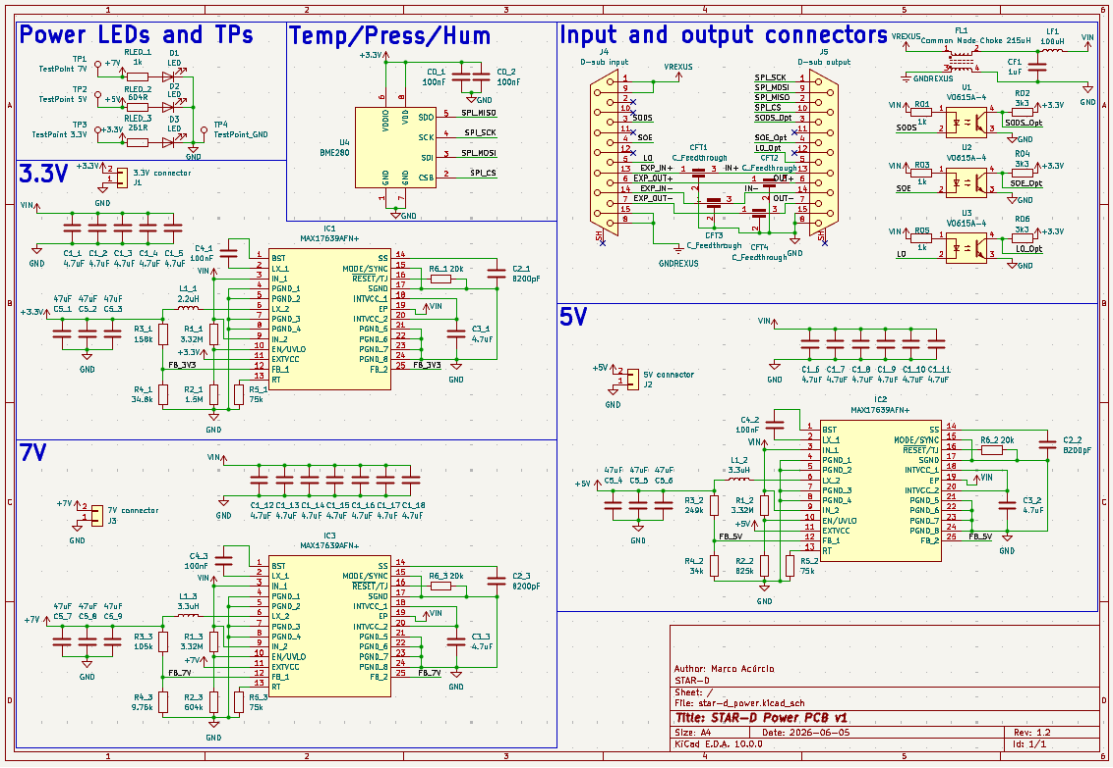


Figure 43: Power PCB schematic

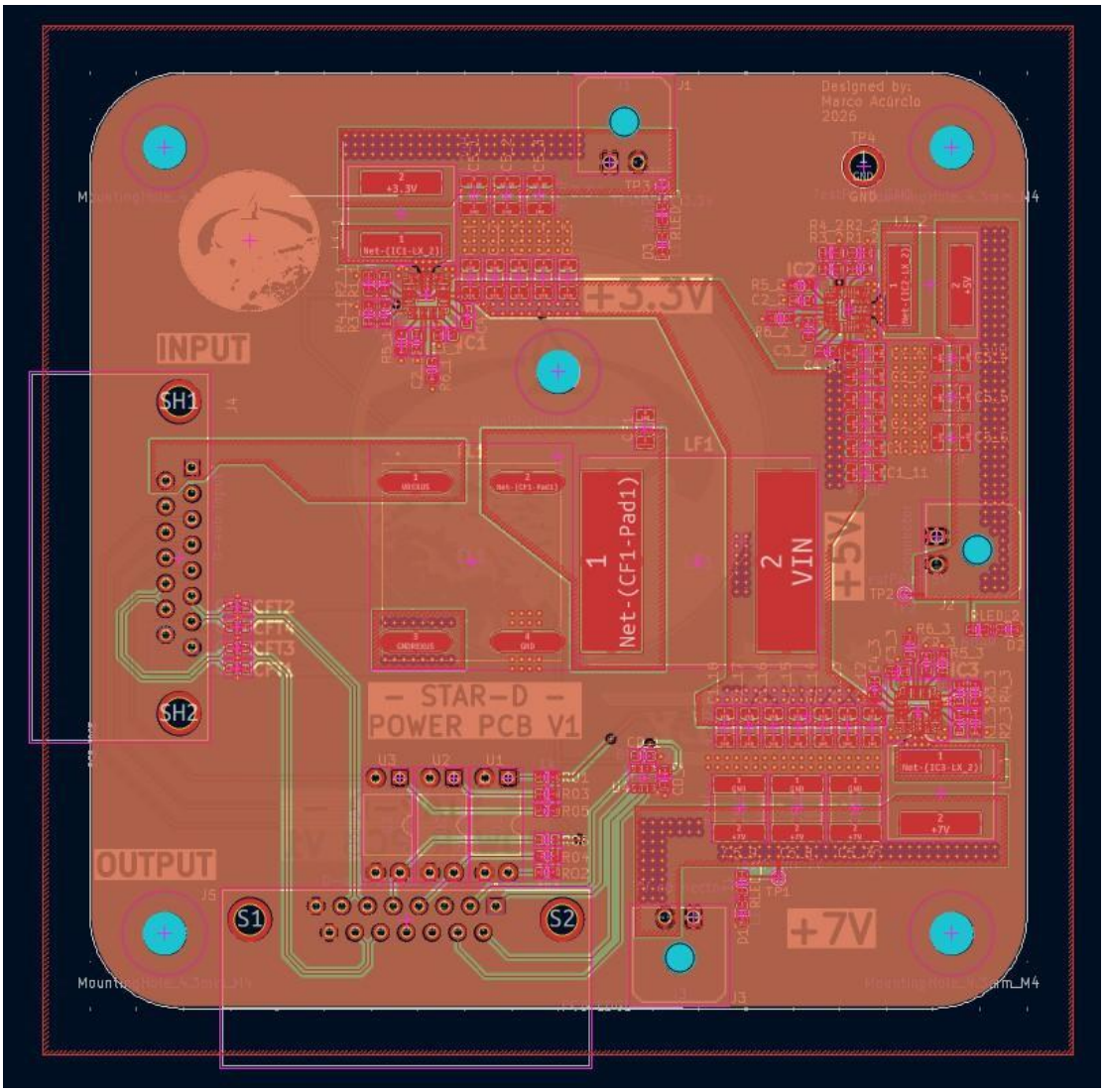


Figure 44: Power PCB Layout

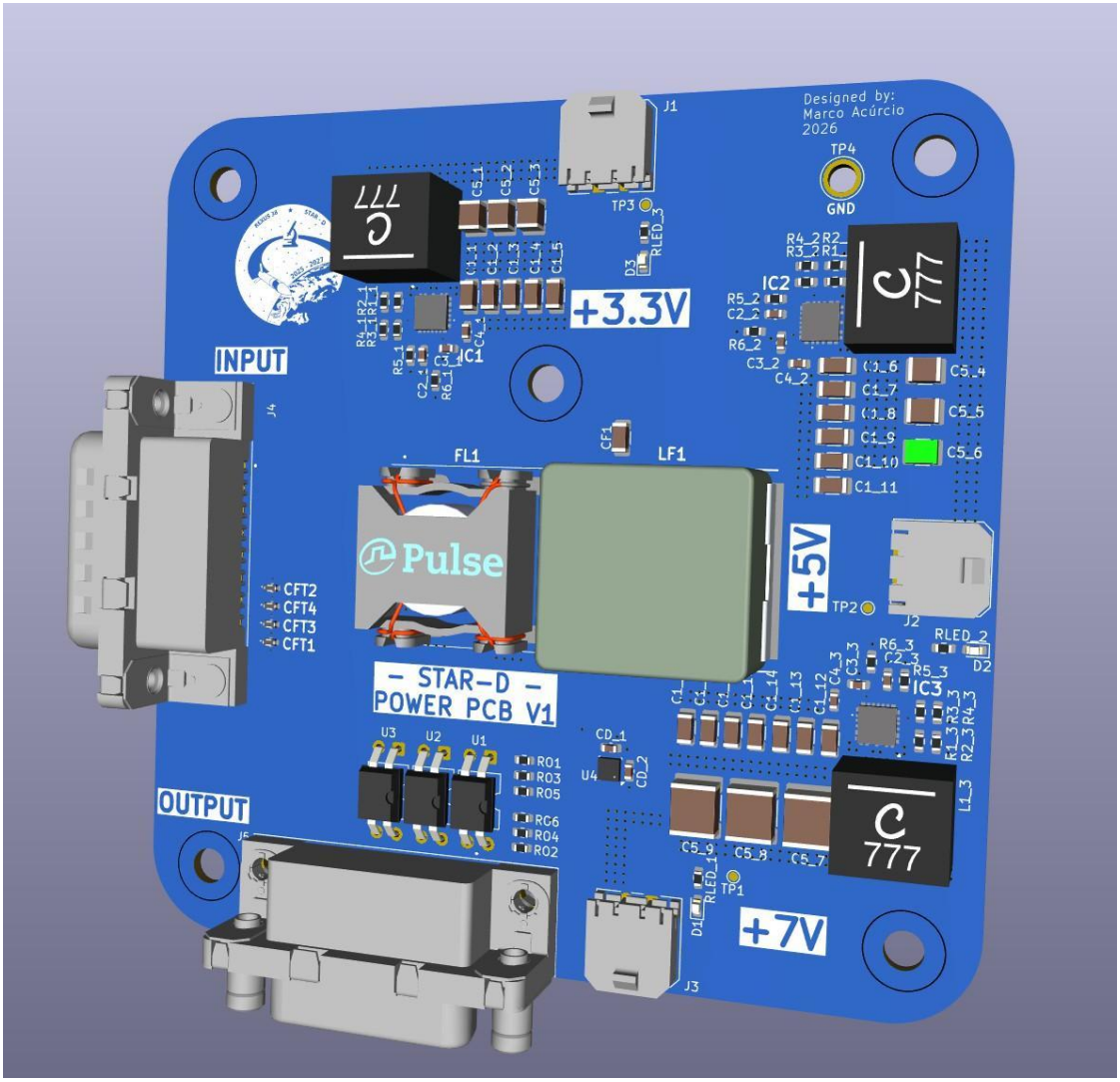


Figure 45: Power PCB top view

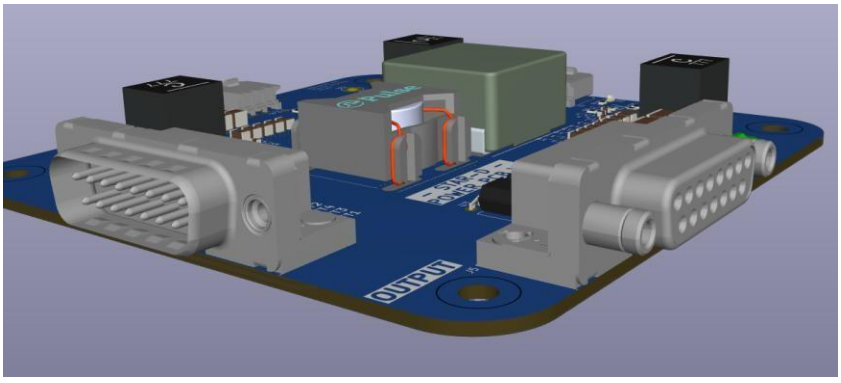


Figure 46: D-Sub interfaces located on the Power Board

Molex Micro-Fit 3.0 connectors are now used for power and higher-current connections, while Molex Micro-Latch connectors are used for signal and lower-power interfaces. These connector families provide dedicated locking or latching mechanisms, reducing the risk of unintended disconnection under vibration. The PCB schematics, footprints and layouts were updated accordingly.

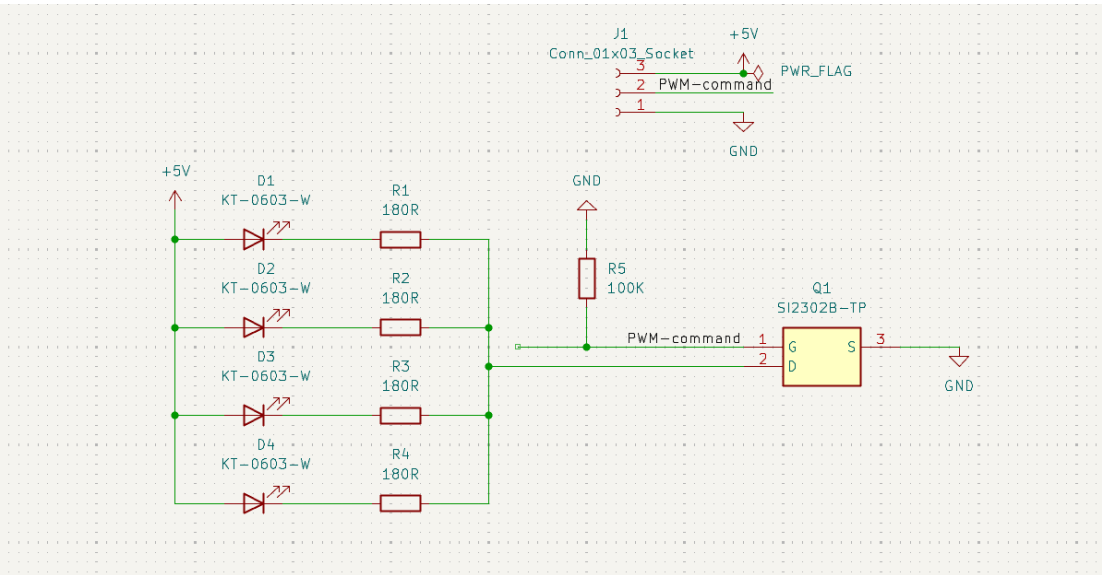


Figure 47: PCB LED Matrix

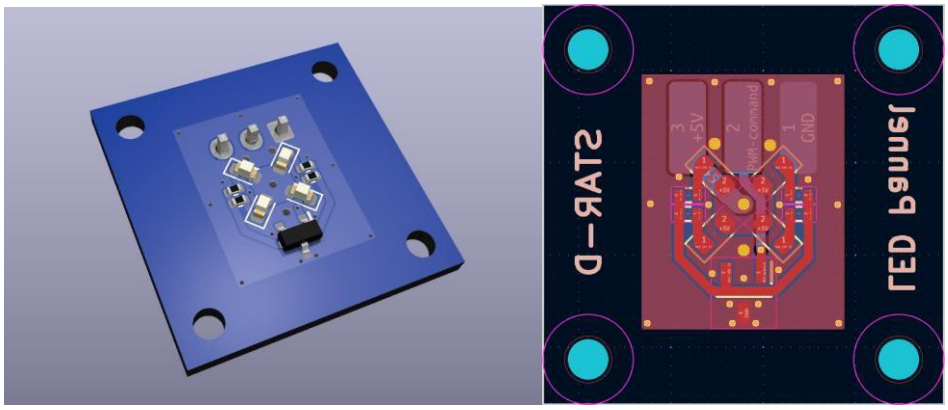


Figure 48: Top and Perspective View of the PCB

5 V Board:

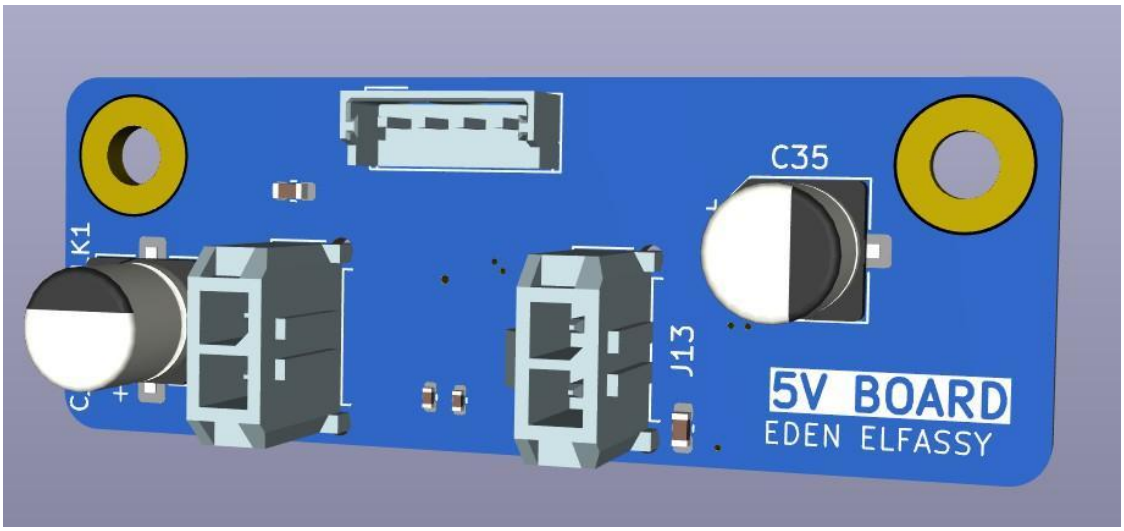


Figure 49: STAR-D 5 V conversion board providing the regulated 5 V rail to the experiment subsystems.

The 5 V board receives the regulated 5 V supply and distributes it to the experiment subsystems that require this voltage, including the Raspberry Pi Compute Module 5 and the microscope LEDs. It provides dedicated connection points and simplifies the routing of the 5 V power lines through locking Molex Micro-Fit 3.0 connectors.

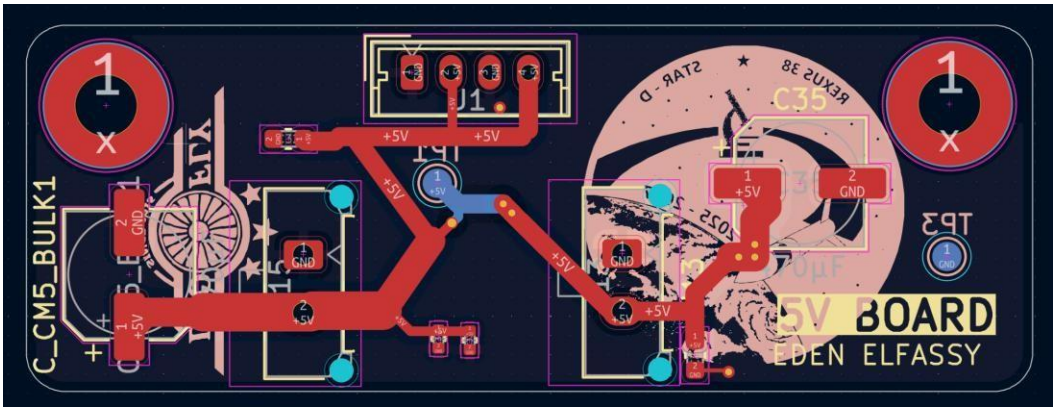


Figure 50: 5 V Board PCB layout

Cabling:

Internal electrical connections are implemented using insulated, flexible wiring suitable for laboratory and aerospace applications. Cable types are selected to ensure reliable signal transmission and mechanical robustness under launch and flight conditions, without the use of specialized or hazardous materials.

TV Channel Interface:

The experiment does not make use of a TV channel. This decision was made for simplification purposes.

The video signal is generated by the onboard Raspberry Pi imaging system and converted into a standard analog composite PAL signal compatible with the REXUS interface.
The resulting signal is transmitted through a coaxial connection to the BNC input of the REXUS service module.

4.6 Thermal Design

Three transient thermal analyses were completed in SolidWorks. All cases began at a uniform temperature of 22°C. The transient thermal response was calculated using the assigned thermal conductivity, density, and specific heat of each material. Bonded thermal contacts were used at physical interfaces, and internal air was not explicitly modeled.

The confidence ratings below are relative and do not represent validation against physical testing.

- Ground, unpowered

The unpowered model represented the payload exposed to a 7°C ground environment for two hours. Convection with a coefficient of 10 $Wm^{-2}K^{-1}$ and surface-to-ambient radiation were applied to the exposed exterior surfaces. No internal heat generation was included.

Confidence: Moderate

This model produced a reasonable cooldown response and has relatively few uncertain inputs. The main uncertainties are the assumed convection coefficient, surface emissivity, and idealized thermal contacts.

Because internal air circulation was not modeled, the microfluidic chip would likely cool slightly faster in the actual hardware than the model predicts. Natural convection inside the payload would provide an additional heat-transfer path from the chip to the colder enclosure.

At the end of the 2h hold the coldest monitored node reaches 21.5°C, corresponding to a cooldown rate of approximately 0.2°C/h. The assembly remains above both the 18°C biological working floor and the 10 °C survival floor throughout the hold, so no active heating is required in this case. Because internal air circulation is not modelled, the microfluidic chip would cool marginally faster in the actual hardware; the predicted result is therefore treated as conservative for this case.

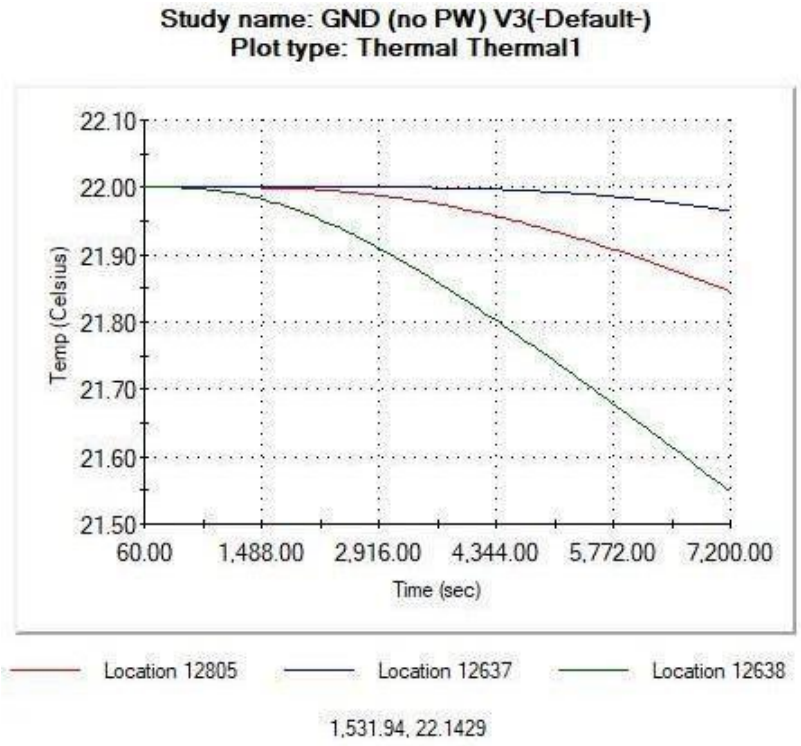
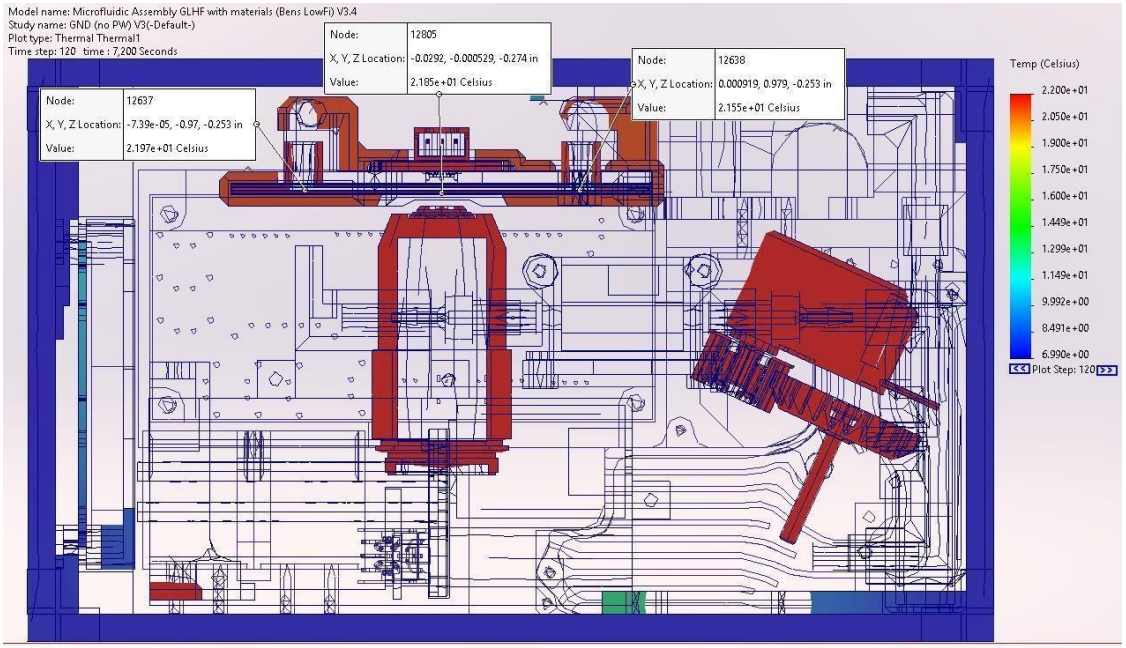
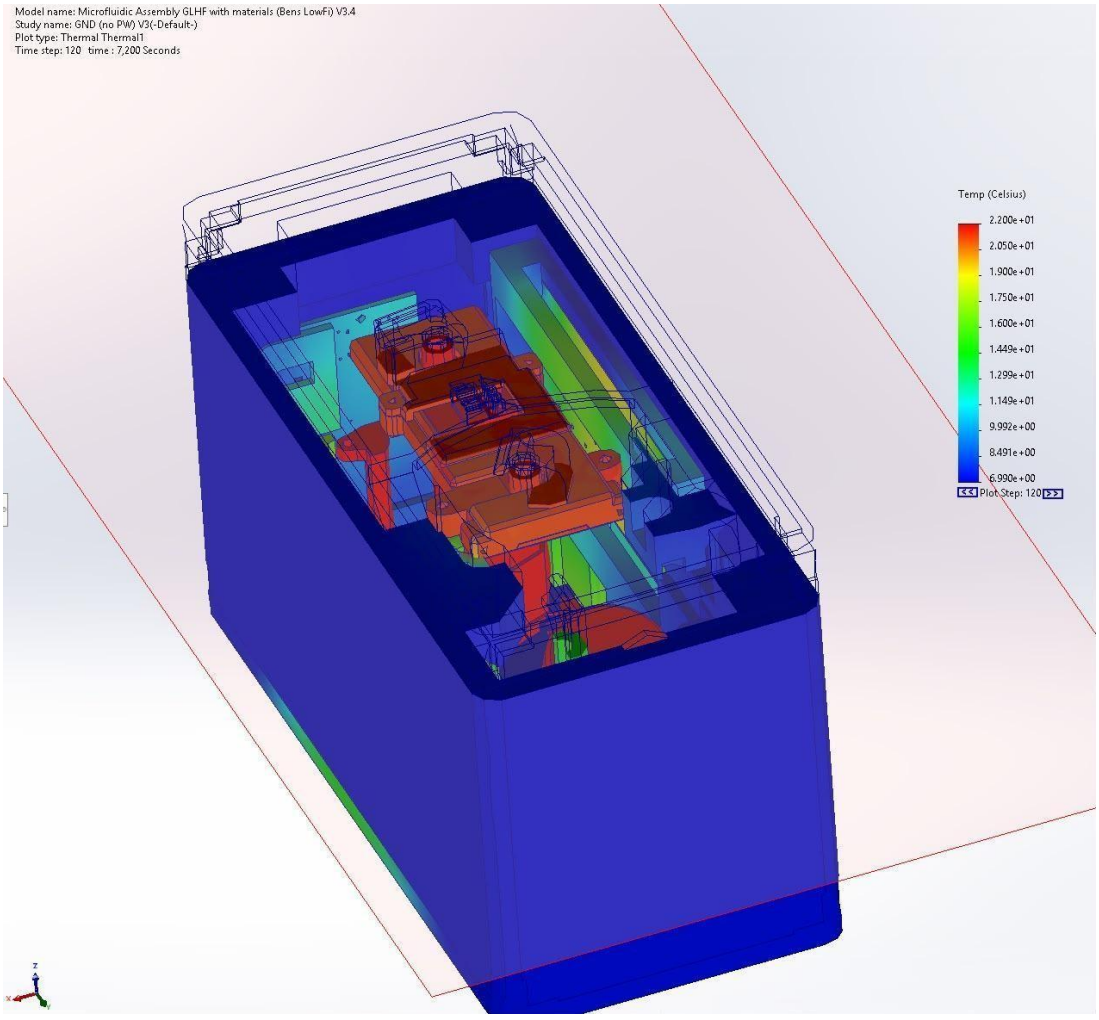
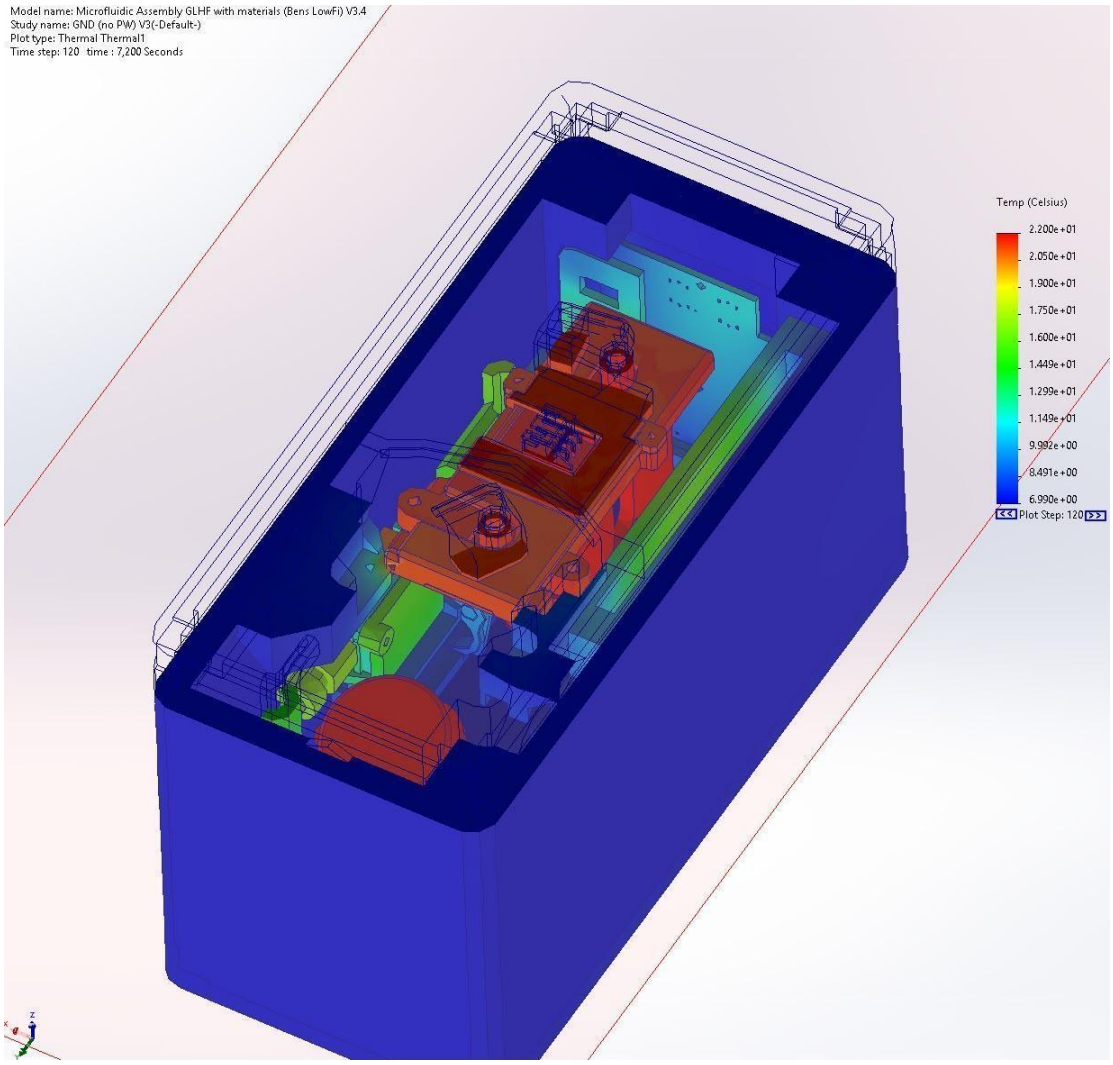


Figure 51: Thermal dissipation on ground with no power


Model name: Microfluidic Assembly GLHF with materials (Bens LowFi) V3.4
Study name: GND (no PW) V3(-Default-)
Plot type: Thermal Thermal1
Time step: 120 time : 7,200 Seconds
Node: 12805
X, Y, Z Location: -0.0292, -0.000529, -0.274 in
Value: 2.185e+01 Celsius
Node: 12638
X, Y, Z Location: 0.000919, 0.979, -0.253 in
Value: 2.155e+01 Celsius
Node: 12637
X, Y, Z Location: -7.39e-05, -0.97, -0.253 in
Value: 2.197e+01 Celsius
Temp (Celsius)
2.200e+01
2.050e+01
1.900e+01
1.750e+01
1.600e+01
1.449e+01
1.299e+01
1.149e+01
9.992e+00
8.491e+00
6.990e+00
Plot Step: 120


Model name: Microfluidic Assembly GLHF with materials (Bens LowFi) V3.4
Study name: GND (no PW) V3(-Default-)
Plot type: Thermal Thermal1
Time step: 120 time : 7,200 Seconds
Temp (Celsius)
2.200e+01
2.050e+01
1.900e+01
1.750e+01
1.600e+01
1.449e+01
1.299e+01
1.149e+01
9.992e+00
8.491e+00
6.990e+00
Plot Step: 120



Figures 52: Thermal analysis in PV when on ground with no power

- Ground, powered

The powered model represented the payload exposed to a −30°C ground environment for two hours. Exterior convection used the same coefficient as the unpowered model of 10 $Wm^{-2}K^{-1}$, with radiation applied to the applicable exposed surfaces.

The internal power loads were:

- Compute Module: 7.5 W

- 3 FIT0845 Heaters: 2 W each

Surface-to-surface radiation was included between the exposed compute module and heater surfaces and the visible inner enclosure surfaces. Bonded contacts represented conduction through the physical mounting interfaces.

Confidence: Low to Moderate

This model is highly sensitive to the effective thermal mass of the compute module and heaters, heat-power distribution, heater duty cycle, thermal contacts, emissivity, and the omission of internal natural convection. The unusually high compute module and heater temperatures have not been fully validated, so the absolute temperatures should be treated as preliminary.

In actual ground conditions, internal air circulation would increase heat transfer from the microfluidic chip and powered components to the cold enclosure. The chip would likely cool faster than the model predicts, while localized compute module and heater temperatures would likely be lower.

Active heating is provided by three FIT0845 resistive heaters (2 W each, 6 W total) mounted [inside the pressure vessel near the microfluidic assembly], supplied from the 5 V rail and switched by the ESP32. The loop holds 22°C using the chip-adjacent sensor, with the heaters below 21°C and off above 23 °C. Specific temperature reaction times when using the PID heater have not yet been specifically established.

Bonded contacts model conduction into the aluminum 6061 shell, which acts as the main heat-sink path for the compute module and heaters. The high compute-module and heater temperatures are localized and preliminary per the confidence rating, but the chip stays within the 18–25 °C band throughout. Internal air and any fan are not credited; a fan would only homogenize the air, not reject heat.

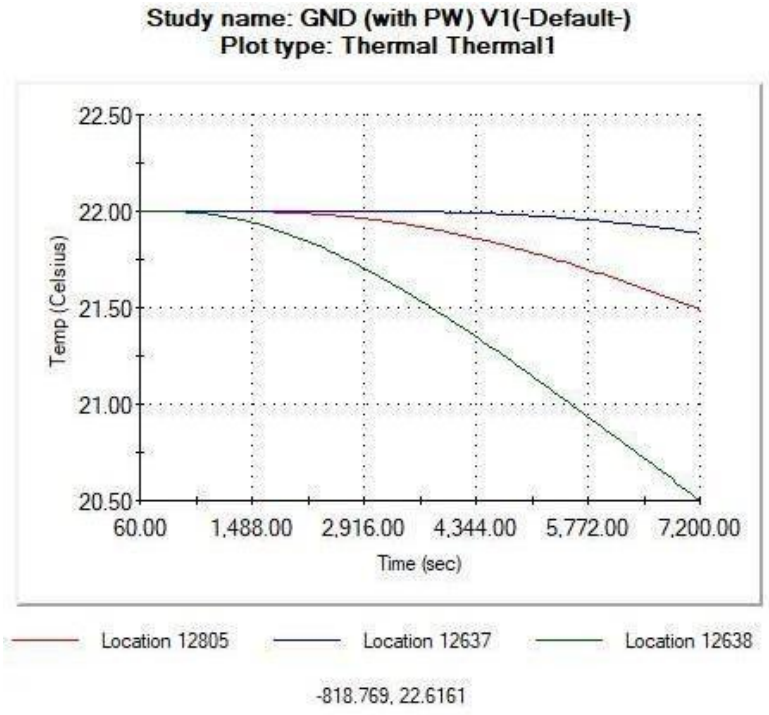
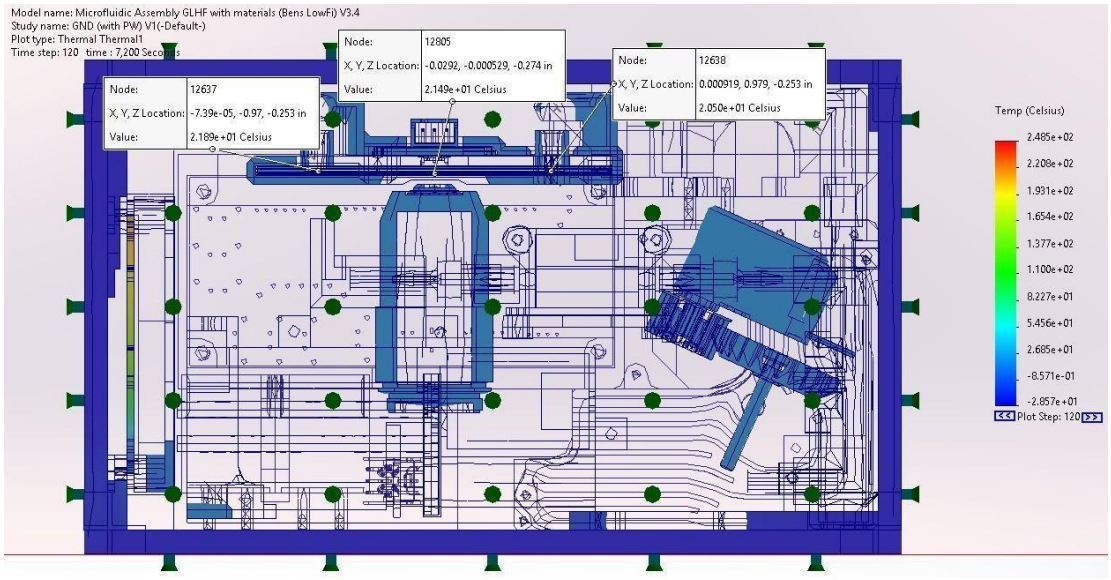
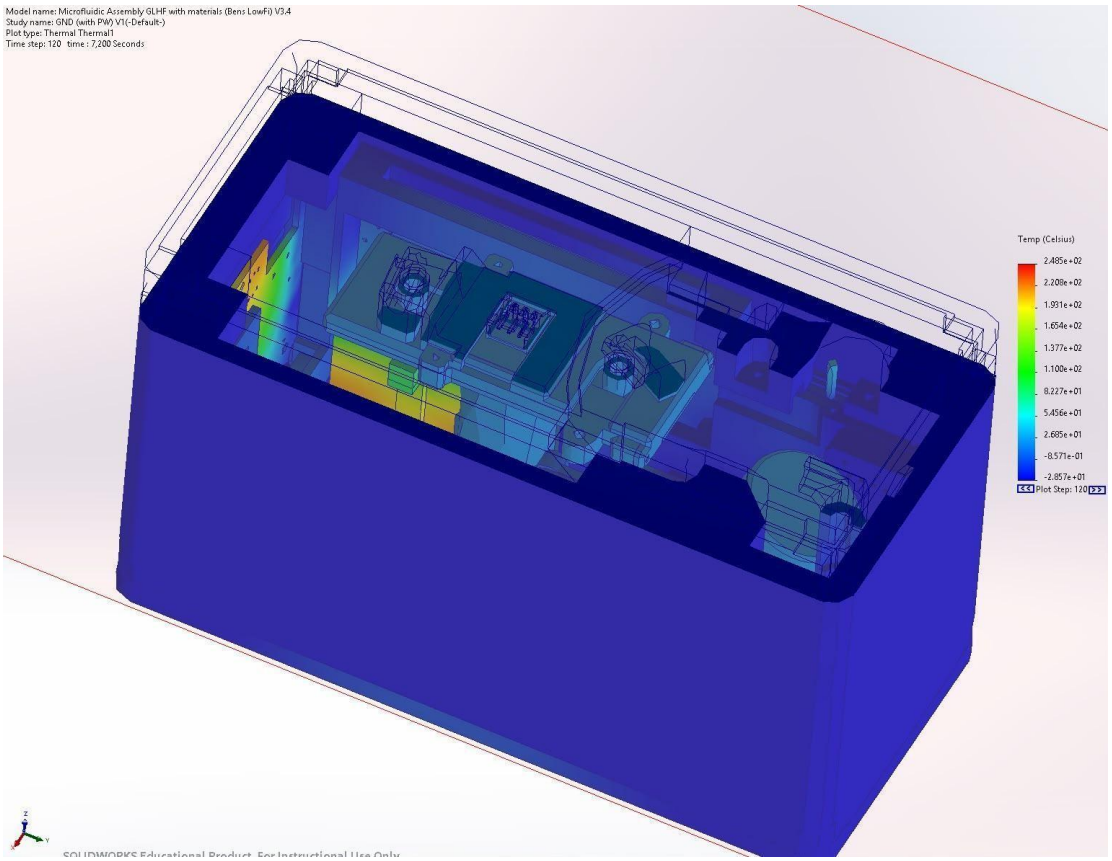


Figure 53: Thermal dissipation on ground with power





Figures 54: Thermal analysis on ground with power

- In-Flight Pressure Vessel Heating

The flight model represented heat transfer from the launch vehicle structure into the pressure-vessel module over 380 seconds. The entire modeled assembly began at 22°C. The launch vehicle's exterior skin temperature increased linearly from 22°C to 110°C during the first 50 seconds and remained at 110°C for the remaining 330 seconds.

Exterior convection was not included because the launch vehicle skin temperature was applied directly in the model. Surface-to-surface radiation was included between the inner launch vehicle skin and the exposed pressure-vessel surfaces. Bonded thermal contacts represented conduction through the physical mounting interfaces.

Confidence: Moderate to High

This is the strongest of the three analyses because the launch vehicle skin temperature was directly specified, the duration was relatively short, and the predicted response followed the expected heat-transfer path. The main

uncertainties are the assumed skin-temperature profile, surface emissivity, and idealized thermal contacts.

Overall, flight analysis has the highest confidence, followed by the unpowered ground analysis. The powered ground analysis has the most uncertainty, so its results should be treated as preliminary.

Over the 380s exposure the internal assembly rises from 22°C to approximately 24.4°C, a rate of about 0.4°C/min, remaining below the 25°C biological working ceiling with roughly 0.6°C of margin. This confirms that the short-duration skin-heating transient does not drive the interior out of band. As above, internal air is not modelled, so the predicted interior rise is treated as conservative.

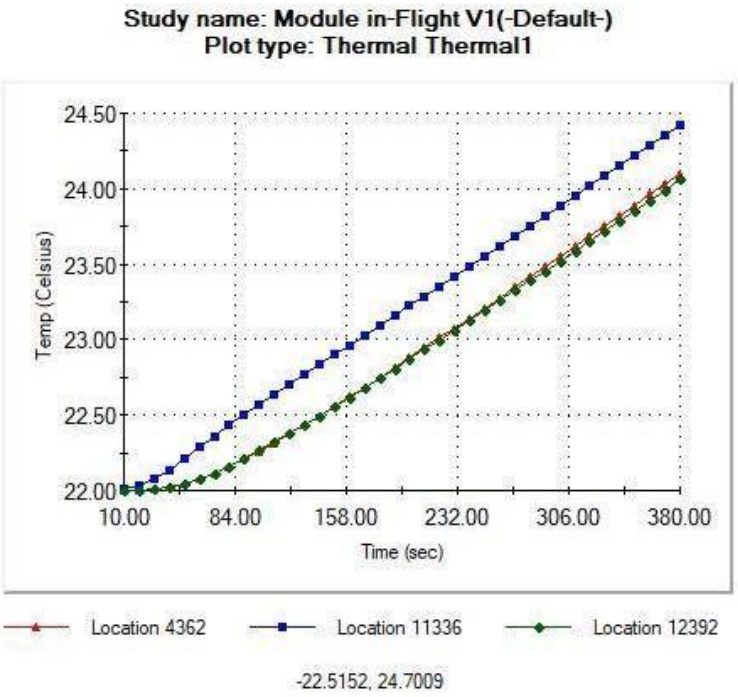
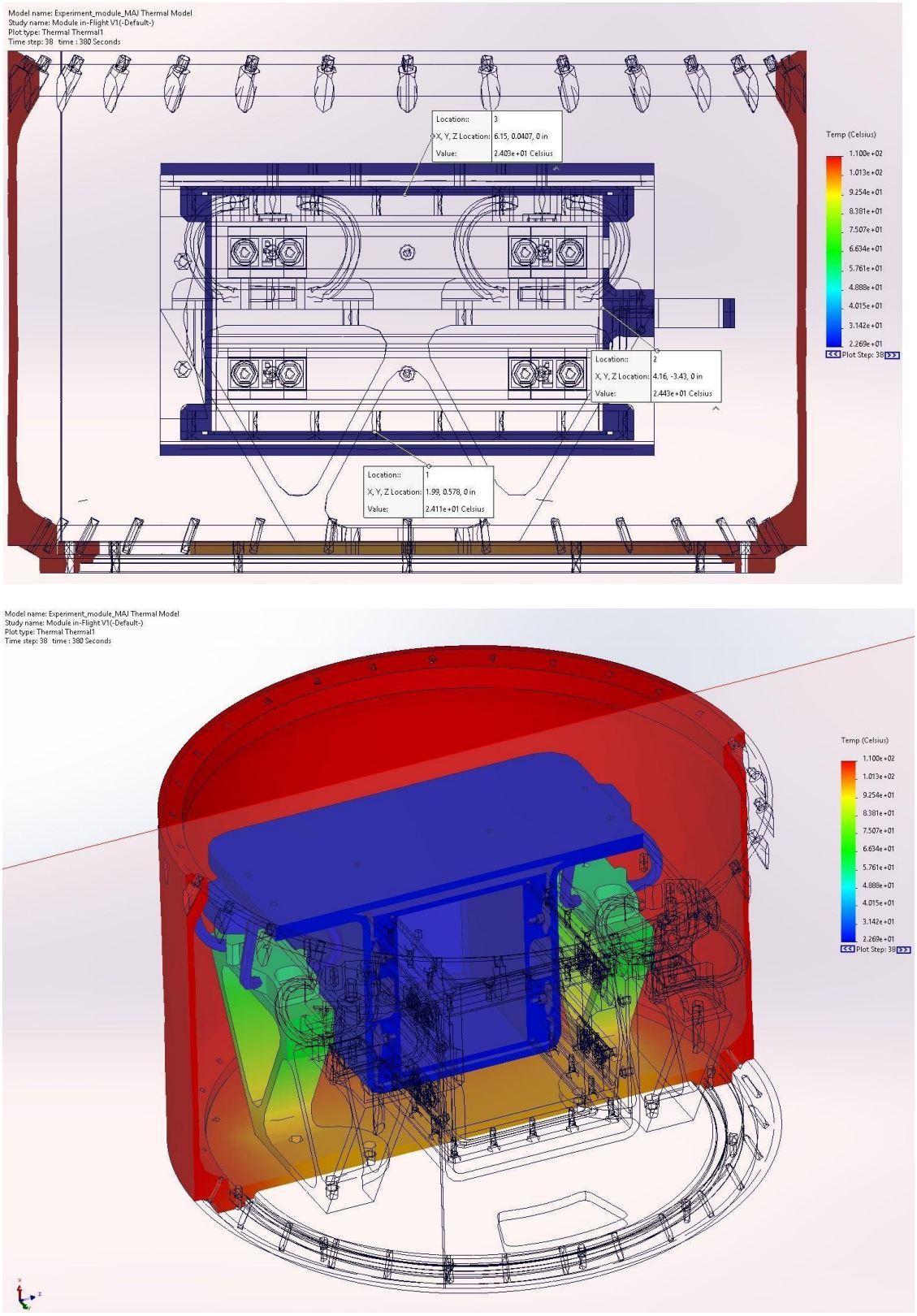


Figure 55: In-flight thermal analysis



Figures 56: Thermal analysis in the EM during flight

4.7 Power System

The experiment is powered through the standard REXUS electrical interface via the 15-pin D-Sub connector. The REXUS service module provides an unregulated DC supply through the experiment power pins, which constitutes the only electrical power input to the STAR-D experiment.

No onboard battery is implemented in the experiment. Consequently, the STAR-D electrical system is entirely dependent on the external power supplied by the REXUS service module during ground operations and flight. The charging pins available on the REXUS connector are not used and are left unconnected.

The incoming supply is routed to a dedicated Power Unit Converter board, which distributes and converts the input voltage into the regulated voltage rails required by the experiment subsystems. These rails include the low-voltage logic supplies and the higher-current supplies required for the Raspberry Pi, sensors, actuators, heating system, and microscope illumination.

The power budget presented in the following tables therefore corresponds to the expected consumption of the experiment from the REXUS supply. It includes the estimated consumption of each electrical subsystem, the power losses dissipated by the DC-DC converters, and the current drawn from the 28 V input in nominal and maximum operating scenarios.

Since no battery is present onboard, no battery capacity, charge requirement, battery isolation verification, or radio-silence procedure related to battery disconnection is applicable.

Table 4.6 - Power consumption of the experiment's electrical components

Component	Quantity	Voltage (V)	Nominal current / unit (A)	Maximum current / unit (A)	Nominal power / unit (W)	Maximum power / unit (W)
ESP32 S3	1	3.3	0.0200	0.1000	0.0660	0.3300
Compute module 5	1	5	0.9000	1.5000	4.5000	7.5000
TMC2209-LA-T Stepper Motor Driver + motor	1	7	0.5000	0.5000	3.5000	3.5000

ULN2003 Stepper Driver Board + motor	1	5	0.5000	0.5000	2.5000	2.5000
BME280	2	3.3	3.60E-06	3.60E-06	1.19E-05	1.19E-05
Kingston NV3 NVMe SSD M.2 2230 (128g)	1	3.3	1.0000	2.5000	3.3000	8.2500
LED PCB	1	5	0.1000	0.1000	0.5000	0.5000
Module Camera v2 8MP	1	3.3	0.2500	0.2500	0.8250	0.8250
Honeywell ABP Pressure Sensor	1	3.3	0.0031	0.0039	0.0102	0.0129
Sensirion SLF3S-0600F Liquid Flow Sensor	1	3.3	0.0045	0.0060	0.0149	0.0198
LSM6DSMTR	1	3.3	0.0005	0.0007	0.0015	0.0021
MAX31865ATP+	1	3.3	0.0020	0.0035	0.0066	0.0116
ISL83491	1	3.3	0.0008	0.0012	0.0025	0.0040
FIT0844 PTC heating element	3	5	0.4000	0.6000	2.0000	3.0000
W25Q01JVZEIQ	1	3.3	0.0000	0.1000	0.0000	0.3300

Table 4.7.1 - Regulator power distribution for the average scenario

Regulator output voltage (V)	Efficiency	Output current (A)	Output power (W)	Input power (W)	Dissipated power (W)
3.3	90%	1.28	4.23	4.70	0.47
5	90%	2.70	13.50	15.00	1.50
7	90%	0.50	3.50	3.89	0.39

Table 4.7.2 - Power budget for the average scenario

	Input power (W)	Input voltage (V)	Input current (A)	Energy (Wh)	Charge (Ah)
TOTAL	23.59	28	0.84	3.93	0.14
+10% margin	**25.94**	28	**0.93**	**4.33**	**0.16**

Table 4.8.1 - Regulator power distribution for the maximum scenario

Regulator output voltage (V)	Efficiency	Output current (A)	Output power (W)	Input power (W)	Dissipated power (W)
3.3	90%	2.97	9.79	10.87	1.09

5	90%	3.90	19.50	21.67	2.17
7	90%	0.50	3.50	3.89	0.39

Table 4.8.2 - Power budget for the maximum scenario

	Input power (W)	**Input voltage (V)**	**Input current (A)**	**Energy (Wh)**	**Charge (Ah)**
TOTAL	36.43	24	1.52	6.07	0.25
+10% margin	**40.07**	24	**1.67**	6.68	0.28

Electromagnetic Compatibility (EMC)

The experiment electronics are designed using low-voltage digital components and moderate switching frequencies. No significant electromagnetic emissions are expected, and no interference with the REXUS service module or other payloads is anticipated. An RC filter and a common-mode choke were added to provide extra protection against emissions from the buck converters.

4.8 Software Design

4.8.1 – Mission overview relevant to software operations

The software's objective is to enable semi-autonomous execution of the experiment for the whole duration of the mission. The software is responsible for data logging, temperature control, and telemetry transmission. Software-related mission objectives are as follows:

- The software autonomously and continuously monitors the temperature inside the experiment module, and enforces temperature corrections
- To collect, format and store camera data that records the microscope's view of the biology experiment
- To correctly format and transmit telemetry data with a fixed and limited packet size
- Acquire, manage, and preserve experimental data
- Provide experiment health monitoring through the telemetry downlink
- Detect and respond to off-nominal conditions
- Ensure modular and fault-tolerant architecture
- Coordinate experiment sequencing and operational mode

Table 4.9 – Mission Phases Overview

Phase	Explanation	Action(s)	Possible actions
Pre-launch, setup	Software upload to EM and startup	Startup and system test	Multiple test startups
Launch countdown	Countdown clock begins	SODS shortly before LO, receive telemetry, temperature control	Telecommands to start/stop SODS, wipe data, take manual control of heating
Launch	Rocket goes into flight	LO, higher frequency of metadata storage	
Ascent	Rocket ascends		
Motor burnout	Coast, microgravity	SOE on	

Descent	End of microgravity	SOE off, SODS off shortly before EM power cutoff	Lower frequency of metadata storage
Recovery	Recovery of data from storage card	Upload storage data into external computer	

Software success is defined by its ability to function autonomously throughout the mission, to transition correctly between modes, to collect and preserve all crucial data from the experiment and sensors, and to continuously maintain the MFC at a strict temperature level. In this section, each success criterion is given verification evidence that is traceable and a criticality factor.

Table 4.10 – Software Success Criteria

ID	Success criterion	Operational phase	Verification evidence	Criticality
SC-01	Software can be uploaded to hardware without errors	Setup	No crash, successful upload observed on electronics LEDs	Critical
SC-02	Software boots nominally after power-on	Setup	Boot logs, electronics LED status	High
SC-03	Transition to flight mode during launch detection/uplink signal	Countdown, launch	Telemetry logs, downlink observation	High
SC-04	Software manages to maintain stable module temperature	Ground tests, flight	Sensor output, downlink observation	Critical
SC-05	Telemetry is correctly formatted and sent	All phases (exc. RS)	Telemetry logs, no bandwidth issues	Medium

SC-06	Experiment begins at uplink signal reception	Microgravity	Telemetry logs, microscope LEDs	Critical
SC-07	No unrecoverable errors	All phases	Error logs	Medium
SC-08	No error shall crash the software	All phases	Error logs, overall system health	Critical
SC-09	Microscope imaging is correctly collected throughout	All flight phases	Telemetry logs, microscope status	High
SC-10	An unexpected crash shall not corrupt previously collected data	All phases	Error logs, check stored data	High
SC-11	Software must correctly receive LO/SOE/SODS	All phases	Experiment status, logged files	Critical

4.8.2 – Software operational concept – State machine

4.8.2.1 Concept of operations

This segment explains how the software will act throughout the mission. The software will be developed in a way that maximizes autonomous functionality. Key characteristics of the software will be:

- Three critical ground commands will be sent during the mission via the RXSM to do the following operations: LO (at lift-off, start onboard clock), SOE (introduce biological particles and medium to microscope observation area), SODS (start recording and storing all data).

- The software will have a deterministic execution regarding error handling and the startup. It will have to transition to the appropriate mode depending on the onboard clock, which starts running at LO.

- The software is set to be fault recoverable but not fault tolerant. All potential errors will, to the best extent possible, be targeted during development, so that the least number of errors occur during flight. The code will have to be able to recover from any faults that occur.

The following schematic shows a simplified overview of the software architecture:

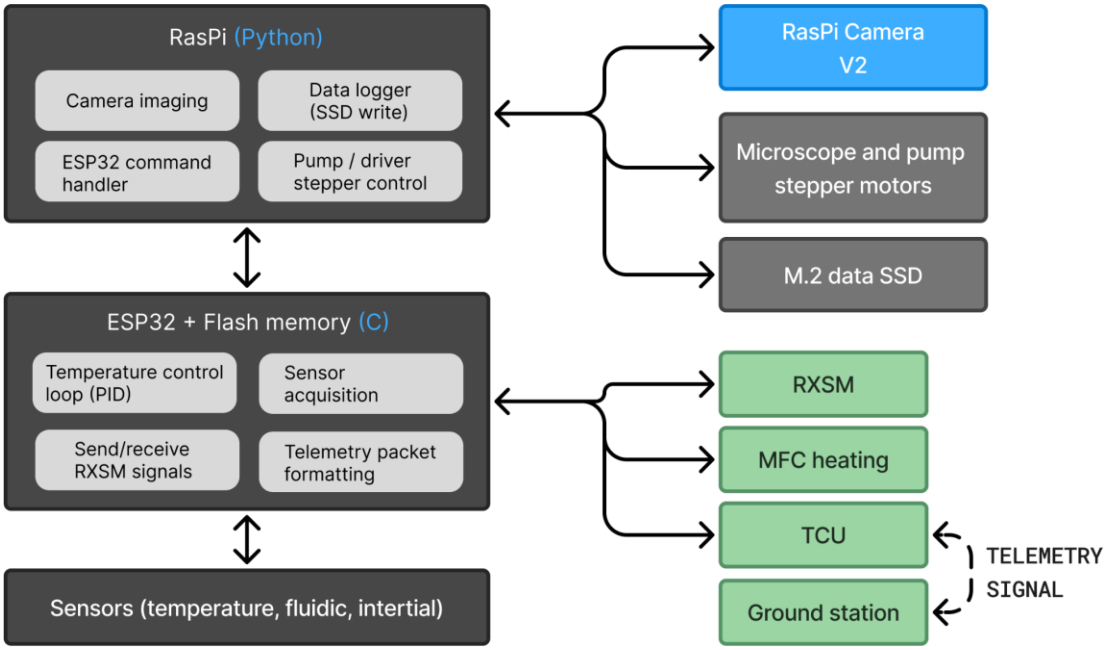


Figure 57: Software architecture overview

The ESP32 will remain in the experiment design alongside the Pi to guarantee deterministic timing for the closed-loop temperature control (PID) and actuator drive signals (heater, stepper motors), functions tied to critical success criterion SC-04 that cannot be reliably guaranteed by the Pi's operating system, particularly given the Pi's own CPU-intensive imaging workload. The ESP-Pi interface is limited to a single, minimal SPI link.

Furthermore, the Pi software will run three concurrent threads (logging, capture, storage write), so that a slow disk write with the imaging does not stall the link with the ESP32. For imaging, the software has a drop-rather-than-block policy, so that a short imaging issue does not stall all imaging.

4.8.2.2 Software modes and state machine

4.8.2.2.1 Modes

Table 4.11 – Definition of operational modes

Mode	Abbreviation	Use/Scope
Startup	SU	Boot experience, software begins running
Test	T	Verify system health
Normal ground	NG	Software is functional on the ground
Normal flight	NF	Software is functional during flight
Normal experiment	NE	Software is functional and experiment is ongoing

The fallback mode, which previously served as a graceful degradation mode in the case of persistent major malfunctions, was removed due to it being overly complex in the scope of the mission. However, several core ideas will be maintained, such as:

- Using simple try/catch logic
- Wrapping sensor reads. Malfunctions are simply logged and skipped, instead of cascading
- Data writes are small, and are stocked in fixed-size files (see section 4.8.3.2.3)
- Error flags are used as indications and logged

4.8.2.1.2 Activation criteria

Table 4.12 – Activation criteria to transition into a given operational mode

Mode	Activation
SU	Once the EM is connected to electrical power
T	By telecommand from ground station
NG	Once the startup phase is complete
NF	After the LO signal is received, & after SOE off
NE	After the SOE signal is received, turns off when SOE is deactivated

4.8.2.1.3 State machine

- **SU:** During the module startup, the main software functions begin working throughout the hardware.
- **T:** If a *test* telecommand is sent from the ground station, a series of tests begin to ensure that the system's health is flight worthy. The tests include:
  - Verify that hardware is functioning (CPU, accessible memory, peripherals and interfaces).
  - Verify that sensors are active and that they transmit coherent data.
  - Verify that there is power running in the system and that the system clock can run correctly (important for mission timestamps).
  - Verify that there is available storage space, ability to upload data into storage (there is a very large amount of storage, but a corrupted data card could quickly invalidate this).
- **NG:** After and if the tests are validated, the normal ground mode starts. No major operations occur; the system is GO for launch. Temperature control is active, and data storage starts at SODS shortly before launch.
- **NF:** Activated at the reception of the LO signal. Data logging frequency is increased, and temperature control continues. Later, after the SOE signal is turned off, NF mode is used for the remainder of the flight and descent.
- **NE:** Activated once the SOE signal is activated. This starts the experiment operations and continues with the previous tasks. Only operational during the experiment phase.

Since the ESP serves as the mode and clock handler, the Pi has no independent mission reference and receives a periodic sync through the UART link poll, computing timestamps locally thereafter. This ensures that imaging timestamps remain valid in the case of a link outage.

4.8.2.1.4 Operational scenario

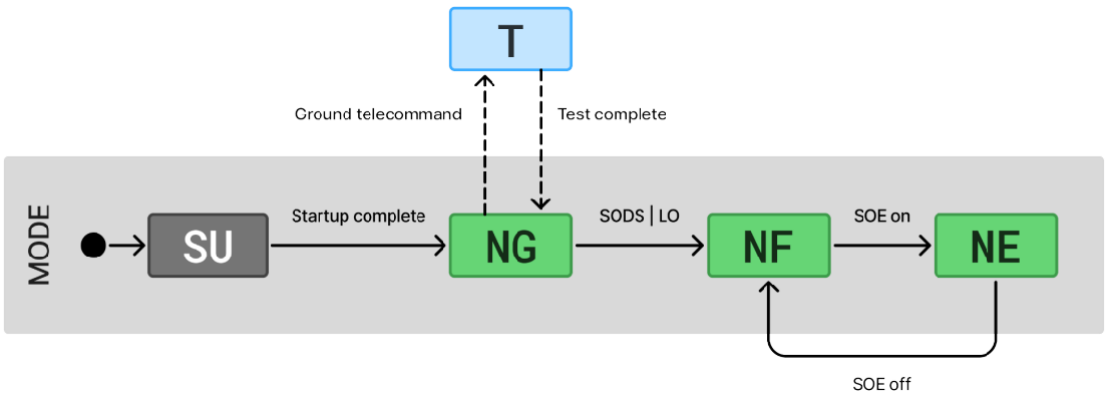


Figure 58: State-transitions

Near the end of the flight, SODS is turned off and the system awaits power cutoff at T+600s. All data is stored in memory by this point, and the EM runs the NF mode until T+600s.

4.8.2.2 Flight computer interfaces

Both flight computers (ESP and Pi) are connected by an SPI interface. SPI will serve as the interface for most components, including both BME temperature sensors, the sensitive PT100 temperature sensor, the LSM6DSM inertial sensor, and the ABP pressure sensor. Due to its design, the SLF3S liquid flow sensor will have an I^2C interface. As shown in *Fig. 59*, a UART interface will communicate with the TCU, and a RS-422 will communicate with the RXSM.

Table 4.13 – ESP-Pi SPI link parameters

Field	Parameter
Clock frequency	1 MHz
SPI mode	Mode 0 (CPOL = 0, CPHA = 0)
Master/Slave	Pi/ESP
Bit order	MSB first
Word size	8 bits
Chip select	Single slave, CS active low
Status exchange frequency	1 Hz
Sensor transfer frequency	10 Hz
Frame validity check	Sequence order + CRC in each ESP response frame
Pi degraded flag activation	3 consecutive invalid or stale frames, cleared after 5 valid frames

4.8.3 – Data flow and management

4.8.3.1 Data list and types

This section describes all the data that is handled by the experiment and explains its purpose, and when said data is to be collected/received/sent.

Table 4.14 – Data sources, type and purpose

Data ID	Source	Type	Category	Purpose	Mission phase
DA-01	ESP-32 external flash memory	Acquired data	Various, temporary storage	Allows ESP-32 to handle more data if needed	All phases
DA-02	BME sensors	Acquired data	Environmental	Characterize experiment environment	All phases
DA-03	Gyroscope (LSM6DSM)	Acquired data	Inertial	Motion data	Launch, ascent, microgravity
DA-04	PT100 temperature sensor	Acquired data	Environmental	Closed-loop temperature control	All phases
DA-05	RasPi Camera V2	Acquired data	Image	Visualization of experiment	Throughout the flight
DA-06	M.2 storage	Acquired data	Metadata, imaging	Mission logs and microscope imaging	All phases
DA-07	RXSM	Acquired data	Experiment signals (LO, SOE, SODS)	Initiate respective mission phases	Countdown, LO, ascent and microgravity
DA-08	TCU to ground station	Transmitted data	Systems data	Flight status downlink	All phases
DA-09	ABP sensor	Acquired data	Biology	Fluid pressure	Experiment
DA-10	SLF35 sensor	Acquired data	Biology	Fluid flow	Experiment
DA-11	ESP32 to TCU	Transmitted data	Telemetry packet	Downlink data	All phases

DA-12	Ground station to TCU	Command	Uplink commands	Wipe data, activate test mode, override heater	Pre-launch
DA-13	Microscope and pump stepper motors	Command	Activate drivers	Inject particles and focus microscope	Experiment

4.8.3.2 Data flow, processing and storage

4.8.3.2.1 Data flow

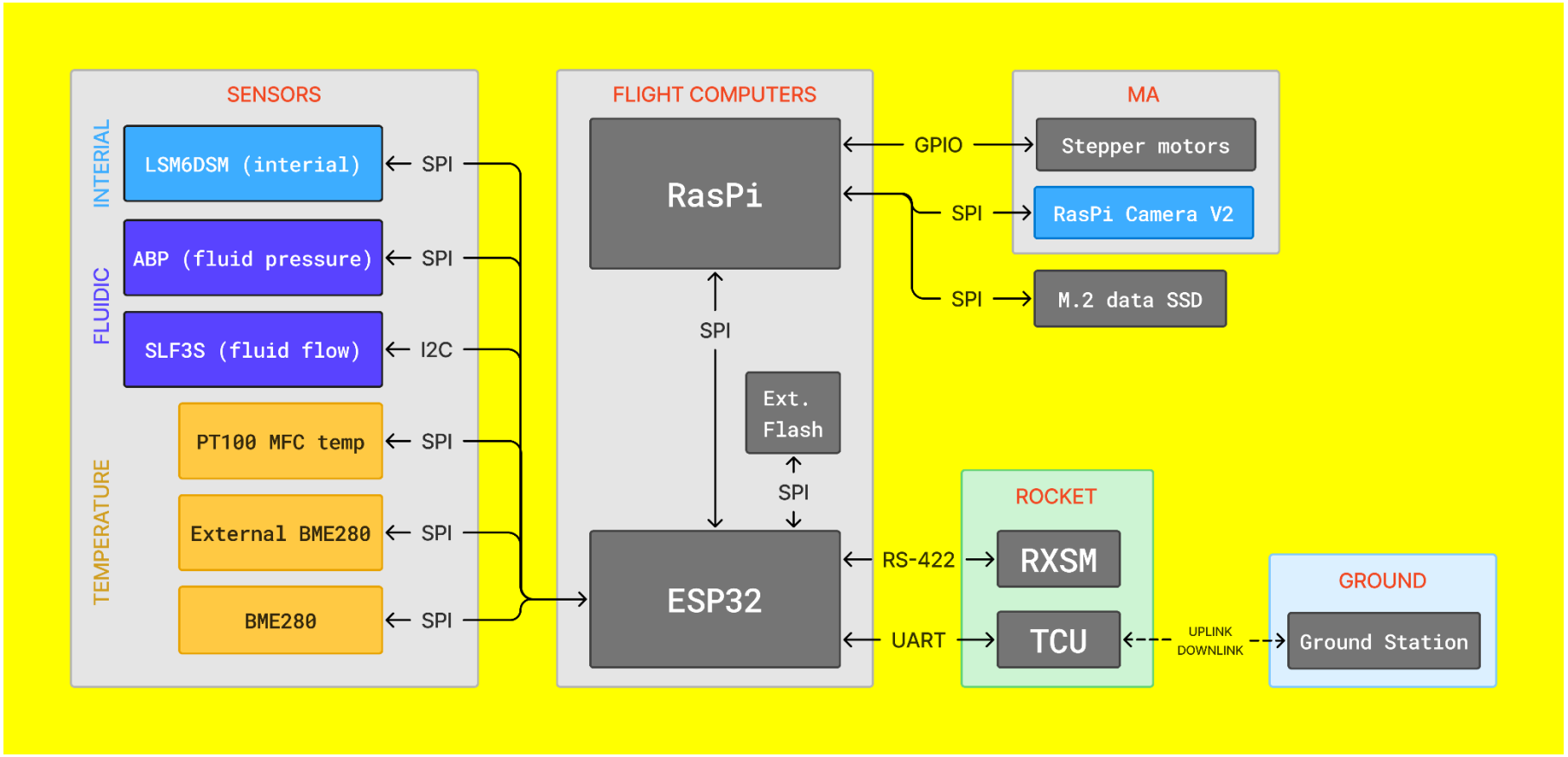


Figure 59: System interfaces visual diagram

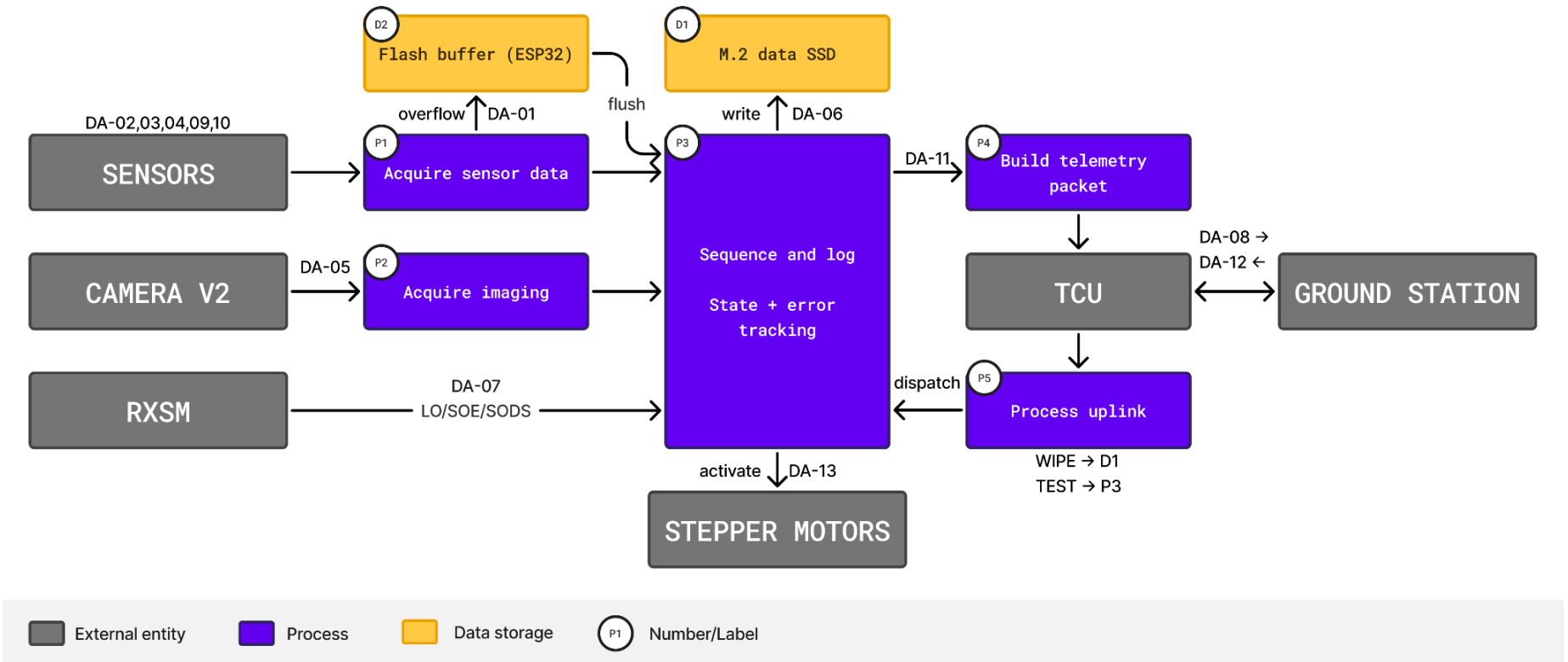


Figure 60: Data flow diagram

4.8.3.2.2 Storage control mechanism

- Write trigger: All logged data is read from an in-memory table that is continuously updated as data arrives. This is also the case for structuring telemetry packets (fig. 60, P4), which are not assembled from stored data. This avoids risks linked with reading files during the mission.
- Category separation: Data is logged into separate files, each with its own category (see table 4.15).
- File rotation policy: File rotation size is smaller during the experiment phase when SOE is active, since experiment data is critical and its loss should be minimized. File rotation validates criterion SC-10 from table 4.10.
- Naming convention: Each file/frame name includes category, sequence number, and a timestamp anchor so files are unambiguously orderable and joinable post-flight even if some are missing. Naming convention now encodes a time reference tag (UNSYNCED / TMINUS / TPLUS) so files from scrubbed countdown attempts are distinguishable from the flight.
- Fault containment: Identically to sensor reads, write malfunctions are simply logged and skipped rather than propagating.

In the case of a prolonged hold before LO, if free SSD space is below 100 GB, the ground team is notified through downlink and can activate the WIPE telecommand. SSD space below 20 GB is generally considered low.

4.8.3.2.3 File/field format table

This section specifies data storage formats and frequencies.

Table 4.15 - Data storage formats and logging frequency

Category	**Format** ***SOE on / SOE off***	**Frequency/sample rate** ***SOE on / SOE off***	**Rotation**	**Rotation (new file every N-lines/frames)** ***SOE on / SOE off***
System state metadata	.csv	10 Hz (unified row logging cadence)	20 s	200/200
Sensor metadata	.csv	10 Hz (native sample rate)	20 s	200/200

Telemetry output	.csv (mirrors downlink packet)	1 Hz (subsampled slice of data above, not independently sampled)	20 s	20/20
Imaging	YUV420 (with sidecar metadata file)	41.85 fps / 4 fps	30 s / 300 s	1256/1200
Uplink commands	.csv	On demand (on ground only)		N/A. One file sufficient
Actuator commands	.csv	Event-triggered		N/A. One file sufficient

File size rotation is set to a reasonable cycle to prevent data loss if one file is corrupted during logging. There is a phase offset of 0, 70 and 14 rows (0, 7 and 14 seconds) between each .csv file. This spreads out the *fsync* signals used each time a new file is created during rotation, which can take a few milliseconds per *fsync*. Keeping them aligned would lead to potential noticeable delays in other Pi tasks. To ensure that imaging is not affected by this, it is handled in its own separate thread.

YUV420 video will be recorded with a frame resolution of 1640x1232 at a frame rate of 41.85fps, which is one of the IMX219 video modes of the Pi. This format is recorded as uncompressed raw imaging, because the Pi has no encoding hardware, which also removes CPU load and the risk of dropping frames. This mode was chosen because its framerate is sufficient, and because it does not crop the image sensor area. Considering that each pixel is 1.5 bytes, and that the NE mode runs for 474s (when imaging is at a high frame rate), this equates to approximately 60GB of video (47% of SSD), or 127MB/s of sustained video.

4.8.3.3 Data output and telemetry

4.8.3.3.1 Camera health definition

To check if the camera is functioning correctly, which is a non-negotiable criterion for experiment results, the software will run two checks to confirm that imaging is in progress:

- The Pi receives timestamped pings from the camera and verifies that the two consecutive timestamps are increasing. This proves that the camera sensor is alive

- The Pi checks that the size of the total imaging write has increased at every code iteration. This confirms that data is being written into storage.
- There is a write-failure indication distinct from the sensor stall, and an explicit “unknown” state that is used when the Pi is unreachable

```
advancing = sensor_ts > last_sensor_ts
last_sensor_ts = sensor_ts

write_ok = size_now > last_written_size
last_written_size = size_now

if not advancing:
    return
CAM_STALLED if not
write_ok:
   return CAM_WRITE_FAIL
```

Figure 61: Camera health software verifications

Each camera flag is defined in a typedef enum and sent as a single byte through downlink.

4.8.3.3.2 Downlink, budget calculation

The telemetry downlink will send all available data to the ground station, except for imaging which requires a very high budget and image loss is highly possible. All telemetry downlink data is identically stored in the M.2 SSD.

Each telemetry packet contains the following items:

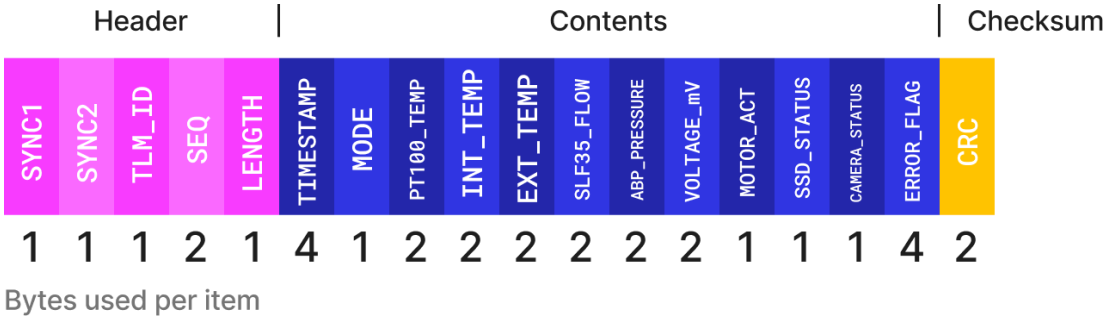


Figure 62: Telemetry downlink protocol

Each packet has a length of 32 bytes, which at a 1 Hz downlink frequency gives a downlink budget of 256 bits/s. By adding a margin of 25% in case additional

fields are added, this gives a downlink budget of 320 bits/s, or 0.320 kbits/s. This value is well below the TCU's downlink limits.

4.8.4 – Software constraints and requirements overview

External constraints imposed by the hardware and the mission environment will bring some limits to the system's capabilities. The software is designed to operate well within the processing limits of the flight computers, to ensure lightweight execution of the code. The software will account for transient faults and will need to detect them to contain the issue and to autonomously switch to a recovery state.

Vibrations, quick temperature variations and radiation effects could have an impact on the system, so tolerance levels will have to be set. Since there is no in-flight access to any onboard systems, autonomy will have to be at the center of developing how the software must act overall.

The software operates under the constraint of a semi-autonomous mission profile. LO, SOE and SODS signals are sent from the RXSM, but all the other operational behavior is pre-defined prior to launch, and the software must run reliably within a fixed mission timeline.

4.8.5 – Verification and validation overview

Software verification will be performed using unit tests during code development, integration tests in a high-fidelity simulation environment, and formal code inspections. Tools such as code coverage will be used extensively.

Validation will be performed by executing representative mission scenarios in a controlled simulation environment, measuring performance against defined mission success criteria.

4.9 Ground Support Equipment

4.9.1 Purpose and scope

The ground station enables the following:

- Continuously monitor all EM parameters, especially temperature values that are critical for the experiment
- Before LO, commands will allow the ground team to activate a test mode, wipe SSD data, and override the MFC heater if needed
- Log a backup of the received telemetry onto the ground station computer

Since no imagery is received by the ground station, and no TV channels are used, it will serve as a control display to see if all parameters and undergoing processes are nominal. This will be sufficient for the ground team to consider that the experiment is unfolding correctly. A flag that confirms that the stepper motor has been activated will be nested on the condition that the SOE signal has been received. This allows the ground station to confirm that bacterial particles have been injected onto the MFC.

4.9.2 Uplink commands

The ground station will be able to send telecommands before LO through uplink to do the following actions:

Table 4.16 - Uplink commands and scope

Uplink command	Use	Logic
TEST	Activates the test mode	Test mode can be run at any time and as many times as desired
WIPE	Wipes data from the M.2 data SSD	In case of prolonged holds, it ensures there is sufficient storage before launch. Manually activated if free space is below 100 GB before launch
HEAT	Overrides the MFC heater PID loop and briefly warms the MFC	If MFC temperature becomes critically low for any reason, it can be reheated manually to keep biological samples alive

The commands above will all require confirmation from the ground team before being executed to avoid any accidental actions.

4.9.3 Downlink handling

The ground station directly decodes the received downlink bytes following the same structure defined by the telemetry packet format. Packet sequence gaps are flagged, but lost packets do not interrupt the reception of subsequent packets.

Telemetry will provide a live indication of any errors that may arise. A *uint32* variable will serve as an error/status flag, sent in each telemetry packet. Possible errors are predefined using a *typedef enum* structure, and if one arises, the corresponding bit number of the flag will be flipped to 1. Once decoded by the ground station, the presence or absence of errors will be visible. This is more efficient than sending an entire error message through downlink.



Figure 63: Error flag representing an error indication based on bit status

4.9.4 Interface, operator display

The UI will be optimized to visibly present the most crucial variables for the ground team to follow. Some variables such as temperature will be used to construct graphs that will help track its evolution.

4.10 Biology

4.10.1 Overview of the Biological subsystem

The STAR-D experiment includes a dedicated biological subsystem interfaced with the microfluidic, optical, thermal, and operational elements of the payload. This subsystem is designed to preserve biological viability and functional activity while remaining fully compatible with system, safety, and mission constraints. The biological subsystem is also designed to ensure reproducibility

and robustness under the specific constraints of sounding rocket missions, including short-duration microgravity, launch-induced mechanical stress, and limited intervention capability. In addition to maintaining cell survival, the biological subsystem enables real-time visualization and quantification of phagocytosis through fluorescence microscopy using FITC-labelled bacterial targets.

All biological components are selected to comply with Biosafety Level 1 (BSL-1) requirements and are fully confined within a sealed microfluidic cartridge. The biological interfaces define how living *Dictyostelium* cells, FITC-labelled Escherichia coli K12 cells, buffers and biological waste interact with the experimental hardware throughout the pre-launch, flight, and post-flight phases

The biological subsystem relies on a single pump, simplifying both the fluidic architecture and the loading procedure. Thanks to the single-chamber design, *Dictyostelium discoideum* cells can be manually loaded into the microfluidic observation chamber during laboratory preparation. The cells are then allowed to adhere to the transparent glass bottom of the chamber, either through their natural adhesive properties or, if necessary, using a cell-adhesion coating such as Ibidiy. This strategy ensures that the cells remain within the microscope field of view during the pre-launch waiting period, including the radio silence phase, and throughout the flight.

4.10.2 Biological Model

The biological model organism is *Dictyostelium discoideum* (strain AX2), a non-pathogenic amoeboid eukaryote widely used to study phagocytosis. Cells are introduced into the microfluidic observation chamber prior to launch, where they adhere to the lower surface and remain motile. *Dictyostelium* is used as an immune-cell analogue, capable of actively recognizing, internalizing, and processing bacterial targets through actin-driven phagocytosis.

Dictyostelium discoideum is extensively used as a model organism to study phagocytosis, chemotaxis, and cytoskeletal dynamics [10], with strong conservation of key mechanisms found in mammalian immune cells. It shares homologous pathways governing actin polymerization, membrane protrusion, and vesicular trafficking, which are essential for particle recognition, engulfment, and phagosome maturation. This makes *Dictyostelium* a robust and well-established proxy for macrophage-like behavior in simplified experimental systems. An overview of the main biological features of *Dictyostelium discoideum*, including its life cycle, chemotactic behavior, and relevance as a biomedical model, is presented in Figure 65.

Beyond its functional similarity to immune cells, *Dictyostelium* is a highly tractable eukaryotic model system combining genetic simplicity with biological relevance. It possesses a haploid genome of approximately 34 Mb encoding around 12,500 proteins, many of which have orthologues in higher eukaryotes,

including humans [11]. This genetic accessibility, together with the availability of large mutant libraries and targeted gene disruption techniques, enables precise investigation of cellular mechanisms involved in motility, phagocytosis, and intracellular trafficking. The underlying processes involve conserved endocytic pathways, vesicle maturation, and intracellular killing mechanisms, including acidification and enzymatic degradation, which closely resemble those observed in mammalian phagocytes. These features make *Dictyostelium* particularly relevant for studying innate immune-like responses in a simplified and controllable system.

Dictyostelium naturally feeds on bacteria through phagocytosis and exhibits highly efficient uptake and degradation mechanisms, consuming large numbers of bacteria per generation.

Importantly, *Dictyostelium* occupies an evolutionary position within the Amoebozoa, diverging early from the lineage leading to animals and fungi, while retaining a wide diversity of ancestral eukaryotic genes [11].

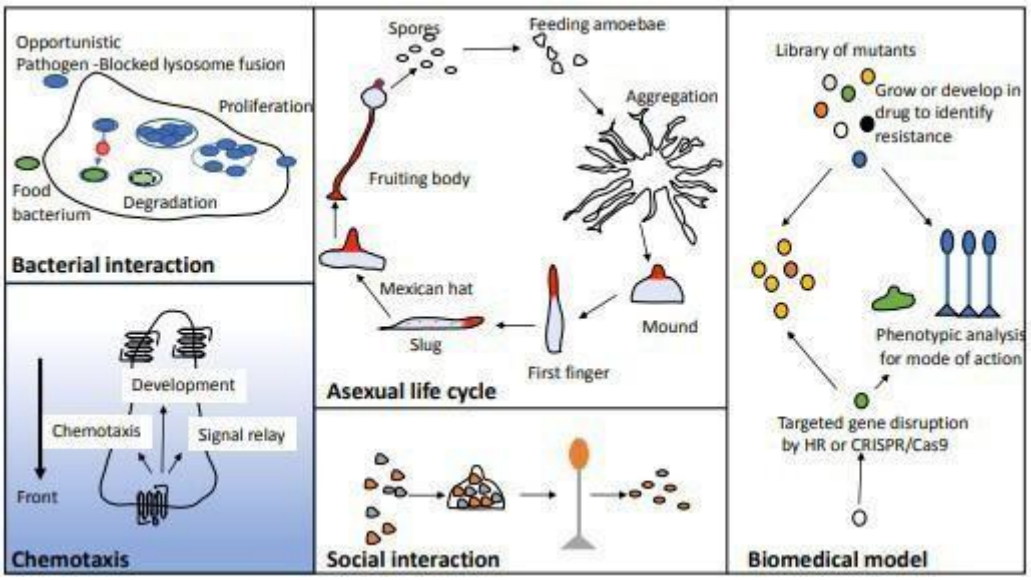


Figure 64: Overview of *Dictyostelium discoideum* as a biomedical model, Reproduced from [12]

4.10.3 *Dictyostelium* Culture Conditions and Biological Constraints

The growth and physiological behavior of *Dictyostelium discoideum* are well characterized and impose specific constraints on experimental design. According to established protocols [13], *Dictyostelium* cells can be cultured either in association with bacteria or under axenic conditions in a nutrient-rich medium such as HL5.

For the present experiment, axenic strains (AX2) are used, allowing growth in the absence of bacteria and ensuring controlled and reproducible conditions. Under these conditions, cells exhibit a doubling time of approximately 8–12 hours depending on temperature and medium composition.

Optimal growth is achieved at temperatures between 21–23°C, while growth is significantly impaired above 25°C, making temperature control a critical parameter for maintaining cell viability and function.

Cell density must be carefully controlled to preserve physiological relevance. Cells should be maintained in the exponential growth phase, typically below 4 × 10^6 cells/mL, as higher densities trigger the onset of development due to the secretion of signaling factors. Conversely, very low densities (<10^4 cells/mL) can result in delayed growth and reduced experimental reproducibility.

Dictyostelium cells do not require strict regulation of CO_2 or nitrogen levels, simplifying experimental handling. However, maintaining sterile conditions is essential to prevent contamination by bacteria, fungi, or yeast.

Importantly, starvation induces developmental processes, which significantly alter cellular behavior. For this reason, cells are transferred into a non-nutritive buffer (KK2) prior to the experiment to control physiological state while preserving motility and phagocytic activity. Although prolonged starvation ultimately induces the multicellular developmental cycle of *Dictyostelium*, the expected pre-launch and flight timelines remain well below the timescale required for aggregation and sporulation. The maximum acceptable holding time will nevertheless be experimentally determined during the biological validation campaign.

Cell density measurements must be performed using direct counting methods (e.g., hemocytometer), as optical density measurements are not reliable due to cell heterogeneity and sedimentation. These biological constraints define the operational window within which the experiment must be conducted and directly inform the design of the bio–microfluidic interface and associated test procedures.

The initial fluorescence strategy was based on pHrodo™, a pH-sensitive fluorophore designed to report phagosome acidification following successful particle internalization. However, the first phase of the biological validation campaign demonstrated that fluorescence activation could occur several tens of minutes after phagocytosis, making this approach incompatible with the short duration of the microgravity phase available during the sounding rocket flight.

To overcome this limitation, the experiment was transitioned to a FITC-based fluorescence strategy. FITC (Fluorescein Isothiocyanate) is one of the most widely used fluorophores in biological imaging due to its high quantum yield, photostability, ease of conjugation and compatibility with standard fluorescence microscopy. The isothiocyanate functional group (-N=C=S) reacts covalently with primary amine groups present on proteins, allowing stable attachment of

FITC to antibodies without significantly altering their biological activity. These FITC-conjugated antibodies are subsequently used to label *Escherichia coli* K12 cells, producing fluorescent bacterial targets suitable for the phagocytosis assay.

Upon excitation with blue light around 495 nm, FITC emits a bright green fluorescence centred near 519 nm, enabling direct visualization of labelled bacteria throughout the experiment. Unlike pHrodo, whose fluorescence depends on phagosomal acidification, FITC is fluorescent immediately after excitation. Although it no longer provides a direct readout of phagosome acidification, it allows continuous observation of cell-particle encounters, bacterial recognition, engulfment, intracellular localization and phagocytic dynamics in real time. This fluorescence strategy is therefore fully compatible with the short duration of the microgravity phase while preserving the core scientific objectives of STAR-D.

To prepare FITC-labelled bacterial targets, E. coli K12 cells are first grown to mid-log phase and collected by centrifugation, then washed to remove any residual culture medium. The bacterial pellet is resuspended in a carbonate buffer adjusted to pH 9?, since mildly alkaline conditions favor the reaction between FITC and the primary amine groups present on bacterial surface proteins. FITC is dissolved in a small volume of DMSO to ensure complete solubilization before being added to the bacterial suspension. Then, the mixture is gently agitated and incubated at room temperature in the dark, allowing the isothiocyanate group of FITC to react covalently with the amines and form stable thiourea bonds. Once the incubation is complete, the cells are washed repeatedly with phosphate-buffered saline to remove any unbound dye, leaving only the fluorescently labeled bacteria. The resulting FITC-tagged cells emit bright green fluorescence when excited around 495 nm, making them immediately detectable under a fluorescence microscope and suitable for short-duration phagocytosis assays in microgravity.

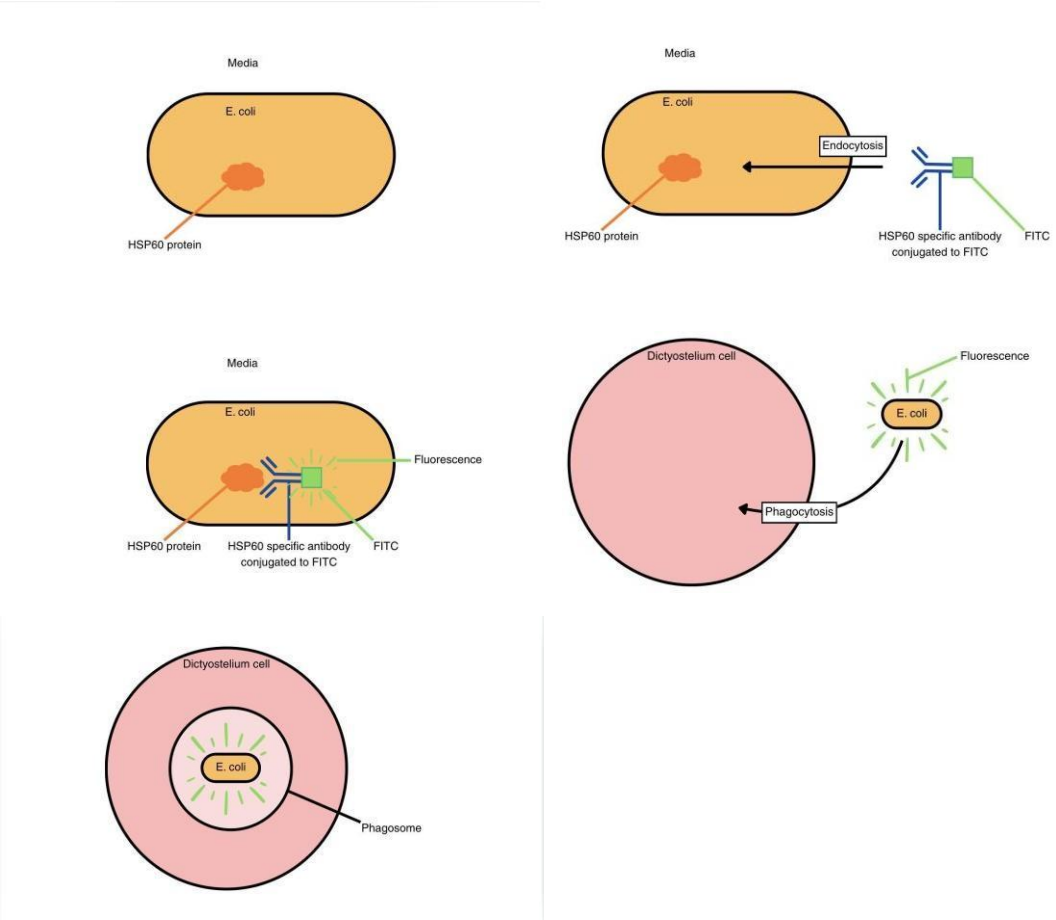


Figure 65: Schematic overview of our FITC-based fluorescence strategy

Table 4.17: *Dictyostelium* Biological Constraints for Experiment Design (based on [13])

Parameter	Value/Range	Impact on experiment
Temperature	21°C – 23°C (optimal), > 25°C inhibitory	Requires controlled thermal environment
Doubling time	8h – 12h (axenic)	Defines preparation timeline
Axenic growth in HL5 medium	8h – 12h (axenic)	Enables reproductible, bacteria-free conditions
Cell density (optimal)	< 4 × 10^6 cells/mL	Prevents unwanted development
Lower density limit	> 4 × 10^4 cells/mL	Avoids lag phase and

		instability
CO_2 / gas control	Not required	Simplifies system design
Development trigger	Starvation	Used intentionally before experiment (KK2 buffer)
Protease peptone 2	5g/L	Feeds *Dictyostelium* cells
Thiotone E peptone	5g/L	Feeds *Dictyostelium* cells
Glucose	10g/L	Feeds *Dictyostelium* cells
Yeast extract	5g/L	Feeds *Dictyostelium* cells
$Na_2HPO_4 \cdot 7H_2O$	0.35g/L	Feeds *Dictyostelium* cells
KH_2PO_4	0.35g/L	Feeds *Dictyostelium* cells

4.10.4 State of the Art of Biological Experiments under Altered Gravity

Ground-based platforms have been extensively developed to study the effects of altered gravity on biological systems. Among the most used systems are rotating wall vessels (RWV), high-aspect rotating vessels (HARV), slow-turning lateral vessels (STLV), clinostats, and random positioning machines (RPM). These devices simulate microgravity conditions by continuously reorienting the gravity vector, thereby reducing its net directional effect on the sample.

These systems have enabled major advances in understanding cellular responses to altered gravity, including changes in gene expression, motility, and chemotactic behavior [15]. However, their physical operating principles introduce important limitations. Fluid dynamics within these devices are highly dependent on system geometry, rotation speed, and fluid properties, leading to variable shear stress distributions and transport conditions. As a result, organisms exposed to different simulated microgravity platforms may exhibit significantly different responses, complicating interpretation and comparison of experimental results [15].

Furthermore, these platforms primarily enable long-duration exposure experiments (hours to days), which are well-suited for studying gene expression, adaptation, and developmental processes. In contrast, short-term cellular responses occurring within seconds to minutes remain less explored, despite evidence that key processes such as cytoskeletal reorganization and signal transduction can respond rapidly to changes in the gravitational vector [14].

Another major limitation concerns real-time imaging under simulated microgravity conditions. While several approaches have been developed, such as microscope-integrated clinostats, digital holographic microscopy coupled to RPM systems, and lab-on-a-chip clinorotation devices, these setups remain technically complex and may require interruptions in rotation or specific configurations to allow imaging. These constraints can limit temporal resolution and complicate the observation of fast and dynamic cellular processes.

In addition to intracellular responses, altered gravity conditions significantly modify the extracellular physical environment. In particular, the reduction or absence of buoyancy-driven convection leads to diffusion-dominated transport regimes, affecting particle distribution, encounter rates, and local concentration gradients. These effects are critical for processes such as phagocytosis, which depend on efficient interactions between cells and target particles.

Despite these advances, few experimental platforms combine (i) true microgravity exposure, (ii) controlled microfluidic environments, and (iii) real-time optical observation of dynamic cellular processes. Most existing studies rely either on simulated microgravity or on long-duration spaceflight experiments, which are not optimized for capturing rapid mechanosensitive responses.

The STAR-D experiment addresses this gap by implementing a compact, autonomous lab-on-chip system capable of performing controlled biological experiments during the short-duration microgravity phase of a sounding rocket flight. By combining a simplified eukaryotic model organism (*Dictyostelium discoideum*), a functional fluorescence-based readout of phagocytosis, and time-resolved microscopy, the system enables direct observation of early cellular responses to altered gravity under well-controlled conditions.

4.10.5 Gravity-Dependent Cellular Mechanisms

Previous studies have demonstrated that actin-dependent processes, including cell motility and membrane remodeling, are sensitive to changes in the gravitational vector. For instance, experiments performed under simulated microgravity conditions in mammalian cells have shown rapid alterations in cytoskeletal organization and cell motility, with transient increases or decreases depending on cell density and environmental constraints [14]. Given the strong conservation of actin-driven mechanisms across eukaryotes, including *Dictyostelium discoideum*, these findings are directly relevant to the present

study. These results suggest that gravity-induced perturbations in cytoskeletal dynamics can directly impact functional processes such as phagocytosis.

At the molecular level, early responses to altered gravity include transient modifications in protein activity, cytoskeletal structure, and intracellular signaling pathways. Hypergravity has been shown to affect metabolic pathways involved in innate immune-like responses, including oxidative burst reactions associated with phagocytic activity in macrophage-like cells [16]. Experimental studies performed under rapidly changing gravity conditions, such as parabolic flights, demonstrated that transitions between microgravity and hypergravity can significantly modulate these pathways, either increasing or decreasing their activity depending on the experimental conditions. Importantly, these effects were reported to be transient and reversible, without inducing lasting cellular damage [16]. However, several studies indicate that these initial perturbations tend to decrease over time as cells adapt to the new gravitational environment [15]

In addition to intracellular processes, microgravity also affects the physical environment surrounding the cells. The suppression of buoyancy-driven convection leads to diffusion-dominated transport, altering particle distribution, encounter rates, and local concentration gradients. Experimental studies on microorganisms have shown that variations in fluid dynamics, including shear stress and mixing behavior, can significantly influence biological response [15]. These effects are particularly relevant for phagocytosis, which depends on effective cell–particle interactions.

Most existing studies focus on long-term microgravity exposure (hours to days), often emphasizing gene expression changes, cellular adaptation, or developmental processes. In contrast, short-term responses occurring within seconds to minutes (such as those accessible during sounding rocket flights) remain poorly characterized. These early responses are critical, as they reflect the immediate mechanical and physical impact of altered gravity on cellular function, before compensatory adaptation mechanisms take place. The STAR-D experiment specifically targets these early, mechanosensitive responses by measuring phagocytosis dynamics in real time during the microgravity phase. By combining a simplified model organism with a direct functional readout, the experiment aims to isolate the contribution of physical factors to phagocytic efficiency under microgravity conditions.

Despite these advances, direct studies investigating the effects of altered gravity on *Dictyostelium discoideum* remain limited, particularly in the context of short-duration microgravity and transient hypergravity phases typical of sounding rocket flights. Most available knowledge is therefore extrapolated from studies on mammalian cells, microorganisms, or long-term spaceflight experiments, as discussed above. While these systems provide valuable insights due to conserved cellular mechanisms, they do not fully capture the specific behavior of *Dictyostelium* under rapidly changing gravitational conditions.

In particular, the impact of short hypergravity phases (up to ~15–20 g for ~20 seconds during launch) is expected to be minimal in terms of cellular damage, as such short-duration exposures are generally well tolerated by eukaryotic cells. However, these conditions may still induce transient mechanical stress affecting cytoskeletal organization or fluid distribution. Previous ground-based studies under hypergravity conditions also suggest that pathways linked to phagocytosis, including focal adhesion kinase (FAK)-ERK signaling, can be transiently modulated by increased gravity levels [17]. These observations support the hypothesis that altered gravity may temporarily affect phagocytic dynamics without permanently compromising cell viability or function. These effects remain poorly characterized and cannot be reliably predicted without experimental validation. Therefore, dedicated experiments are required to assess both the robustness of the biological system and the potential influence of these transient phases on the measured phagocytic response.

In addition, dedicated ground validation experiments will be performed prior to flight to evaluate the effects of short-term hypergravity exposure on *Dictyostelium discoideum* cells under controlled conditions. These tests will aim to verify that any gravity-induced perturbations remain minor, transient, and reversible within the operational conditions expected during launch.

4.10.6 Bio–Microfluidic Interface

The interface between the cells and the microfluidic chip must ensure:

- sufficient adhesion to prevent cell drifting in microgravity,

  This requirement is critical as *Dictyostelium* cells rely on substrate adhesion to generate traction forces required for actin-driven motility and efficient phagocytosis. In physiological conditions, these processes depend on dynamic interactions between the cytoskeleton and the substrate, which are well characterized in *Dictyostelium* as a model for cell motility and chemotaxis [12] *Dictyostelium* naturally feeds on bacteria through phagocytosis and is capable of ingesting and degrading large numbers of bacteria per generation, using conserved intracellular killing mechanisms including phagosome acidification, enzymatic degradation, and reactive oxygen species production

- preservation of actin-driven motility and phagocytic behavior,

- compatibility with optical imaging (low cellular autofluorescence),

- stable viability and activity during pre-launch standby and experiment execution.

Cells are maintained in a non-nutritive physiological buffer (KK2) to prevent uncontrolled proliferation and to minimize optical background while preserving motility and phagocytic capacity.

The experiment uses FITC-labelled Escherichia coli K12 cells as phagocytic targets. The bacteria are fluorescently labelled using FITC-conjugated antibodies prior to loading and stored in a dedicated fluidic reservoir. During the microgravity phase, they are injected into the observation chamber where they interact with *Dictyostelium* cells.

The particle interface must ensure:

- homogeneous suspension of bioparticles prior to injection,
- minimal aggregation during storage and delivery,
- controlled flow behavior under microgravity conditions,

  In microgravity conditions, the suppression of buoyancy-driven convection leads to a diffusion-dominated transport regime, which significantly alters particle distribution, encounter rates, and local concentration gradients. As *Dictyostelium* relies on chemotaxis and active motility to locate and ingest bacterial targets, these physical changes can directly influence phagocytosis efficiency and interaction dynamics between cells and particles [12]

- stable fluorescence intensity throughout the experiment.

FITC-labelled bacteria remain fluorescent throughout the experiment, enabling continuous visualization of bacterial transport, cell–particle encounters, engulfment and intracellular localization. While FITC fluorescence may vary slightly with changes in pH, it primarily serves as a constitutive fluorescent label, allowing real-time monitoring of bacterial uptake and phagocytosis dynamics throughout the microgravity phase.

Biological fluids interface directly with the microfluidic channels, syringes, and waste reservoirs. To preserve biological function and system stability, the following constraints apply:

- buffers and additives must exhibit low autofluorescence in the selected optical wavelength range,
- additives (e.g., BSA, optional methylcellulose) must reduce adhesion to channel walls and limit bioparticle aggregation,
- all fluids must be degassed prior to loading to prevent air bubble formation,
- flow rates must preserve cell adhesion, motility, and viability.

All biological material, including cells, E. coli bioparticles, buffers, and waste fluids, remains confined within the sealed microfluidic cartridge for the entire mission lifecycle. Used fluids are collected in internal waste reservoirs.

This interface ensures:

- no release of biological material into the payload or environment,
- full compliance with BSL-1 safety standards,
- safe post-flight handling and disposal of biological materials.

4.10.7 Laboratory Facilities and Implementation Plan

The biological experimental activities will be conducted at FABLAB UNIVERSITÉ PARIS-SACLAY, at the following address: Bât 660 (Digiteo Labs Moulon), 4 Rue René Thom, 91190, Gif-Sur-Yvette, France. This fablab has agreed to lend us the required infrastructure for cell culture, microscopy, and biological testing under controlled laboratory conditions. The list of material lent by FABLAB UNIVERSITÉ PARIS-SACLAY can be found in Appendix E.

The laboratory is equipped to support all core experimental operations, including *Dictyostelium discoideum* axenic culture, sample preparation, and fluorescence-based imaging. Standard microbiological equipment (incubation systems, sterile handling environment, pipetting systems, and microscopy platforms) is available and compatible with the requirements of the STAR-D experiment.

The first phase of the biological validation campaign is currently underway, with priority given to the early-stage verification tests:

- BV-T01: Standalone biological validation (phagocytosis functionality)

- BV-T02: Pre-launch holding time validation (viability and motility)

These experiments represent a key milestone of the biology work package. Beyond validating the biological proof of concept, they are intended to determine several operational parameters that cannot be reliably established from the literature alone. In particular, they will define the maximum acceptable holding time between sample preparation and launch, evaluate how long *Dictyostelium* cells remain viable and phagocytically active after loading into the microfluidic chamber, assess fluorescence stability over time, and characterize the operational limits of the biological subsystem under representative experimental conditions. These results will directly inform the launch campaign procedures and the final experimental timeline.

In parallel, a comprehensive list of required biological equipment, consumables, and reagents has already been established, including supplier identification and cost estimation. This list covers all materials required for the biological validation campaign, together with redundant consumables and backup reagents to ensure the robustness and continuity of the experimental programme.

The validation campaign also includes matched 1g control experiments, allowing direct comparison with future flight data. These controls will help distinguish gravity-induced effects from biological variability and experimental artefacts while providing a quantitative reference for data interpretation.

At the same time, coordination with the host laboratory is ongoing to finalize experimental procedures, confirm equipment availability, and optimize the integration of the biological protocol with the laboratory infrastructure.

Overall, the combination of an operational laboratory, an ongoing validation campaign, secured biological resources, and dedicated control experiments substantially reduces technical and biological risks while providing the quantitative parameters required to finalize the flight protocol and prepare the launch campaign.

Table 4.18 - Biology Experiment Interfaces Overview.

Interface Category	Component	Description	Key Constraints
Biological Model	*Dictyostelium discoideum* AX2	Living amoeboid cells performing phagocytosis	BSL-1, 18°C – 25°C, adhesion required
Target Particles	FITC-labelled *Escherichia coli* K12 cells (heat inactivated)	FITC-labelled bacterial cells used as fluorescent phagocytic targets	Homogeneous suspension, low aggregation, preservation of pH-dependent fluorescence signal
Buffer System	KK2 buffer	Physiological medium for observation	Low autofluorescence, pH stability
Additives	BSA, methylcellulose (optional)	Reduce adhesion and sedimentation	Optical neutrality, biological compatibility
Fluidic Interface	Syringes, tubing, chip channels	Delivery of cells and particles	Bubble-free, controlled flow
Optical Interface	Fluorescence + brightfield imaging	Visualization of ingestion events	Blue excitation (~495 nm), green emission (~519 nm), low background fluorescence
Waste Interface	Internal reservoirs	Collection of used biological material	Fully sealed, no leakage
Safety Interface	Cartridge containment	Isolation from payload and crew	BSL-1 compliance

5 EXPERIMENT VERIFICATION AND TESTING

5.1 Verification Matrix

Table 5.1 - Verification table

Req. ID	Requirement (abbreviated)	Procedure (Test)	Method
Functional Requirements (F)			
F.1	Sample container holds cells	FV-T01	**R, T**
F.2	Mix cells with phagocytosis target	FV-T02	**R, T**
F.3	Video recording of sample cells	FV-T03	**R, T**
F.5	Bright-field illumination	FV-T04	**R, T**
F.6	Epi-fluorescent illumination	FV-T04	**R, T**
F.7	Measure phagocytosis target flow	FV-T05	**R, T**
F.8	Control phagocytosis target flow	FV-T02	**R, T**
F.9	Measure phagocytosis target pressure	FV-T06	**R, T**
F.10	Control phagocytosis target pressure	FV-T02	**R, T**
F.11	Measure sample container temperature	FV-T07	**R, T**
F.12	Control sample container temperature	FV-T07	**A, T**
F.13	Keep sample container at constant pressure	PV-T01	**T**
F.14	Measure ambient pressure outside container	PV-T01	**R, T**
F.15	Measure experiment vibration	PV-T02	**R, T**
Performance Requirements (P)			
P.1	Sample holder: 500 µL capacity	PV-T03	**R, T**
P.2	Cells aligned with camera for 180 s	PV-T03	**R, T**
P.3	Mix 2 mL phagocytosis target	FV-T02	**R, T**
P.4	Video record for 400 s	FV-T03	**R, T**
P.5	Video at 1080p	FV-T03	**R, T**
P.6	Video at 3 fps	FV-T03	**R, T**
P.7	Video zoom ×40	FV-T03	**R, T**
P.8	Bright field 0.1–1 W/mm²	FV-T04	**R, T**
P.9	Epi-fluorescence at 560 nm (± tol – Ch.2)	FV-T04	**R, T**
P.10	Measure pressure 0–0.1 bar	FV-T06	**R, T**
P.11	Pressure precision 1 mBar	FV-T06	**R, T**
P.12	Pressure sample rate 0.5 s	FV-T06	**R, T**
P.13	Mix target 0–500 mBar	FV-T02	**R, T**
P.14	Measure flow 0–500 µL	FV-T05	**R, T**
P.15	Flow precision 50 µL	FV-T05	**R, T**
P.16	Flow sample rate 0.5 s	FV-T05	**R, T**

Req. ID	Requirement (abbreviated)	Procedure (Test)	Method
P.17	Mix target 0–500 µL (lower bound – Ch.2)	FV-T02	**R, T**
P.18	Temperature range −30 to 30 °C	FV-T07	**R, T**
P.19	Temperature precision 0.5 °C	FV-T07	**R, T**
P.20	Temperature sample rate 0.5 s	FV-T07	**R, T**
P.21	Control temperature 18–25 °C	FV-T07	**R, T**
P.22	Container at 1 bar (range – Ch.2)	PV-T01	**R, T**
P.23	Ambient pressure 0–1 bar	PV-T01	**R, T**
P.24	Ambient pressure precision 0.1 bar	PV-T01	**R, T**
P.25	Ambient pressure sample rate 0.5 s	PV-T01	**R, T**
P.26	Vibration 0–1 g, 3 axes (split per axis – Ch.2)	PV-T02	**R, T**
P.26.1	On x-axis	PV-T02	**R, T**
P.26.2	On y-axis	PV-T02	**R, T**
P.26.3	On z-axis	PV-T02	**R, T**
P.27	Vibration precision 0.1 g	PV-T02	**R, T**
P.28	Vibration sample rate 0.1 s	PV-T02	**R, T**
Design Requirements (D)			
D.1	EuroLaunch D-sub bracket 20 mm below radax	DV-I02	**R, T**
D.2	Bulkhead open for 4× D-SUB 15	DV-I02	**R, T**
D.3	Hatch flush with module skin	DV-I01	**R, I**
D.4	Hatch follows module shape	DV-I01	**R, I**
D.5	Hatch not protruding into airflow	DV-I01	**R, I**
D.6	Hatch offset max 0.5 mm inward	DV-I01	**R, T**
D.7	Hatch cut-out max 156×156 mm	DV-I01	**R, T**
D.8	Hatch thickness 4 mm	DV-I01	**R, T**
D.9	Hatch has supporting frame	DV-I01	**R, I**
D.10	Frame thickness min 4 mm	DV-I01	**R, T**
D.11	Gap 0.1–0.5 mm	DV-I01	**R, T**
D.12	Rivets 4 mm or 5 mm diameter	DV-I01	**R, T**
D.13	Frame–module overlap 2.5× thickness	DV-I01	**R, T**
D.14	Frame–hatch overlap 2.5× thickness	DV-I01	**R, T**
D.15	Aluminium, similar stiffness	DV-I02	**R**
D.16	Absorbent material top & bottom (>2 mL), absorbent material 20x fluid volume	DV-T01	**R, T**
D.17	Pressure systems 2:1 safety factor	DV-T01	**A, T**
D.18	Withstand 20 g, 3 axes	DV-A01	**A**
D.19	REXUS User Manual vibration profile	DV-A01 + DV-T02	**A, T**
D.20	CG close to longitudinal axis	DV-A01	**A**
D.21	BNC connector present	DV-I02	**R, I**
D.22	Outer wall ≤ T_amb + 10 °C	DV-T03	**R, T**
D.23	Feed-through cable ≤ 70 °C	DV-T03	**R, T**

Req. ID	Requirement (abbreviated)	Procedure (Test)	Method
D.24	Inter-module face ≤ 50 °C	DV-T03	**R, T**
D.25	Convective air ≤ T_amb + 10 °C	DV-T03	**R, T**
D.26	D-SUB 15 male connector	DV-I02	**R, I**
D.27	Downlink baud ≤ 80% (30 kbit/s)	DV-T05	**R, T**
D.28	Downlink 8N1 format	DV-T05	**R, T**
D.29	Uplink baud ≤ 80% (30 kbit/s)	DV-T05	**R, T**
D.30	Uplink 8N1 format	DV-T05	**R, T**
D.31	Tolerate voltage steps	DV-T04	**R, T**
D.32	Peak current ≤ 3 A	DV-T04	**R, T**
D.33	Mean current ≤ 1 A (~30 W)	DV-T04	**R, T**
D.34	Ripple ≤ 100 mVpp on power line	DV-T04	**R, T**
D.35	Optocouplers on control lines	DV-I03	**I, T**
D.36	Radio silence: no current during window	DV-T05	**R, T**
D.37	Power cables twisted	DV-I03	**I**
D.38	Data cables twisted	DV-I03	**I**
D.39	GPS band ≤ −115 dBm (1573–1578 MHz)	DV-T05	**A, T**
D.40	TC bands free ≤ −100 dBm	DV-T05	**A, T**
D.41	Power emissions ≤ 100 mVss on 28 V	DV-T04	**A, T**
Operational Requirements (OP)			
OP.1	Safely contain biological samples all flight phases	FV-T01	**R, T**
OP.2	Autonomous operation, no ground intervention	OV-T01	**R, T**
OP.3	Biological system withstands hypergravity	OV-T02	**T**
OP.5	PV inserts into EM with no resistance	DV-I03	**T**
OP.6	PV has no displacement once fixed	DV-I03	**T**

5.2 Verification Plan

Table 5.2 - Verification plan

Test ID	Test Objective	Model	Setup	Date / Timing	Requirements Verified	Method	Pass / Fail Criteria	Expected Output	Status
2. Functional Verification (FV)									
FV-T01	Verify the sample container holds cells and remains biologically sealed throughout handling and simulated flight phases.	FM cartridge	Assembled flight cartridge; handling + vibration-conditioned sample, sealed-volume leak/dye check.	Planned from 22/09	F.1 (sample container), O.1 (biological containment)	**R, T**	PASS: no leakage / no biological escape across all handling & simulated flight steps; seal integrity maintained. FAIL: any breach.	Seal/containment test log + post-test inspection report.	Open
FV-T02	Verify the fluidic system can mix cells with phagocytosis target at required volume and pressure range.	EM fluidic rig	Fluidic bench with syringe drivers, flow & pressure sensors, representative cartridge.	Open, planned at final integration mid-October	F.2 (mixing), F.8 (flow control), F.10 (pressure control), P.3 (2 mL), P.13 (0–500 mBar), P.17 (0–500 µL)	**R, T**	PASS: mixing achieved within 2 mL, 0–500 mBar, 0–500 µL. FAIL: any parameter outside range.	Mixing log (volume/pressure traces) + imaging confirmation.	Open
FV-T03	Verify recording duration, resolution, frame rate and zoom in a single imaging test; footage reviewed against acceptance criteria.	EM imaging chain	Microscope + camera on optical bench, target slide, full acquisition pipeline.	Ongoing	F.3, P.4 (400 s), P.5 (1080p), P.6 (3 fps), P.7 (×40 zoom)	**R, T**	PASS: ≥400 s continuous 1080p @ 3 fps at ×40, readable. FAIL: any spec unmet.	Video file meeting stated parameters, retained for review.	Open
FV-T04	Verify bright-field intensity range and epi-fluorescence wavelength in one illumination session.	EM illumination	LED matrix + optics, calibrated photometer / spectrometer.	Planned from 22/09	F.5, P.8 (0.1–1 W/mm²); F.6, P.9 (560 nm ± tol)	**R, T**	PASS: bright-field within 0.1–1 W/mm²; epi-fluorescence centred at 560 nm within defined tolerance. FAIL: outside band.	Intensity + wavelength measurement record.	Open
FV-T05	Single closed-loop test: inject known volumes, verify measurement range, precision and sample rate simultaneously.	EM fluidic rig	Fluidic bench, calibrated reference volumes, flow sensor DAQ.	Open, planned at final integration mid-October	F.7 (flow meas.), P.14 (0–500 µL), P.15 (50 µL precision), P.16 (0.5 s rate)	**R, T**	PASS: measured vs injected within 0–500 µL, 50 µL precision, 0.5 s rate. FAIL: any spec unmet.	Flow log (injected vs measured) + precision/rate statistics.	Open
FV-T06	Apply and hold pressures across range on phagocytosis	EM fluidic rig	Pressure-controlled fluidic bench, reference	Open, planned at final integration mid-October	F.9, P.10 (0–0.1 bar), P.11 (1 mBar precision), P.12 (0.5 s rate); F.8 (control)	**R, T**	PASS: pressure measured/held across 0–0.1 bar, 1 mBar	Pressure log across range + accuracy/stability statistics.	Open

Test ID	Test Objective	Model	Setup	Date / Timing	Requirements Verified	Method	Pass / Fail Criteria	Expected Output	Status
	line; verify measurement accuracy and control stability.		gauge, sensor DAQ.				precision, 0.5 s. FAIL: outside spec.		
FV-T07	Thermal-chamber test: verify sensor range, precision, sampling rate and active control band.	EM thermal setup	Thermal chamber, heater + control loop, reference probe, sensor DAQ.	Open, planned by end of October	F.11, P.18 (−30 to 30 °C), P.19 (0.5 °C precision), P.20 (0.5 s rate); F.12, P.21 (18–25 °C control)	**A, T**	PASS: sensor accurate over −30..30 °C at 0.5 °C / 0.5 s; control holds 18–25 °C band. FAIL: outside spec.	Temperature log + control-band trace.	Open
2b. Performance Verification – Environmental Sensors									
PV-T01	Verify constant ~1 bar maintained in container and ambient outside pressure measured across full range.	EM / FM vessel	Pressure vessel in chamber, reference gauge, internal + ambient sensors.	Ongoing	F.13 (constant pressure); P.22 (1 bar); F.14, P.23 (0–1 bar ambient), P.24 (0.1 bar precision), P.25 (0.5 s rate)	**R, T**	PASS: internal held near 1 bar (per revised P.22 range); ambient measured 0–1 bar at 0.1 bar / 0.5 s. FAIL: outside spec.	Pressure dataset (internal + ambient) + precision statistics.	Open
PV-T02	Single data-acquisition test on shaker or flight data; verify range, precision and sample rate across all 3 axes.	STM / EM	Shaker table or recorded flight data, 3-axis IMU DAQ.	Open, planned at final integration mid-October	F.15, P.26 (0–1 g, 3 axes – split per axis pending Ch.2), P.27 (0.1 g precision), P.28 (0.1 s rate)	**R, T**	PASS: vibration measured on each axis over 0–1 g at 0.1 g / 0.1 s. FAIL: any axis outside spec.	3-axis vibration dataset + precision/rate statistics.	Open
PV-T03	Verify sample-holder volume capacity and camera alignment duration by inspection and timed imaging test.	EM holder	Sample holder + camera rig, reference volume, timer.	Open, planned at final integration mid-October	P.1 (500 µL capacity), P.2 (180 s aligned)	**R, T**	PASS: 500 µL capacity confirmed; alignment achieved within 180 s. FAIL: capacity or timing unmet.	Capacity measurement + alignment timing record.	Open
3a. Structural / Hatch Interface									
DV-I01	Single session: flush/offset, gap, cut-out size, thickness, frame overlap, rivet diameter.	FM structure	Flight hatch/frame, callipers + gauges (measurement).	Planned by end of September	D.3–D.14 (flush, shape, no protrusion, ≤0.5 mm inward, 156×156, 4 mm, frame, 4 mm frame, 0.1–0.5 mm gap, 4/5 mm rivets, 2.5× overlaps)	**R, T**	PASS: every dimension within stated tolerance. FAIL: any out of tolerance.	Dimensional inspection report (measured vs tolerance).	Open
DV-I02	Verify D-sub bracket placement, bulkhead feed-through, material, BNC connector, D-SUB 15 male.	FM structure	Flight interface, callipers, material certificate, connector check.	Planned by end of September	D.1 (bracket 20 mm below radax), D.2 (4× D-SUB 15 feed-through), D.15 (aluminum), D.23 (BNC), D.28 (D-SUB 15 male)	**R, T**	PASS: bracket position within tolerance; correct connectors present & keyed; material confirmed. FAIL: any mismatch.	Interface / connector inspection report.	Open
DV-I03	Verify insertion and fixation of PV inside the EM, and solidity of bolting PV in place	PV structure	Materia	Planned by end of September	OP.4, OP.5	**T**	PASS: PV inserts with no resistance. FAIL: PV inserts irregularly	Displacement inspection	Open
3b. Fluid Safety									

Test ID	Test Objective	Model	Setup	Date / Timing	Requirements Verified	Method	Pass / Fail Criteria	Expected Output	Status
DV-T01	Verify 2:1 pressure safety factor and absorbent-material capacity (20× fluid volume).	EM fluid rig	Pressurization rig to 2× MEOP; absorbent sample + measured fluid load.	Open, planned at final integration mid-october	D.16 (absorbent ≥2 mL), D.17 (20× capacity), D.18 (2:1 safety factor)	**A, T**	PASS: withstands 2× working pressure w/o failure; absorbent retains ≥20× fluid volume. FAIL: leak/rupture or under-capacity.	Pressure-safety + absorbent-capacity test report.	Open
3c. Structural / Mechanical									
DV-A01	FEM analysis for 20 g quasi-static + REXUS User Manual vibration qualification profile; CG position verified from mass budget.	CAD / FE model	Finite-element model with flight materials & masses; REXUS load cases.	December at ZARM ITW	D.19 (20 g, 3 axes), D.20 (REXUS User Manual vibration profile), D.21 (CG on axis)	**A, T**	PASS: positive margins under 20 g + REXUS profile; CG on axis within tolerance. FAIL: negative margin or CG out of tolerance.	FE analysis report (stress margins, modal, CG).	Open
DV-T02	Shake-table test to REXUS User Manual qualification profile; post-test inspection for structural integrity. Lateral displacement of the isolated assembly shall remain below 2.3 mm under the nominal test profile; no uncontrolled impact, permanent deformation, loosening or damage shall occur if an end-stop is contacted.	STM / FM	Shaker table with REXUS profile; accelerometers + displacement measurement.	December at ZARM ITW	D.20 (vibration test, physical – REXUS profile), cross-links D.19	**A, T**	PASS: survives REXUS qualification profile; lateral displacement < 2.3 mm; no deformation/loosening/damage. FAIL: any exceedance.	Vibration test report (response, displacement, post-test inspection).	Open
3d. Thermal									
DV-T03	Single thermal-chamber run with thermocouples at outer wall, feed-through cable, inter-module face and air outlet; four acceptance limits from one dataset.	EM / FM	Thermal chamber, powered payload, 4× thermocouples + DAQ.	December at ZARM ITW	D.24 (outer wall ≤ Tamb+10 °C), D.25 (feed-through ≤70 °C), D.26 (inter-module ≤50 °C), D.27 (air ≤ Tamb+10 °C)	**R, T**	PASS: all four locations within stated limits. FAIL: any limit exceeded.	Thermal dataset at 4 locations + margin assessment.	Open
3e. Electrical / Harness									
DV-I03	Verify twisted pairs on power and data cables; verify optocoupler isolation on control lines (functional check, not presence only).	FM harness	Flight harness; visual inspection + continuity/isolation test set.	Ongoing	D.37 (optocouplers), D.39 (power cables twisted), D.40 (data cables twisted)	**I, T**	PASS: twisted pairs confirmed; optocoupler galvanic isolation verified functionally. FAIL: missing twist or isolation failure.	Harness inspection + continuity/isolation record.	Open
3f. Power									

Test ID	Test Objective	Model	Setup	Date / Timing	Requirements Verified	Method	Pass / Fail Criteria	Expected Output	Status
DV-T04	Powered bench test: measure peak current, mean current and 28 V line ripple in one run.	EM power	Power bench at 28 V, electronic load, current probe + oscilloscope.	January at bench test DLR/OPH	D.33 (voltage steps), D.34 (peak ≤3 A), D.35 (mean ≤1 A), D.36 (ripple ≤100 mVpp), D.43 (emissions ≤100 mVss on 28 V)	**R, T**	PASS: peak ≤3 A, mean ≤1 A, ripple ≤100 mVpp, emissions ≤100 mVss, voltage steps correct. FAIL: any limit exceeded.	Power dataset (peak/mean/ripple) + emissions plot.	Open
3g. Communications / EMC									
DV-T05	Single EMC session: spectrum analyzer for GPS + telecommand bands; current probe for power-line noise; link format & baud verified by loopback.	FM avionics	EMC bench, spectrum analyzer, current probe, loopback link.	January at bench test DLR/OPH	D.29 (downlink baud ≤80%), D.30 (8N1), D.31 (uplink baud ≤80%), D.32 (8N1), D.38 (radio silence), D.41 (GPS ≤−115 dBm), D.42 (TC ≤−100 dBm)	**A, T**	PASS: GPS ≤−115 dBm, TC ≤−100 dBm, baud ≤80%, 8N1, radio silence held. FAIL: any breach.	EMC scan records + link-format / baud log.	Open
4. Operations Verification (OV)									
OV-T01	Simulate full flight timeline (radio silence, LO, SECO, µg phase) with experiment running autonomously from on-board sequencer; no ground intervention.	FM avionics + SW	Integrated payload, simulated service signals, full flight sequence.	Planned shortly before EAR	O.2 (autonomous operation)	**R, T**	PASS: full timeline executes autonomously, all events triggered, no ground intervention. FAIL: any missed event or required intervention.	Timeline execution log / sequencer trace.	Open
OV-T02	Biological samples subjected to representative hypergravity profile; viability assessed post-test. NOTE: centrifuge test method to be formally added to SED.	Biological samples	Centrifuge, representative hypergravity profile, post-test viability assay.	Planned	Hypergravity viability requirement – ID to be created in Ch.2 (currently unnamed)	**T**	PASS: sample viability remains above defined threshold after hypergravity. FAIL: viability below threshold.	Viability assessment report.	Open
5. Biology verification (BV)									
BV-T01	Verify that *D. discoideum* AX2 performs phagocytosis of FITC-labelled *E. coli* under the nominal assay conditions, establishing the biological proof of concept.	Lab	FABLAB Paris-Saclay: axenic AX2 culture, FITC-labelled target, fluorescence microscope, matched 1 g control.	Open	F.2 (mixing), F.3 (video recording)	R, T	PASS: measurable phagocytic uptake vs. negative control; phagosome acidification detectable in fluorescence	Phagocytosis assay report + representative image set	Open
BV-T02	Determine the maximum pre-launch holding time over which AX2 cultures retain viability and motility in the flight-representative configuration.	Lab (flight-representative cartridge)	Loaded cartridge held at representative temperature and pressure; viability and motility assessed at defined intervals.	Open	OP.3, P.21 (18–25 °C)	T	PASS: viability and motility remain within defined acceptance thresholds across the target holding duration.	Holding-time curve; recommended maximum holding time (h).	Open

BV-T04	Verify that fluorescence intensity remains stable under nominal imaging conditions over the full acquisition duration (photobleaching assessment).	Lab + flight optics	Flight-representative epifluorescence path, 560 nm ±5 nm excitation, time-lapse at flight acquisition parameters.	TBD - with optics integration	P.9 (560 nm ±5 nm), P.2 (180 s alignment)	T	PASS: signal decay below the defined threshold over the ~3–5 min acquisition; no artefact mistakable for a biological event.	Photobleaching curve + acceptance statement.	Open
BV-T07	Determine the maximum acceptable sample holding time after cartridge loading and PV integration, under representative pre-launch conditions, to size the scrub / delay strategy.	Lab (flight cartridge, integrated)	Loaded and integrated PV held under representative launcher conditions; endpoint viability, motility and phagocytic capacity.	TBD - before campaign	OP.1, OP.3	T	PASS: defined maximum holding time established with margin covering the launch window.	Maximum post-loading holding time (h); closes the TBD in Table 6.11.	Not started

5.3 Verification Results

Table 5.3 - Example of a test result overview

Verification number	*DV-T01*
Type of test	*Pressure Test*
Facility	*University workshop*
Verified item	*D.18*
Verification description	*We need to verify the leak rate during the flight time in the vacuum (5mins)*
Expected results	*0.02bar/min*
Obtained Results	*0.06bar/min*
Conclusions	Due to damaged seals the target was not meant, new seals were order to re pass it.

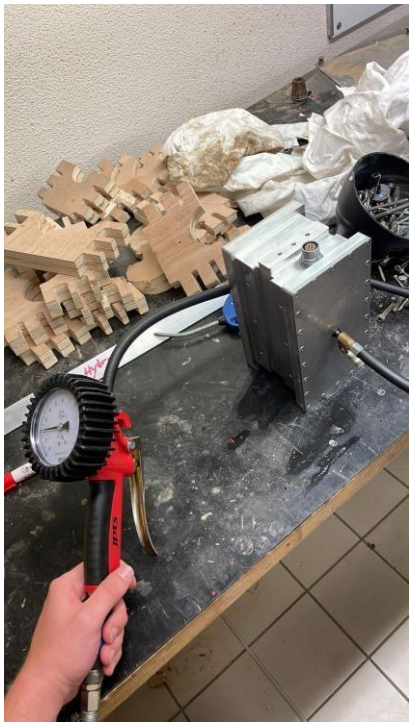
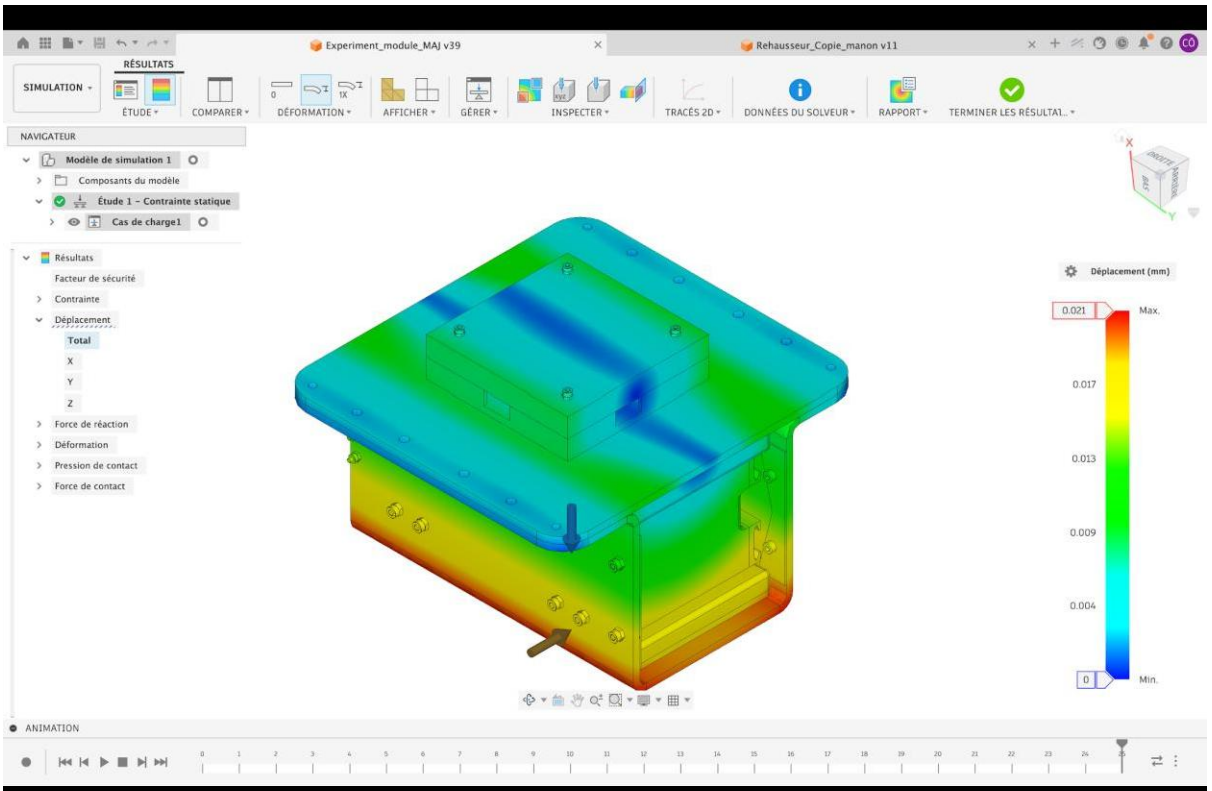
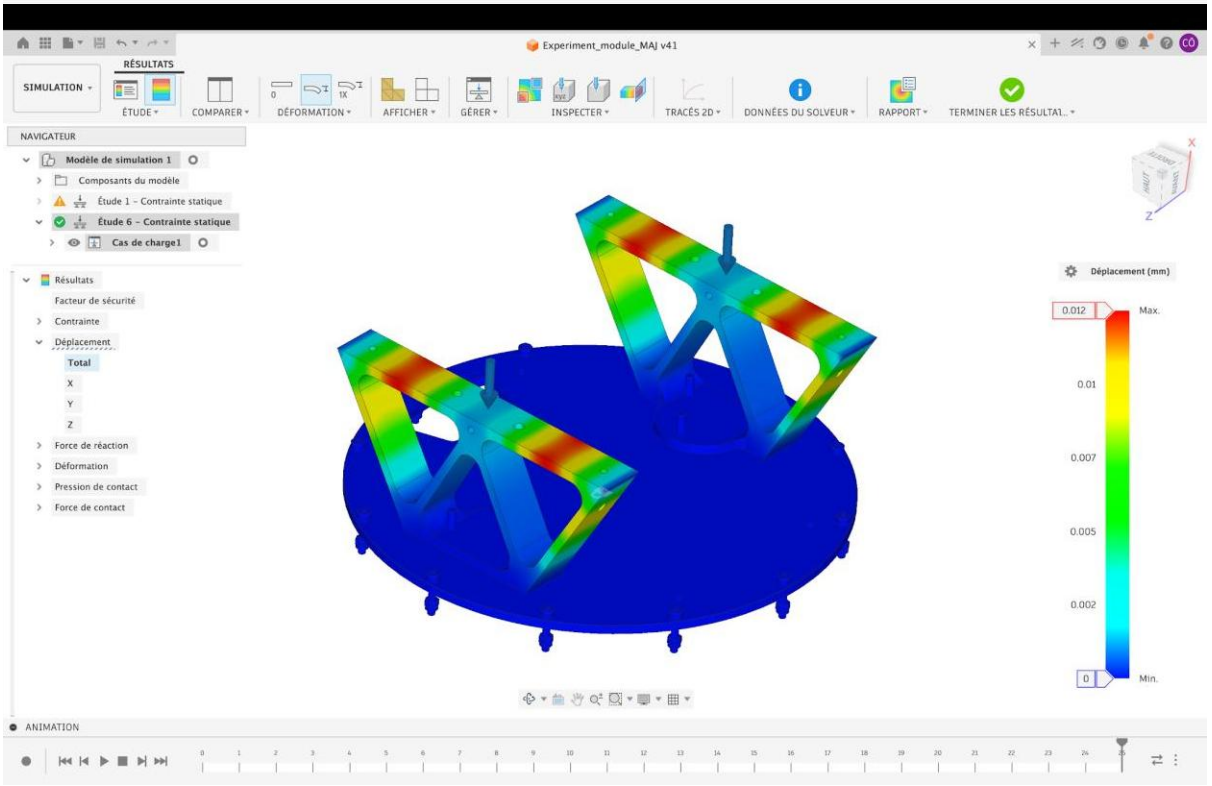
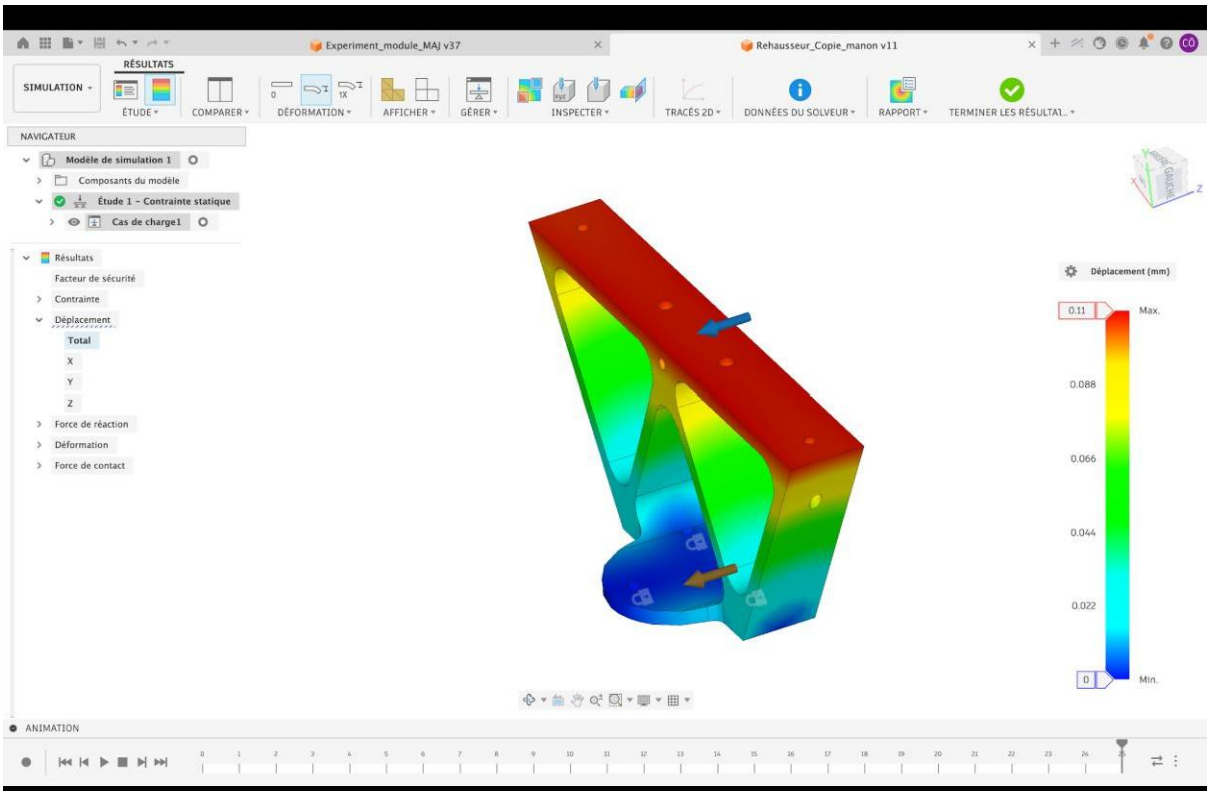
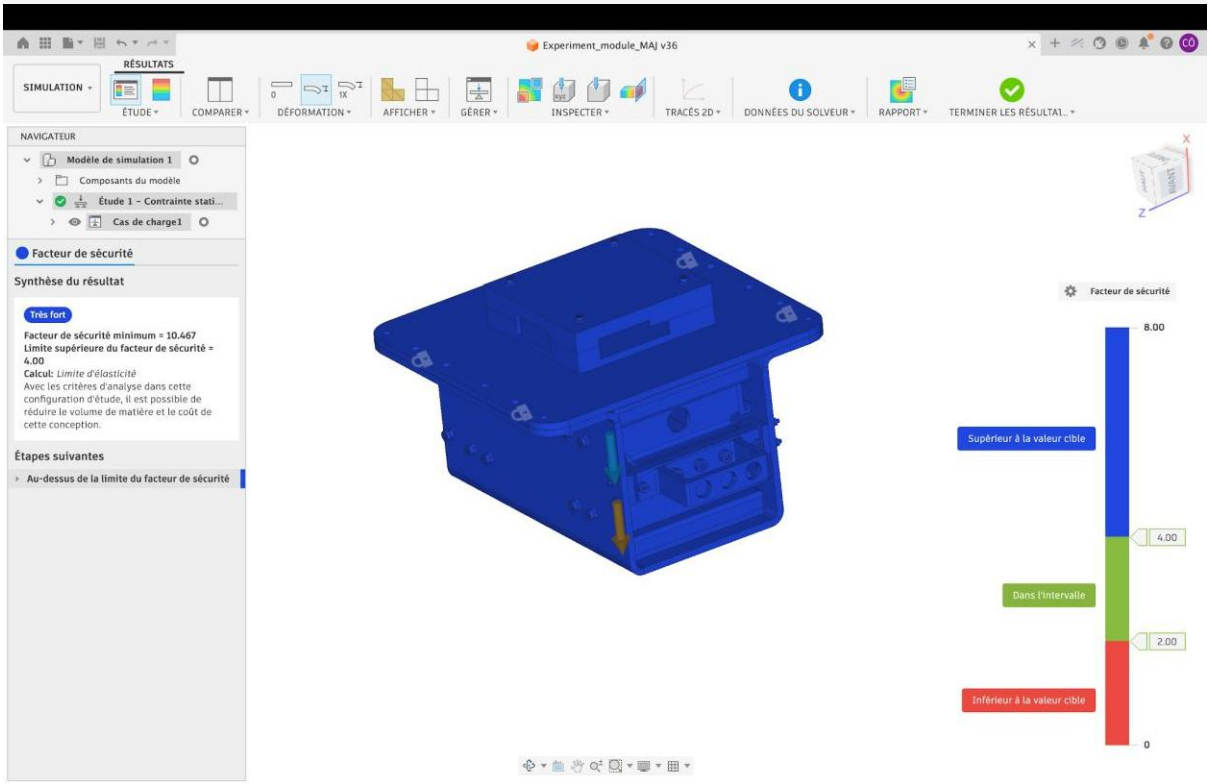


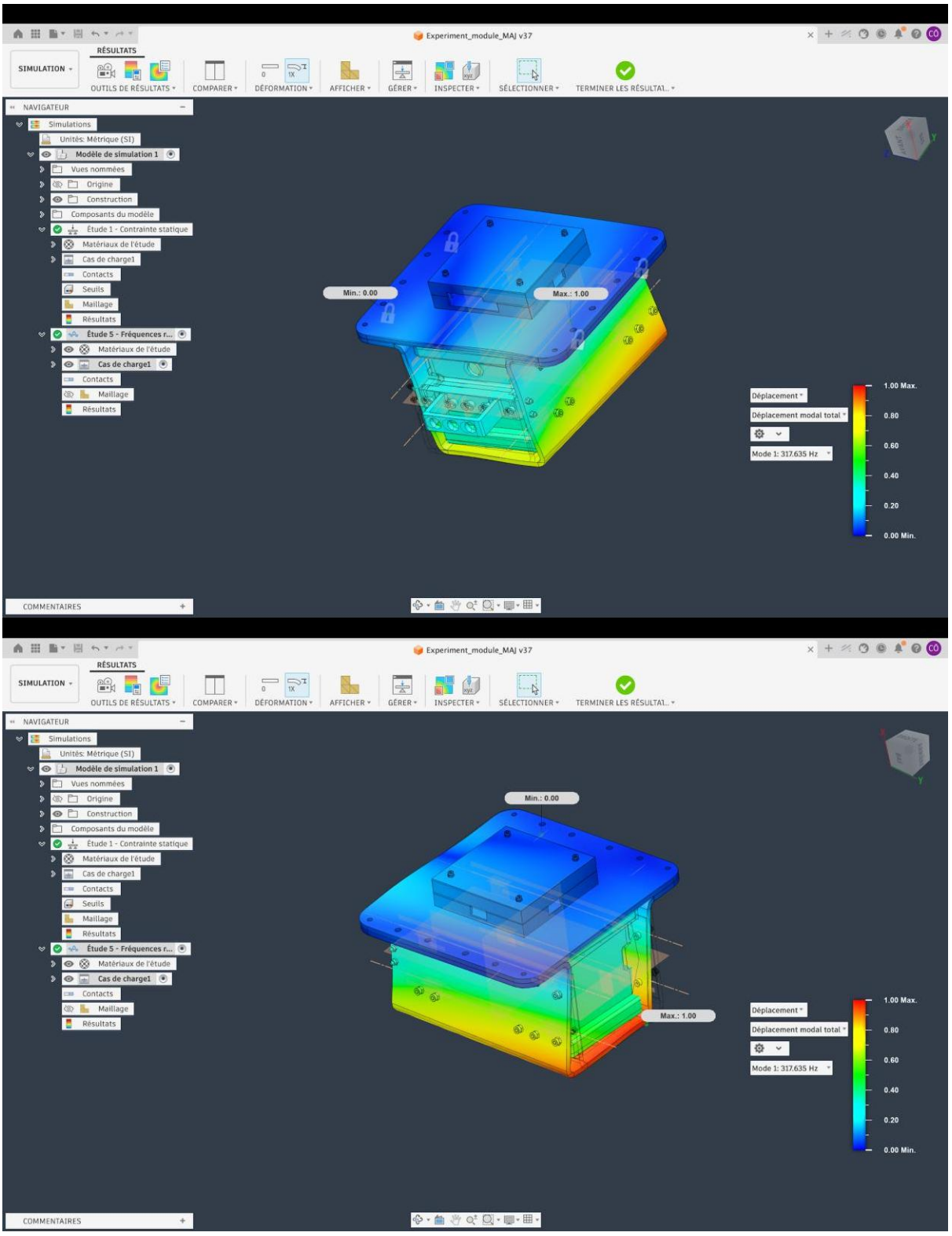
Figure 66: pressure testing


Experiment_module_MAJ v39
Rehausseur_Copie_manon v11
SIMULATION
RÉSULTATS
ÉTUDE
COMPARER
DÉFORMATION
AFFICHER
GÉRER
INSPECTER
TRACÉS 2D
DONNÉES DU SOLVEUR
RAPPORT
TERMINER LES RÉSULTA1...
NAVIGATEUR
Modèle de simulation 1
Composants du modèle
Étude 1 – Contrainte statique
Cas de charge1
Résultats
Facteur de sécurité
Contrainte
Déplacement
Total
X
Y
Z
Force de réaction
Déformation
Pression de contact
Force de contact
Déplacement (mm)
0.021
Max.
0.017
0.013
0.009
0.004
0
Min.
ANIMATION


Experiment_module_MAJ v41
SIMULATION
RÉSULTATS
ÉTUDE
COMPARER
DÉFORMATION
AFFICHER
GÉRER
INSPECTER
TRACÉS 2D
DONNÉES DU SOLVEUR
RAPPORT
TERMINER LES RÉSULTA1...
NAVIGATEUR
Modèle de simulation 1
Composants du modèle
Étude 1 – Contrainte statique
Étude 6 – Contrainte statique
Cas de charge1
Résultats
Facteur de sécurité
Contrainte
Déplacement
Total
X
Y
Z
Force de réaction
Déformation
Pression de contact
Force de contact
Déplacement (mm)
0.012
Max.
0.01
0.007
0.005
0.002
0
Min.
ANIMATION


Experiment_module_MAJ v37
Rehausseur_Copie_manon v11
SIMULATION
RÉSULTATS
ÉTUDE
COMPARER
DÉFORMATION
AFFICHER
GÉRER
INSPECTER
TRACÉS 2D
DONNÉES DU SOLVEUR
RAPPORT
TERMINER LES RÉSULTAT...
NAVIGATEUR
Modèle de simulation 1
Composants du modèle
Étude 1 – Contrainte statique
Cas de charge1
Résultats
Facteur de sécurité
Contrainte
Déplacement
Total
X
Y
Z
Force de réaction
Déformation
Force de contact
Déplacement (mm)
0.11
Max.
0.088
0.066
0.044
0.022
0
Min.
ANIMATION


Experiment_module_MAJ v36
SIMULATION
RÉSULTATS
ÉTUDE
COMPARER
DÉFORMATION
AFFICHER
GÉRER
INSPECTER
TRACÉS 2D
DONNÉES DU SOLVEUR
RAPPORT
TERMINER LES RÉSULTAT...
NAVIGATEUR
Modèle de simulation 1
Composants du modèle
Étude 1 – Contrainte stati...
Cas de charge1
Facteur de sécurité
Synthèse du résultat
Très fort
Facteur de sécurité minimum = 10.467
Limite supérieure du facteur de sécurité = 4.00
Calcul: Limite d'élasticité
Avec les critères d'analyse dans cette configuration d'étude, il est possible de réduire le volume de matière et le coût de cette conception.
Étapes suivantes
Au-dessus de la limite du facteur de sécurité
Facteur de sécurité
8.00
Supérieur à la valeur cible
4.00
Dans l'intervalle
2.00
Inférieur à la valeur cible
0



Figures 67: Mechanical resistance analysis on various module components

6 LAUNCH CAMPAIGN PREPARATION

6.1 Input for the Campaign / Flight Requirement Plans

6.1.1 Dimensions and Mass

Table 6.1 - Experiment Mass and Volume

Experiment mass (in kg):	Estimated 12.235kg REXUS module included)
Experiment dimensions (for BEXUS) or module size (for REXUS) (in m):	0.220m
Experiment expected COG (center of gravity) position:	X: 90.457mm Y: 0.001mm Z: -5.019mm

The mass stays constant during the three flight stages, liftoff, de-spin and re-entry.

6.1.2 Ground and Flight Safety Risks

Table 6.2 - Ground and flight safety risks

Risk	Key Characteristics	Mitigation
Electrical heater	FIT0845 PTC heating element, 5 V, 0.4 Amp 60 degrees Celsius max	Heater inaccessible to personnel once the pressure vessel is closed.
Pressure vessel	Self-designed to operate at 1bar differential pressure at apogee	The pressure vessel was designed with a security factor of 2

6.1.3 Electrical Interfaces

Table 6.3 - Electrical interfaces applicable to REXUS

REXUS Electrical Interfaces		
Service module interface required? Yes/No		
	Number of service module interfaces:	**1**
	TV channel required? If yes, when is it required:	**NO**
Up-/Downlink (RS-422) required? Yes/No		
	Data rate – downlink (max and average): **YES**	**0.320 kbit/s (32-byte packets with 8-byte margin)**
	Data rate – uplink (max and average): **YES**	**0.320** kbit/s
Power system: Service module power required? **YES**		
	Peak power and current consumption:	**40W, 1.7A @24 V**
	Average power and current consumption:	**26W, 0.9A @28 V**
	Total power and current consumption after lift-off (until T+600s) :	**4.3Wh, 0.16Ah**
	Power ON :	**600s before lift-off**
	Power OFF :	**600s after lift-off**
	Battery recharging through service module:	**NO**
Experiment signals: Signals from service module required?		
	LO: **YES**	
	SOE: **YES**	**T+66s**
	SODS: **YES**	**T-180s**
	Pyro(s):	**NO**

6.1.4 Launch Site Requirements

Although the biological validation campaign is still required to determine precisely which biological materials and reagents must be transported to Esrange, as well as the quantities required to cover the complete launch window and potential launch delays, the laboratory infrastructure and general equipment required to perform the biological operations can already be estimated. Based on the current experimental workflow, the following preliminary list identifies the main laboratory equipment, facilities and general consumables that should be available at Esrange during the campaign. This list will be progressively refined as the biological validation tests are completed, but it already provides a basis for coordinating laboratory requirements with SSC and identifying equipment that should be available before the arrival of the STAR-D biology team.

Table 6.4 - Launch Site Requirements

Category	Item	Amount
Biology - Facilities	Access to biolab	1
	Clean bench or equivalent clean workspace for biological sample and cartridge handling	1
	Refrigerated storage (compatible with validated biological storage conditions)	1
	Secure storage area for biological hardware and material prior to integration	1
	Micropipette P20	1
Biology - Equipment	Micropipette P200	1
	Micropipette P1000	1
	Centrifuge compatible with 15 mL and 50 mL tubes	1
	Microscope for cell condition and viability checks	1
	Hemocytometer / Neubauer counting chamber	1 + 1 backup if available
	Insulated cool box	1-2
Biology - Consumable	Sterile pipette tips compatible with P20 / P200 / P1000	≥1 box per size

s	Sterile 15 mL tubes	≥50
	Sterile 50 mL tubes	≥50
	Sterile 1.5-2 mL microtubes	≥100
	Disposable laboratory gloves	2-3 boxes
	Laboratory wipes / absorbent material	2 packs
	70% ethanol or equivalent surface disinfectant	≥1 L
	Biological waste containers / bags	Sufficient for complete campaign
	Milli-Q or distilled water	Several litres

6.1.5 Flight Requirements

Table 6.5 - Flight Requirements applicable to REXUS

De-spun/spinning	**Yes**
Weather conditions during countdown	**None**
Module temperature during countdown	**18 to 25 degrees**
Launch window	**None**

Table 6.6 - Flight Requirements on timeline events applicable to REXUS

Yo-yo de-spin	**Yes, required for stability and microscope imaging**
Motor separation	**None**
Nosecone ejection	**None**

6.1.6 Accommodation Requirements

The closest to the center of gravity of the rocket the better to reduce the

acceleration forces caused by the tumbling rocket during the microgravity phase.

6.2 Preparation and Test Activities at Esrange

Table 6.7 - List of planned activities

Time/Day	Main Task	Description	Responsible	Duration [h:m]	Comments
Arrival day	Biological hardware check	Visual inspection of biological microfluidic cartridge; verification of labels, seals, and absence of visible damage or leaks	Biology + System team	0:30	
Arrival day	Storage verification	Verification that biological cartridges are stored at 4 °C prior to integration	Biology team	0:15	
Integration day	Pre-integration handling	Transfer of cartridge from storage to integration area; confirma	Biology + System team	0:30	

		tion of acceptable temperature range (18–25 °C)			
Integration day	Pre-launch holding time monitoring	Tracking elapsed standby time between biological loading and experiment activation to ensure it remains within validated limits	Biology team	Ongoing	
Integration day	System-level compatibility check	Coordination with system tests to ensure biological constraints (temperature, timing) are respected	Biology + System team	0:30	

Prior to Launch (H-1h40)	**Integration of the experiment**	**First, the Early Access Hatch is removed by unscrewing**	**Mechanical Team**	**0:10**	

		the fastening bolts extracting the hatch from the module structure. **Once access is available, the Pressure Vessel is inserted into the Experiment Support Frame by engaging it with the guiding rails. The PV is then slid along the rails until it reaches the end-stop, ensuring correct positioning and alignment.** **After positioning, the PV is secured by fastening screws located on the PV handle are then tightened to ensure**			

		full			

		constraint of the PV. **The electrical connection is established by connecting the interface cable to the Pressure Vessel. The connector is then secured using its locking mechanism to prevent disconnection during flight.** **Finally, the Early Access Hatch is reinstalled and fastened using the original bolts.**			
Optional	**Functional check with surrogate fluids**	**Coordination with system tests to ensure biological constraints (tempera**	**System team**	**0:45**	

		ture, timing) are respecte d			

6.3 Timeline for Countdown and Flight

Table 6.8 - Example of timeline of the experiment events

Time [s]	EVENT	Function
T - 600s	POWER ON	Start electronics
T - 1h40	Early access	Placing the PV inside the module
T - 180s	SODS ON	Begin data logging and imaging
T + 0s	LO	Start internal clock
T + 66s	SOE ON	Bioparticle injection to trigger the start of the biological reaction
T + 540s	SOE OFF, SODS OFF	Stop data logging
T + 600s	POWER OFF	Power cutoff

Sample Preparation and Operational Timeline

1) Operations at Launch Site

Dictyostelium discoideum AX2 cells will be cultured and expanded in advance in a dedicated laboratory facility under controlled conditions.

Cells will be maintained in HL5 medium in exponential growth phase to ensure optimal physiological state. Cell density will be controlled below 4
× 10^6 cells·mL^{-1} to prevent the onset of development and preserve a homogeneous vegetative population [Fey et al., 2007].

Prior to transport to the launch site, cells will be prepared and maintained under controlled conditions to preserve viability and physiological stability.

Final biological sample preparation and loading operations will be conducted at the launch site under controlled laboratory conditions.

Upon arrival, cells will be harvested and resuspended in KK2 buffer to suppress proliferation while preserving motility and phagocytic capability. Starvation conditions are known to maintain *Dictyostelium* in a motile, chemotactically responsive state while preventing uncontrolled growth [Fey et al., 2007; Martín-González et al., 2020].

Transport conditions (temperature and duration) will be controlled to ensure maintenance of cell viability and physiological state.

At the time of writing, the biological validation campaign is ongoing. While the overall preparation workflow has already been established, several operational parameters cannot yet be defined with sufficient accuracy from the literature alone. In particular, the validation tests will determine the maximum acceptable holding time between sample preparation and launch, the duration over which *Dictyostelium* remains viable, motile and phagocytically active after loading, the stability of the FITC fluorescence signal, and the optimal timing for sample preparation and cartridge integration. Until these quantitative limits are experimentally established, the operational sequence presented below should be regarded as a preliminary workflow based on current biological knowledge and expected launch operations. The final timeline will be updated using the results of the biological validation campaign.

A focus position verification step will occur after the microfluidic assembly is sealed into the pressure vessel. The electronic team will connect to the camera to verify the current focus position and store it in memory. The microscope will then be put at the maximum travel length of the focus axis. The microscope will stay in this position with the additional mechanical connection during launch and will return to its stored focus position after the propulsive phase.

Table 6.9 describes the sequence, detailed actions and responsibilities

associated with biological operations, while Table 6.11 provides the corresponding preliminary time estimates. Both tables should therefore be considered together when planning biological operations during the launch campaign

Table 6.9 - Preliminary Biological Operational Timeline for the Launch Campaign

Time	Operation	Responsible	Operational Verification Checklist
Weeks before launch	Dicty culture	Biology Team	☐ Culture visually healthy ☐ No visible contamination ☐ Cell density within validated range ☐ Exponential growth maintained ☐ Backup culture available

Weeks before launch	E. coli preparation	Biology Team	☐ Correct bacterial material used ☐ Heat-inactivation completed ☐ FITC labelling completed ☐ Fluorescence detectable ☐ No excessive aggregation ☐ Backup batch available
Days before launch	Biological validation	Biology Lead / Biology Team	☐ AX2 morphology acceptable ☐ Motility acceptable ☐ Phagocytic activity confirmed ☐ FITC fluorescence detectable ☐ No abnormal bacterial aggregation ☐ Validation results recorded

Launch campaign	Culture renewal	Biology Team	☐ Nominal culture healthy ☐ Cell density acceptable ☐ Fresh HL5 available ☐ Backup culture healthy and stored separately ☐ Culture status recorded
Launch campaign	FITC-labelled target preparation	Biology Team	☐ Validated target batch selected ☐ Suspension homogenized ☐ Fluorescence detectable ☐ No excessive aggregation ☐ Protected from unnecessary light exposure

Launch day	Cell harvesting	Biology Team	☐ Healthy culture selected ☐ Required growth phase confirmed ☐ Required volume collected ☐ Sterile equipment used ☐ Sample identification recorded
Launch day	Washing and KK2 transfer	Biology Team	☐ HL5 removed ☐ Washing completed ☐ Cells resuspended in KK2 ☐ Suspension homogeneous ☐ No obvious contamination

Launch day	Cell counting and concentration adjustment	Biology Team	☐ Cell count performed ☐ Concentration calculated ☐ Concentration adjusted to validated target ☐ Final concentration recorded
Launch day	Observation chamber loading	Biology Lead / Biology Team	☐ Correct sample loaded ☐ Required volume loaded ☐ Chamber correctly filled ☐ No visible air bubbles ☐ No visible leakage ☐ No obvious contamination

Launch day	Cell adhesion	Biology Team	☐ Adhesion period completed ☐ Environmental conditions maintained ☐ Sufficient cells attached ☐ Cell morphology acceptable ☐ Adhesion completion time recorded
Launch day	FITC target reservoir preparation and filling	Biology Lead / Biology Team	☐ Suspension homogenized ☐ Required volume loaded ☐ Correct reservoir used ☐ Fluidic line filled ☐ No visible bubbles ☐ No leakage

Launch day	Cartridge sealing and final biological inspection	Biology Lead / Fluidics or Mechanical Lead	☐ Cartridge correctly closed ☐ Seals checked ☐ Connections secured ☐ No visible leakage ☐ No unacceptable bubbles ☐ Biological condition acceptable ☐ Preparation/loading time recorded
Launch day	Microscope focus measurement	Electronics lead	☐ Microscope is in focus position ☐ Focus position is stored ☐ Microscope focus part set to its lowest position

Early access	PV integration	Biology Lead / Mechanical Lead	☐ Correct cartridge/PV configuration confirmed ☐ PV correctly inserted on rails ☐ Locking bracket secured ☐ Connections verified ☐ Mechanical fixation verified ☐ Integration completed within allowed time ☐ Biological holding time not exceeded
Flight	Automatic bacterial injection	On-board system (automatic)	☐ Injection command/event recorded ☐ Pump activation recorded ☐ Flow/pressure data recorded ☐ Injected volume consistent with requirement ☐ No system error reported

Flight	Microscopy and image acquisition	On-board system (automatic)	☐ Acquisition started ☐ Brightfield images recorded ☐ Fluorescence images recorded ☐ Required acquisition duration achieved ☐ Image/data files stored ☐ No critical imaging error reported

Recovery	Post-flight inspection	Biology Lead / Mechanical Lead	☐ PV condition inspected ☐ Cartridge condition inspected ☐ No leakage/loss of containment ☐ Mechanical damage documented ☐ Biological material safely handled ☐ Data retrieved ☐ Sample disposition completed

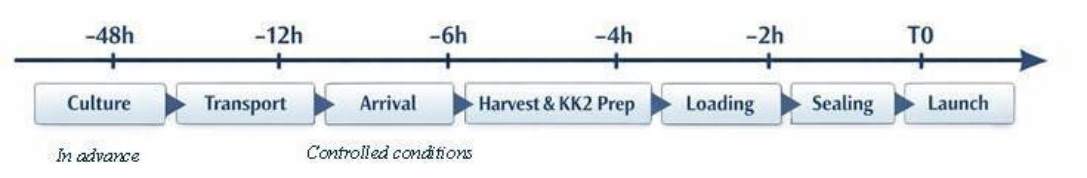


Figure 68: Estimated Operational Timeline

An additional operational challenge arises from the launch campaign itself, as the exact launch day within the launch window cannot be predicted in advance. Depending on weather conditions, technical constraints, or range operations, the experiment may be launched on any day within this predefined period. This uncertainty has important implications for the biological subsystem, particularly regarding sample storage, culture renewal, reagent availability, and the overall organization of laboratory activities at the launch site. Fresh, flight-compatible biological samples must remain available throughout the entire launch window while ensuring that cell viability, motility, and phagocytic activity remain within validated limits. Dedicated operational procedures are therefore currently under development to manage these different launch scenarios.

The ongoing biological validation campaign will provide the quantitative data required to define the maximum acceptable holding time, storage conditions, sample renewal strategy, and preparation schedule. More specifically, the validation campaign is intended to answer several key operational questions that cannot be reliably addressed from the literature alone:

- What quantity of biological material (*Dictyostelium* cultures, FITC-labelled E. coli, buffers and consumables) should be brought to the launch site?

- Which biological reagents and consumables require redundancy throughout the launch window?

- How should the on-site culture workflow be organized to ensure that a flight-compatible culture is always available, regardless of the launch day?

- When should cultures be renewed or replaced to remain within the validated biological activity window?

- What backup strategy should be implemented in case of contamination, unexpected launch delays, or biological failure?

- What is the maximum acceptable delay between sample preparation, cartridge loading, and launch while maintaining full biological functionality?

Once these experimental data have been obtained, a dedicated operational protocol and detailed biological timeline for the launch campaign will be established. This document will describe the complete on-site workflow, from culture maintenance and sample preparation to cartridge loading, backup management, and contingency procedures, ensuring that the biological subsystem remains flight-ready throughout the entire launch window.

Table 6.10 presents a preliminary estimate of the biological material expected to be transported to Esrange during the launch campaign. As the biological validation campaign is still ongoing, the quantities and number of backup samples remain provisional and will be refined based on the experimentally determined culture viability, holding time, storage conditions, and launch campaign constraints. All quantities are preliminary estimates and will be updated following the biological validation campaign. This list should not be considered as the final Esrange logistics manifest. It represents a preliminary operational estimate established prior to completion of the biological validation campaign. Final quantities, storage strategy, redundancy level, and renewal schedule will be defined once the experimental characterization of culture viability, storage duration, and launch campaign constraints has been completed.

Table 6.10 - Preliminary Biological Material and Consumables Required for the Launch Campaign

Category	Item	Purpose	Preliminary Quantity*	Remarks
Living cultures	*Dictyostelium discoideum* AX2 cultures	Biological model for the experiment	Several independent cultures	Enables culture renewal and backup during the launch window
	Fresh Dicty cultures	Flight samples	To be determined	Prepared during the launch campaign
Bacterial targets	*E. coli* K12 culture	Preparation of fluorescent targets	1 master culture + backups	Used to produce fresh labelled bacteria
	FITC-labelled *E. coli* suspensions	Phagocytosis targets	Several independent batches	Freshly prepared before loading

Culture media	HL5 medium	Dicty culture and expansion	Sufficient for complete campaign	Quantity to be determined experimentally
	KK2 buffer	Washing and observation medium	Sufficient for complete campaign	Includes reserve volume
Consumables	Sterile Falcon tubes	Cell preparation	Several dozen	Includes contingency stock
	Sterile microcentrifuge tubes	Sample preparation	Several dozen	Backup included
	Pipette tips	Liquid handling	Large excess	Standard laboratory consumable
	Syringes	Loading microfluidic system	Multiple sterile units	Spare units included
	Sterile filters (if required)	Solution preparation	Backup stock	If needed during campaign
	Hemocytometer / counting chamber	Cell concentration adjustment	1–2	Reusable
Microfluidics	Loaded microfluidic chips	Flight hardware	Flight unit + spare units	Spare chips in case of loading failure
	Tubing and connectors	Fluidic assembly	Spare set	Replacement if contamination occurs
Safety & cleaning	70% ethanol	Sterilization	1 bottle	Routine decontamination
	Laboratory wipes	Cleaning	Several packs	Standard laboratory practice

	Disposable gloves	Sterile manipulation	Multiple boxes	Throughout campaign
Storage	Cool boxes / refrigerated storage	Preserve biological material	As required	Temperature maintained according to validation results
Backup material	Frozen Dicty stock	Biological backup	1–2 cryostocks	Only if compatible with campaign logistics
	Additional FITC-labelled bacteria	Backup flight samples	To be determined	Depends on validated holding time

The exact biological operations performed at Esrange will depend on the results of the ongoing validation campaign, particularly the experimentally determined culture growth time, post-loading holding time, storage conditions and phagocytic activity window. Until these parameters are fully characterized, the following organization is proposed as a preliminary operational strategy.

The main objective is to ensure that at least one fresh, flight-compatible *Dictyostelium discoideum* AX2 culture is always available during the launch window, while maintaining independent backup cultures in case of contamination, unexpected launch delays or preparation failure.

Biological Validity Window

Once loaded into the sealed microfluidic cartridge and integrated into the pressure vessel, the biological system must remain functional until experiment activation.

Dictyostelium discoideum cells maintained in non-nutritive buffer conditions remain motile and capable of phagocytosis, as starvation induces a physiological state characterized by active chemotaxis and cytoskeletal dynamics [Fey et al., 2007; Microbe Profile, 2020].

However, starvation also progressively triggers developmental pathways and alters cellular physiology over time [Fey et al., 2007]. As a result, the duration

over which cells maintain optimal experimental functionality is limited and must be experimentally characterized.

Based on literature-established culture constraints and on the known starvation-induced physiology of *Dictyostelium*, the acceptable pre-launch holding time is conservatively assumed to be on the order of several hours and will be experimentally validated.

To ensure robustness, this time window will be experimentally validated through dedicated ground tests (V-BIO-02, V-BIO-03, V-BIO-07), assessing cell viability, motility, and phagocytic activity as a function of pre-launch delay.

Within this time window:

- cell viability remains high,
- actin-driven motility is preserved,
- phagocytic capacity is maintained.

Beyond this period, risks include reduced motility, altered responsiveness, and decreased phagocytosis efficiency.

Therefore, launch operations will be scheduled to minimize the delay between biological loading and experiment execution, ensuring that all measurements are performed within the validated biological activity window.

This acceptable holding time will be experimentally verified through ground validation tests (V-BIO-02, V-BIO-03, V-BIO-07).

Biological Sample Preparation Time and Operational Lifetime

At the time of writing, the biology work package is experiencing delays compared with the initial validation schedule. Several biological materials and reagents required for the experimental campaign had to be ordered from external private suppliers, and longer-than-expected procurement and delivery times have prevented the team from performing the full set of biological validation tests according to the initially planned schedule. Consequently, only a limited part of the experimental validation campaign has been performed so far.

The team is actively addressing this issue. The delayed biological supplies are now being received progressively, allowing the experimental campaign to proceed and biological

testing to accelerate. In parallel, a backup strategy is currently being developed to reduce dependence on individual suppliers and to ensure continuity of the biological validation campaign in case of further procurement difficulties. Alternative sources of biological material, reagents, laboratory support and experimental approaches are being assessed where applicable.

As discussed previously, several operational parameters of the biological subsystem can only be determined reliably through experimental validation. In particular, the exact preparation time of a flight-ready sample, the required cell adhesion time, the maximum acceptable holding time after loading, the stability of the FITC-labelled bacterial targets, and the time required to prepare a replacement sample following a launch scrub cannot yet be considered experimentally validated values.

Nevertheless, preliminary time estimates are required to support launch campaign planning and to evaluate the feasibility of biological operations at Esrange. Based on the current experimental workflow, literature-derived biological constraints, preliminary observations, and discussions with colleagues experienced in biological laboratory procedures, the following indicative time scales are currently considered reasonable for operational planning.

These values are preliminary estimates rather than verified operational requirements. They are intended to provide an initial basis for campaign organization and will be progressively refined and replaced by experimentally validated values as the biological validation campaign progresses, in particular through BV-T02 and BVT07. Thes estimations are shown in table 6.11

Table 6.11 : Preliminary Biological Sample Preparation and Operational Time Estimates

Biological operation	**Preliminary time estimate**	**Current status / rationale**
Maintenance of flight-compatible Dictyostelium discoideum AX2 culture	Continuous during campaign	Fresh nominal and backup cultures maintained throughout the launch window
Cell harvesting	20 min	Preliminary operational estimate - to be experimentally

		validated
Washing and KK2 transfer	15 - 30 min	Includes centrifugation and resuspension steps; to be experimentally validated
Cell counting and concentration adjustment	10 - 20 min	Hemocytometer-based counting and concentration adjustment; to be experimentally validated
Observation chamber loading	10 - 20 min	Dependent on final validated loading procedure
Cell adhesion	30 - 60 min	Preliminary estimate; adhesion duration to be experimentally optimized
FITC target preparation/resuspension	15 - 30 min	Assumes FITC-labelled heat-inactivated E. coli K12 targets are already available
FITC target reservoir filling	10 - 15 min	Preliminary microfluidic handling estimate; to be validated

Cartridge sealing and final biological inspection	15 - 30 min	Includes sealing, bubble/leak inspection and final biological verification
Total final sample preparation on launch day	~1.5 - 3 h targeted	Preliminary operational envelope assuming partial overlap of compatible operations; final duration to be established through representative preparation tests
PV integration	<10 min targeted	Final value to be established through early-access rehearsals
Maximum sample holding time after loading	TBD through BV-T07	Maximum acceptable holding time to be experimentally established under representative pre-launch conditions
Preparation of replacement sample following scrub	~1.5 - 3 h targeted	Assumes a flight-compatible AX2 culture and FITC-labelled bacterial targets are immediately available; scrub procedure to be validated

Preparation of a new flight-compatible AX2 culture from an earlier culture stage	TBD - longer timescale	Depends on validated on-site culture renewal strategy

For the Maintenance of flight-compatible *Dictyostelium discoideum* AX2 culture, cells are maintained in exponential growth in HL5 medium under appropriate temperature conditions. Cultures are regularly monitored and diluted or transferred to fresh HL5 medium to prevent overgrowth and maintain cell density below approximately 4×10^6 cells/mL. Fresh cultures are maintained throughout the launch campaign to ensure that viable and physiologically active cells are continuously available for nominal or backup sample preparation.

6.4 Biological Sample Strategy in Case of Launch Scrub or Relaunch

In the event of a launch delay or scrub, the biological samples cannot be considered usable indefinitely, as prolonged storage in the experimental configuration may affect *Dictyostelium discoideum* viability, adhesion, motility, and phagocytic activity. The maximum acceptable holding time between sample preparation and launch will therefore be experimentally determined during the biological validation campaign, in particular through BV-T07. Until these tests are completed, the exact maximum holding time remains TBD.

If a launch is delayed but remains within the experimentally validated holding time, the loaded biological sample may be kept in the experiment under controlled and validated environmental conditions and used for the rescheduled launch. The time elapsed since sample preparation and loading will be recorded to ensure that the validated operational lifetime is not exceeded.

If the delay exceeds the validated holding time, or if the biological condition of the sample can no longer be guaranteed, the sample will be considered unsuitable for flight and will be replaced. Fresh *Dictyostelium discoideum* AX2 cultures will therefore be maintained throughout the launch campaign in HL5 medium, allowing a new sample to be prepared when required. Backup FITC-labelled heat-inactivated *E. coli* K12 targets will also be maintained in sufficient quantity to prepare a replacement experimental configuration.

Based on the current preliminary operational estimates, preparation of a complete replacement sample is expected to require approximately 1.5 - 3 hours, provided that a flight-compatible *Dictyostelium* culture and FITC-labelled bacterial targets are immediately available. This estimate includes cell harvesting, washing and transfer to KK2, cell counting and concentration adjustment, chamber loading and adhesion, bacterial reservoir preparation, sealing, and final biological and fluidic verification. This duration is preliminary and will be refined through representative preparation and integrated tests.

To increase robustness against launch delays and scrubs, biological preparation will not

rely on a single culture or sample batch. Several independent AX2 cultures will be maintained in parallel throughout the launch campaign, with their maintenance and renewal staggered in time to reduce the risk of all cultures reaching an unsuitable physiological state simultaneously. FITC-labelled bacterial targets will similarly be maintained as separate nominal and backup batches. On launch day, the nominal flight sample will be prepared first, while flight-compatible backup biological material will remain available for rapid preparation of a replacement sample if required. In the event of a scrub, the loaded sample will be retained only if the rescheduled launch remains within its experimentally validated operational lifetime; otherwise, a replacement sample will be prepared from an available backup culture and bacterial target batch. **T**he exact number of parallel cultures and backup batches, their preparation and renewal intervals, and the final staggering strategy will be defined following BV-T07 and representative launch-day preparation tests and incorporated into the final launch operations timeline.

This strategy aims to ensure that a launch delay or scrub does not result in the use of a biologically degraded sample and that a replacement configuration can be provided within a predictable and experimentally validated timeframe.

Biological Material Transport to Esrange

Biological materials and reagents required for the STAR-D experiment will be transported from the team's laboratory facilities in the Paris area to Esrange Space Center before the launch campaign. The transport strategy is currently being finalized and will depend on the specific storage, temperature, safety and regulatory requirements of each biological material.

Two main transport solutions are currently considered. The first option is transportation by STAR-D team members travelling from Paris to Esrange. For biological materials transported by air, the team will contact the relevant airline in advance to confirm acceptance conditions and required documentation. Appropriate biological material datasheets and supporting documentation will be prepared and carried with the samples where required. Requirements related to airport security, customs and border control will also be verified before departure.

The second option, particularly for materials requiring controlled transport conditions, is the use of a specialized courier service. The team will investigate available shipping solutions capable of maintaining the required temperature and handling conditions throughout transportation to Esrange.

Prior to shipment, the biological materials will be classified according to their transport requirements. Particular attention will be given to *Dictyostelium discoideum* AX2 cultures, heat-inactivated *E. coli* K12 material, FITC-labelled bacterial targets, culture media and other biological reagents required during the campaign. Appropriate packaging, containment and temperature conditions will be defined for each category.

Biological materials will be transported in sealed primary containers placed inside leak-

proof secondary containment and a protective insulated transport box. Temperature-sensitive materials will be transported using insulated containers equipped with cold packs or another appropriate cooling system when required. FITC-labelled material will additionally be protected from light. A temperature logger may be included for critical biological samples to verify that acceptable transport conditions have been maintained throughout the journey. The target transport temperature for each material, including AX2 cultures and FITC-labelled bacterial targets, will be defined according to validated storage conditions and supplier recommendations before the campaign.

Whenever possible, redundancy will be included in the transport strategy. Critical biological materials will be divided into independent batches or backup stocks to reduce the risk of a single transport incident compromising the biological campaign. The possibility of transporting critical materials through separate routes or maintaining backup stocks will be assessed as part of the final logistics plan.

A Biological Transport Documentation Package will be prepared before transportation to Esrange. The package is currently expected to include a Biological Material Transport Manifest listing all transported biological materials and reagents; individual Biological Material Information Sheets for the main biological samples, including *Dictyostelium discoideum* AX2 and heat-inactivated/FITC-labelled *E. coli* K12; relevant Safety Data Sheets (SDS) and supplier Technical Data Sheets (TDS); documentation confirming the BSL-1 status and non-pathogenic nature of the organisms where applicable; documentation confirming the heat-inactivated status of the *E. coli* material; and a Packing and Containment Description specifying the primary containers, secondary leak-proof containment, insulated transport box and temperature-control system used.

For temperature-sensitive material, a Transport Temperature Sheet will specify the required transport temperature range, maximum acceptable transport duration and actions to be taken in case of a temperature excursion. Where a temperature logger is used, its record will be retained as part of the transport documentation. The package will also include an Emergency Contact Sheet identifying the responsible STAR-D biology team members, sending laboratory and relevant Esrange contact.

Depending on the selected transport method, additional carrier-specific documentation may be required. For transport as accompanied airline baggage, the team plans to obtain written confirmation from the airline that the biological materials and associated packaging are accepted for carriage and to retain this confirmation in the package. For shipment by specialized courier, the package will include the courier shipment documentation and any required biological-material declaration. Any additional documentation requested for airport security, customs or border control will be added once the final transport route and carrier have been selected.

The envisaged documentation package therefore includes, at minimum: Biological Material Transport Manifest; Biological Material Information Sheets; SDS/TDS where applicable; BSL-1/non-pathogenicity supporting documentation; heat-inactivation documentation for *E. coli*; Packing and Containment Description; Transport Temperature Sheet; Emergency Contact Sheet; and carrier acceptance or shipping documentation. The exact list will be finalized after consultation with the selected airline and/or courier service.

The final transportation procedure, including the selected transport method, packaging configuration, required documentation and temperature conditions, will be established before the campaign once the requirements of the airline and/or courier service have been confirmed.

Table 6.12 : Preliminary Biological Material Transport Plan

Material	Planned transport condition	Documentation / constraints
D. discoideum AX2	TBD - compatible with validated culture/storage conditions	Biological material documentation; airline/courier requirements to be confirmed
Heat-inactivated E. coli K12 / FITC-labelled targets	TBD according to validated storage conditions	Biological material documentation; protect FITC-labelled material from light
HL5 / KK2 media	Appropriate sealed containers; temperature according to product requirements	Transport and volume restrictions to be checked
FITC and biological reagents	According to manufacturer storage requirements	SDS and supplier documentation where applicable
Biological consumables	Ambient unless otherwise specified	Standard baggage/courier requirements

6.4 Post-Flight Activities

- Hardware: When retrieving the EM, the PV is to be taken out of the EM by unbolting the locking mechanism. If the locking mechanism cannot be removed, the EM should be unbolted from the adjacent team modules / RX38 payloads so that work can be done on the individual module without damaging or interacting with other experiments. The PV shall then be opened, and the M.2 data SSD is to be retrieved.
- Biology: The biological samples do not need to be retrieved after the flight since they are only useful during the experiment window. Microscope imaging is what will show experiment results, and not the retrieved biological sample itself. In terms of sample disposal, the fluids from the microfluidic system shall be removed and disposed into the adequate lab bin. The samples do not need to be contained when disposed since they do not pose any high-risk contamination hazards.
- Data: Data shall be extracted from the M.2 data SSD without corrupting files or accidentally mishandling them. The retrieved files will be correlated with their respective timestamps, and the data can then be analyzed.

7 DATA ANALYSIS AND RESULTS

7.1 Data Analysis Plan

The data analysis plan for the STAR-D experiment is designed to remain simple, robust, and compatible with the limited duration and autonomous nature of a sounding rocket mission. The objective of the analysis is to quantify phagocytosis activity under microgravity conditions and to compare it with ground reference experiments performed under identical biological and optical settings.

Nature of the Data

The experiment generates time-resolved image data acquired by an onboard microscope during the microgravity phase. Two image channels are recorded simultaneously:

- Brightfield images, used to observe cell morphology, adhesion, and motility.
- Fluorescence images, used to visualize FITC-labelled heat-inactivated *E. coli* K12 cells and monitor their interaction, engulfment, and intracellular localization within *Dictyostelium discoideum* cells.

Each dataset consists of bacterial target injection and subsequent interaction with *Dictyostelium discoideum* cells.

Definition of a Phagocytosis Event

A phagocytosis event is defined as the internalization of a FITC-labelled *E. coli* K12 target by a *Dictyostelium* cell, identified by the appearance and persistent localization of a fluorescent bacterial signal within the boundaries of the cell over successive image frames.

Because FITC-labelled bacteria are fluorescent both before and after internalization, fluorescence intensity alone cannot be used as an unambiguous indicator of phagocytosis. Identification of ingestion events will therefore rely on the spatial colocalization and persistence of the fluorescent bacterial target within the cell boundaries, supported by the corresponding brightfield images and the temporal evolution of the interaction.

Primary Analysis Metrics

The main quantitative parameters extracted from the data are:

- Number of phagocytosis events per cell over the observation period.

- Total number of ingestion events within the field of view.

- Time delay between bacterial injection and the first confirmed internalization event, providing information on phagocytosis kinetics.

- Fraction of active cells, defined as cells exhibiting at least one ingestion event during the experiment.

These metrics allow direct comparison between microgravity and ground-control experiments.

Image Acquisition Parameters

Image acquisition is performed as a time-resolved sequence during the microgravity phase.

- **Acquisition mode**: time-lapse imaging (brightfield + fluorescence channels)

- **Acquisition frequency**: one image every **1–2 seconds** (adjustable during ground testing)

- **Total acquisition duration**: approximately **2 minutes**, corresponding to the microgravity phase duration

This acquisition strategy is selected to ensure a balance between:

- **Temporal resolution**: sufficient to capture dynamic processes such as particle internalization and fluorescence activation kinetics (typically occurring on the order of seconds to minutes),

- **Data volume constraints**: compatible with onboard storage limitations and processing capabilities,

- **Signal stability**: limiting photobleaching and excessive light exposure during fluorescence imaging.

Fluorescence signal stability is a critical parameter for reliable quantification of phagocytosis dynamics. In particular, excessive illumination can lead to **photobleaching of the fluorophore (pH-sensitive FITC)**, resulting in a progressive decrease of fluorescence intensity over time that is unrelated to biological processes.

To mitigate this effect, the imaging strategy is designed to:

- **Limit excitation light exposure** by using time-lapse acquisition instead of continuous video, reducing cumulative photon dose,
- **Optimize illumination intensity and exposure time** to maintain sufficient signal-to-noise ratio while minimizing fluorophore degradation,
- **Restrict total acquisition duration** to the microgravity phase (~3–5 minutes), thereby limiting long-term signal decay.

In addition, ground-based validation tests (BV-T04) include **photobleaching assessment**, ensuring that fluorescence intensity remains stable under nominal imaging conditions over the expected acquisition time.

Maintaining signal stability is essential to ensure that observed fluorescence variations reflect **true biological events (particle internalization and phagosome acidification)** rather than optical artefacts. However, given the short duration of the microgravity phase (~3–5 minutes), photobleaching is not expected to significantly affect signal interpretation.

Analysis Methodology

Data analysis is performed post-flight using standard image analysis tools (e.g., ImageJ/Fiji or equivalent open-source software). The analysis procedure follows these steps:

1. Identification of *Dictyostelium* cells using brightfield images.
2. Overlay of fluorescence images to detect intracellular fluorescence signals.
3. Manual or semi-automated counting of phagocytosis events.
4. Extraction of timing information from image timestamps.
5. Normalization of results by the number of cells in the field of view.

Given the limited dataset size and the exploratory nature of the experiment, the analysis relies primarily on event counting and temporal comparisons rather than complex statistical modeling.

Ground Control Comparison

All biological parameters (cell density, particle concentration, buffer composition, optical settings) are replicated during ground experiments. This enables direct comparison between microgravity and 1-g conditions, isolating the effect of gravity on phagocytosis behavior.

Data Quality Assessment

Before quantitative analysis, datasets are screened for:

- Image stability (focus, vibration artifacts),
- Absence of air bubbles in the observation area,
- Acceptable signal-to-noise ratio in fluorescence images.

Only datasets meeting predefined quality criteria are included in the final analysis.

Expected Outcomes

The data analysis will determine whether microgravity affects:

- the frequency of phagocytosis events,
- the timing of ingestion and phagosome acidification,
- the proportion of active phagocytosing cells.

This approach ensures that the biological conclusions are directly supported by observable, quantifiable events, while remaining compatible with the constraints of a student-led sounding rocket experiment.

Experiment Recovery

The experiment does not require immediate post-flight recovery for biological preservation, as all relevant scientific data (fluorescence images and time-lapse sequences) are recorded onboard during flight.

Biological samples (*Dictyostelium* AX2 cells and fluorescent *E. coli* bioparticles) are confined within a sealed microfluidic cartridge and are not intended to be reused or analyzed post-flight. Therefore, no time-critical biological constraints are associated with sample retrieval.

Standard REXUS recovery procedures are considered sufficient for this experiment. The payload can be recovered together with the rocket module via

the nominal recovery chain (including helicopter retrieval and ground crew operations), without the need for priority handling.

Post-flight recovery is primarily required for:

- Retrieval of onboard stored data,
- Inspection of hardware integrity,
- Optional verification of containment (no leakage).

The experiment is therefore classified as non-time-critical with respect to recovery operations.

8 ABBREVIATIONS AND REFERENCES

8.1 Abbreviations

AIT	Assembly, Integration and Test
CDR	Critical Design Review
COG	Centre of Gravity
CRC	Cyclic redundancy check (packet checksum)
DLR	Deutsches Zentrum für Luft- und Raumfahrt
EAR	Experiment Acceptance Review
ECTS	European Credit Transfer System
EIT	Electrical Interface Test
EM	Experiment module
ESA	European Space Agency
Esrange	Esrange Space Center
fps	Frames per second
IPR	Integration Progress Review
LO	Lift Off
MA	Microfluidic assembly
MFC	Microfluidic chip
MORABA	Mobile Raketen Basis (DLR, EuroLaunch)
OP	Oberpfaffenhofen, DLR Center
PCB	Printed Circuit Board (electronic card)
PDR	Preliminary Design Review
RS	Radio silence
RXSM	REXUS service module
SED	Student Experiment Documentation
SNSA	Swedish National Space Agency
SODS	Start of Data Storage
SOE	Start of Experiment
SSC	Swedish Space Corporation
T	Time before and after launch noted with + or -
TBD	To be determined
TCU	Telemetry central unit
WBS	Work Breakdown Structure
YUV	Video file format
ZARM	Zentrum für Angewandte Raumfahrttechnologie und Mikrogravitation

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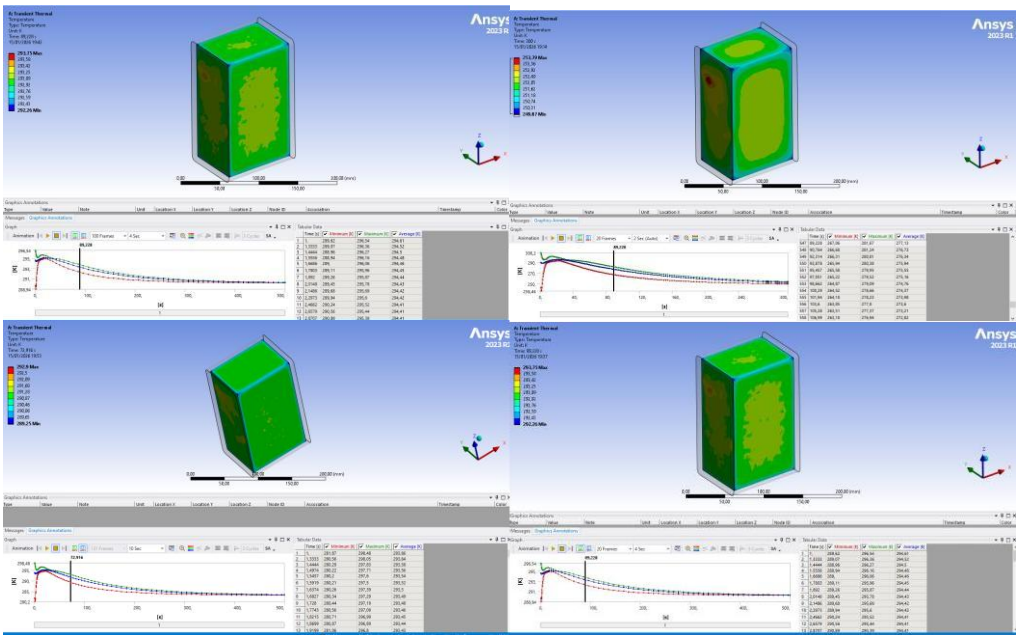
Appendix A - EXPERIMENT REVIEWS

Preliminary Design Review – PDR

Critical Design Review – CDR

Appendix B – OUTREACH AND MEDIA COVERAGE

Appendix C – ADDITIONAL TECHNICAL INFORMATION



Appendix D – CHECKLISTS

Appendix E – LIST OF MATERIAL LENT BY FABLAB UNIVERSITÉ PARIS-SACLAY

Type of equipment	Number of items
QPCR	1
Electrophoresis kit	1
Kit Make: zero to engineering hero	2
Bioplastics kit	2
Genetically modified organisms kit	8

Genetic modification of bacteria via DNA kit	8
Kit to create paint with living organisms	8
Decimal scale	3
DNA playground machine	1
Hemolysis tube support	1
PCR tube support	1
Eppendorf tube support	1
Topless hemolysis tubes	26
13mL tubes with tops	3
2-200 µL pipettes tips	96
pH paper roll	1
1.5 mL microcentrifuge tube	400
Absorption tanks	100
100 µL PCR strip tubes	500
12L Portable Autoclave	1
150 mL beaker	20
250 mL Beaker	20
600 mL Beaker	20

1L Beaker	5
2L Beaker	1
Universal incubator	1
Bacteriological incubator	1
Heated stirrer	1
Foldscope instruments	_
White pouches of Foldscope microorganism accessories	38
Black pouches of Foldscope microorganism accessories	12
Pink pouches of Foldscope microorganism accessories	1
Orange pouches of Foldscope microorganism accessories	3
Purple pouches of Foldscope microorganism accessories	6
Green pouches of Foldscope microorganism accessories	3
Red pouches of Foldscope microorganism accessories	3

Blue pouches of Foldscope microorganism accessories	6
Gray pouches of Foldscope microorganism accessories	19
X pouches of Foldscope microorganism accessories	20
Foldscope microscopic paper assembly and observation kit (complementary to the gray pouches)	97
Foldscope microscopic paper observation kit (complementary to the X pouches)	100
Complementary kit for white microscopic paper observation pouches	207
Individual Foldscope kit	4
Foldscope slides, group A	4
Sterilized needles, 21G x 1 1/2" (0.8 x 40 mm)	85
Sterilized needles, 18G x 1 1/2" 1.2 x 40 mm	87
Sterile 23G x 1 1/2" 0.6 x 25 mm needles	95

Sterile 19G x 1 1/2" x 1.1 x 40 mm needles	95
20 mL syringe without needle	1
Sterile scalpel	2
1 mL syringe without needle	1
10 mL syringe without needle	6
5 mL syringe without needle	1
20 mL syringe without needle	5
50 mL syringe without needle	3
Sterile 18 GA 1.16IN 1.3 x 30 mm needles	2
Sterile 20 GA 1.16IN 1.1 x 30 mm needles	3
Metal curette	_
Wooden tongue depressors	100
Single-way gravity infusion set with air vent	1